

CompTIA Server+

Server and Application Hardening Techniques

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Introduction

Server+

Firewall Configuration

Driver Updates

Disable Unused Services

Host Intrusion Detection System (HIDS)

Host Intrusion Prevention System (HIPS)

Change Management

Patches

BIOS Password

Boot Order

Welcome to the **Server and Application Hardening Techniques** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Technology applications, systems, infrastructure, firmware, and other areas can be made more secure using a set of methods known collectively as system hardening. By removing potential points of entry for attackers and minimizing the attack surface, systems hardening can help lower the system's overall security risk. To reduce the number of entry points for hackers and viruses in your IT infrastructure, users should disable unused accounts, functions, ports, apps, rights, access, etc. In this module, some common techniques used to secure the servers and host operating systems will be discussed.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Server and Host OS Hardening
- Exercise 2 - Host Security
- Exercise 3 - Hardware Hardening
- Exercise 4 - Application Hardening

After completing this module, you should be able to:

- Configure Services
- Apply Driver Updates
- Apply Firewall Configuration
- Explore Windows Defender Antivirus
- Disable AutoRun and AutoPlay

After completing this module, you should have further knowledge of:

- Installing only Required Software
- Third-Party Antivirus and Antimalware
- Host Intrusion Detection System (HIDS) and Host Intrusion Prevention System (HIPS)
- Setting BIOS Password
- Setting Boot Order
- Installing Updates, Patches and Deployment
- Change Management
- Disabling Unneeded Services, Roles, or Features

Exam Objectives

The following exam objectives are covered in this module:

3.5 Given a scenario, apply server hardening methods

- OS hardening
- Application hardening
- Host security
- Hardware hardening
- Patching

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Select **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.7 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)r

Select **Next** to proceed to the first exercise.

Exercise 1 - Server and Host OS Hardening

Even the most meticulously engineered systems may have security flaws. An operating system's default configuration is designed to appeal to the largest intended customer base. Hardening the operating system can increase computer security and limit the attack surface. When a system's attack surface is hardened, the potential for exploits, malware injections, and unauthorized access is also reduced. A vulnerability assessment should be conducted to identify weak areas in the operating system and guide choices on whether they can or should be hardened. Following the vulnerability assessment, the amount of necessary hardening will be determined. Based on the results, efforts must be focused on the areas with the most impact and

the least effort. Too much hardening can make the system less user-friendly, leading to dissatisfaction and users finding workarounds unless extra precautions are taken.

In this exercise, you will learn how to configure services and apply driver updates and firewall configuration. The importance of installing only the required software will also be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Configure Services
- Apply Driver Updates
- Apply Firewall Configuration

After completing this exercise, you should have further knowledge of:

- Installing only Required Software

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)

Task 1 - Configure Services

When discussing computers, services often refer to programs that monitor and respond to network traffic. Other services, such as web servers, file servers, FTP servers, email, proxy servers, Remote Desktops, Open ports, and share access to your computer and data, allow direct access to your machine. Numerous systems are compromised by threat actors that take advantage of security flaws or vulnerabilities in these protocols. The more services you have active on the computer, the greater the risk that it may be compromised in some way.

Administrators will use many methods to harden the system and shrink the attack surface, like using scripts or creating a baseline image that has been hardened.

In this task, you will learn how a service in the local machine can be configured.

Step 1

Connect to **PLABDC01**.

The **Server Manager** window is displayed.

Click the **Tools** menu and select **Services**.

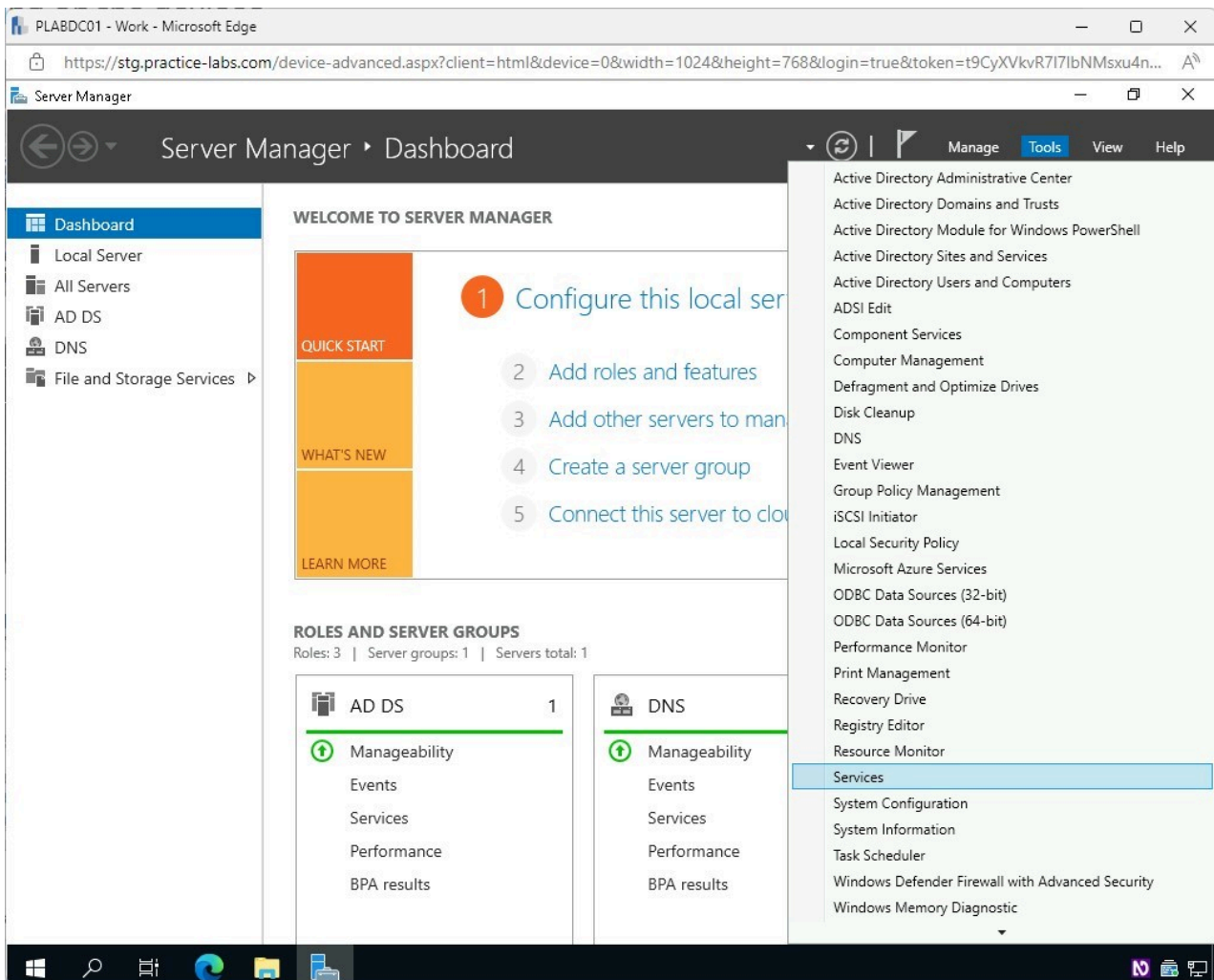


Figure 1.1 Screenshot of PLABDC01: Displaying the Server Manager window and selecting Tools > Services.

Note: Windows 10 does not have a Server manager, but simply selecting the Windows button and searching and selecting the Services app will provide the same features.

Step 2

In the **Services** window, configuring services that are specific to this machine is indicated by **Services (Local)**.

In the right pane, scroll through the list and double-click on **Print Spooler**.

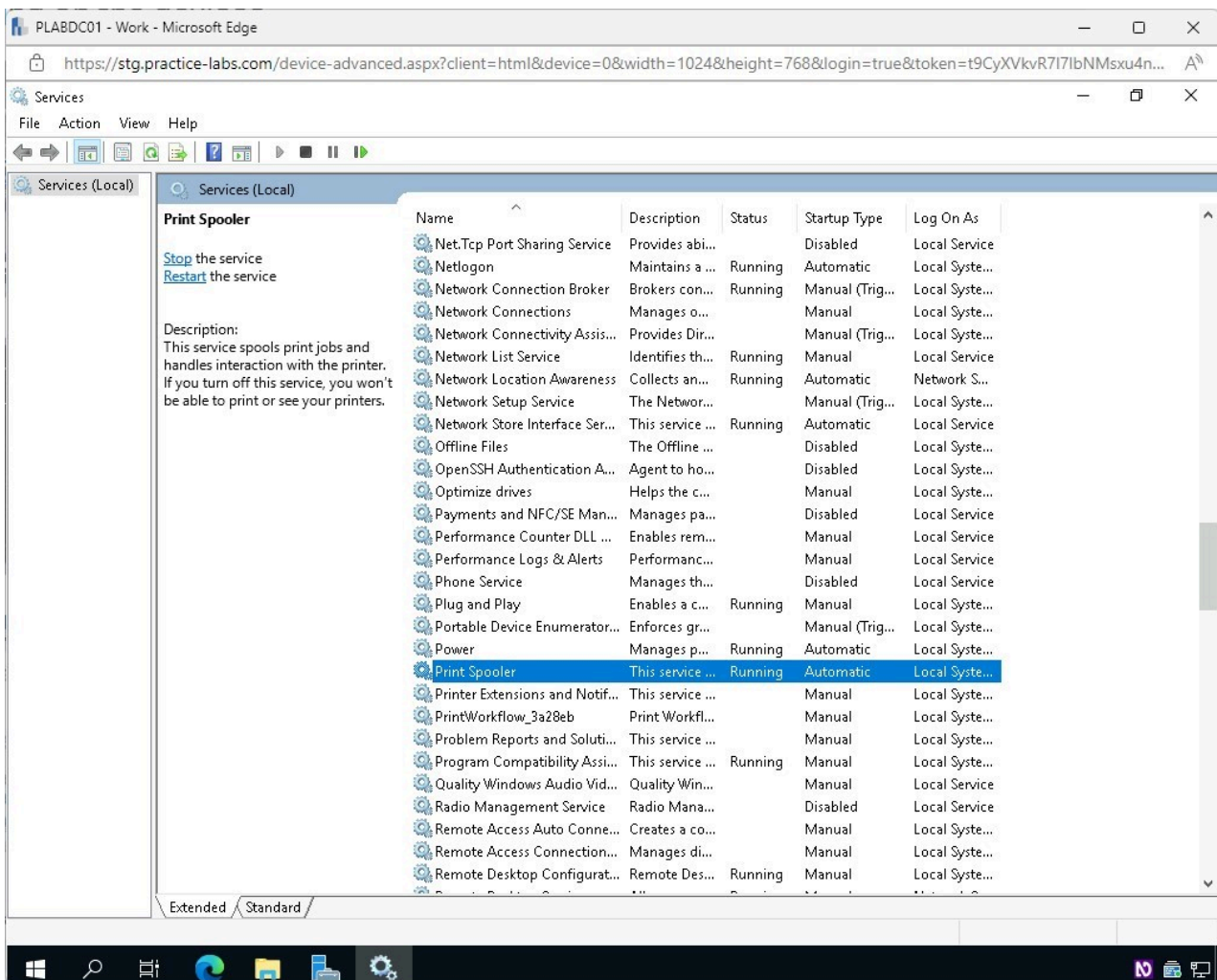


Figure 1.2 Screenshot of PLABDC01: Displaying the Services window, showing Print Spooler selected.

Step 3

In the **Print Spooler Properties (Local Computer) - General** tab, you can view the **Services name**, **Display name**, **Description**, **Path to executable**, **Start-up type**, and **Service status**.

The **Path to executable** is where the file needed to start the service resides.

Start-up type: tells the service how it is to be initiated. The options are:

- **Automatic:** Starts automatically with the operating system.
- **Automatic Delayed:** Starts automatically once the Automatic Services are all started. Helps reduce bottlenecks and speeds up the boot process.
- **Manual:** Needs to be executed to start, not started with the operating system.
- **Disabled:** The service is shut down and will not start.

Note: You can visit <https://learn.microsoft.com/en-us/windows-server/security/windows-services/security-guidelines-for-disabling-system-services-in-windows-server> for a list

of services and Microsoft's recommendations.

Select the **Log On** tab.

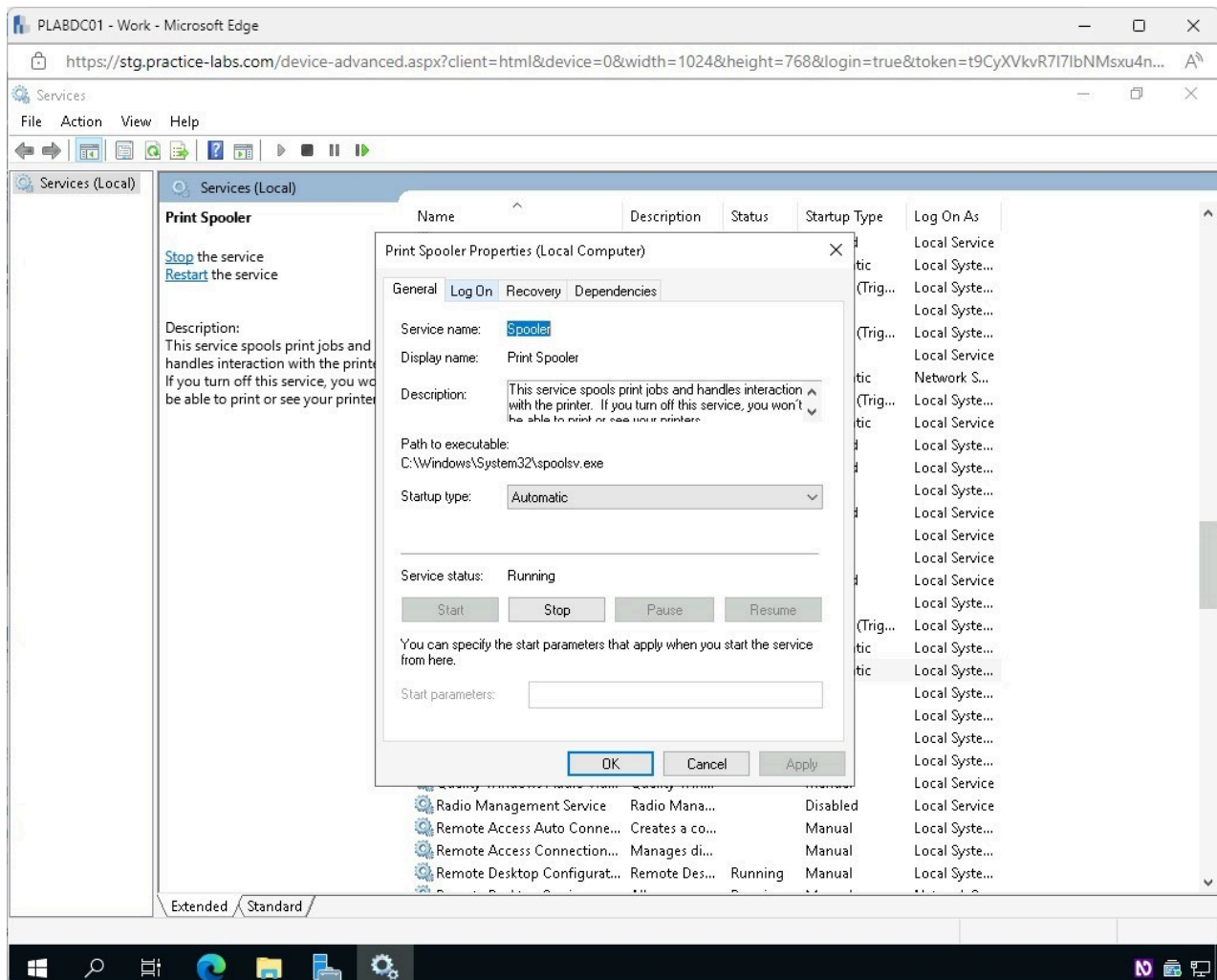


Figure 1.3 Screenshot of PLABDC01: Displaying the Print Spooler Properties (Local Computer) - General tab, showing the Log On tab selected.

Step 4

The **Print Spooler Properties (Local Computer) - Log On** tab is displayed.

Because Windows Services do not end when a user logs out, access to these resources remains a capability. If you want the service to be able to access a certain user's protected resources even when they aren't signed in, you must supply the user credentials under the service's **Log On** tab.

Select the **Recovery** tab.

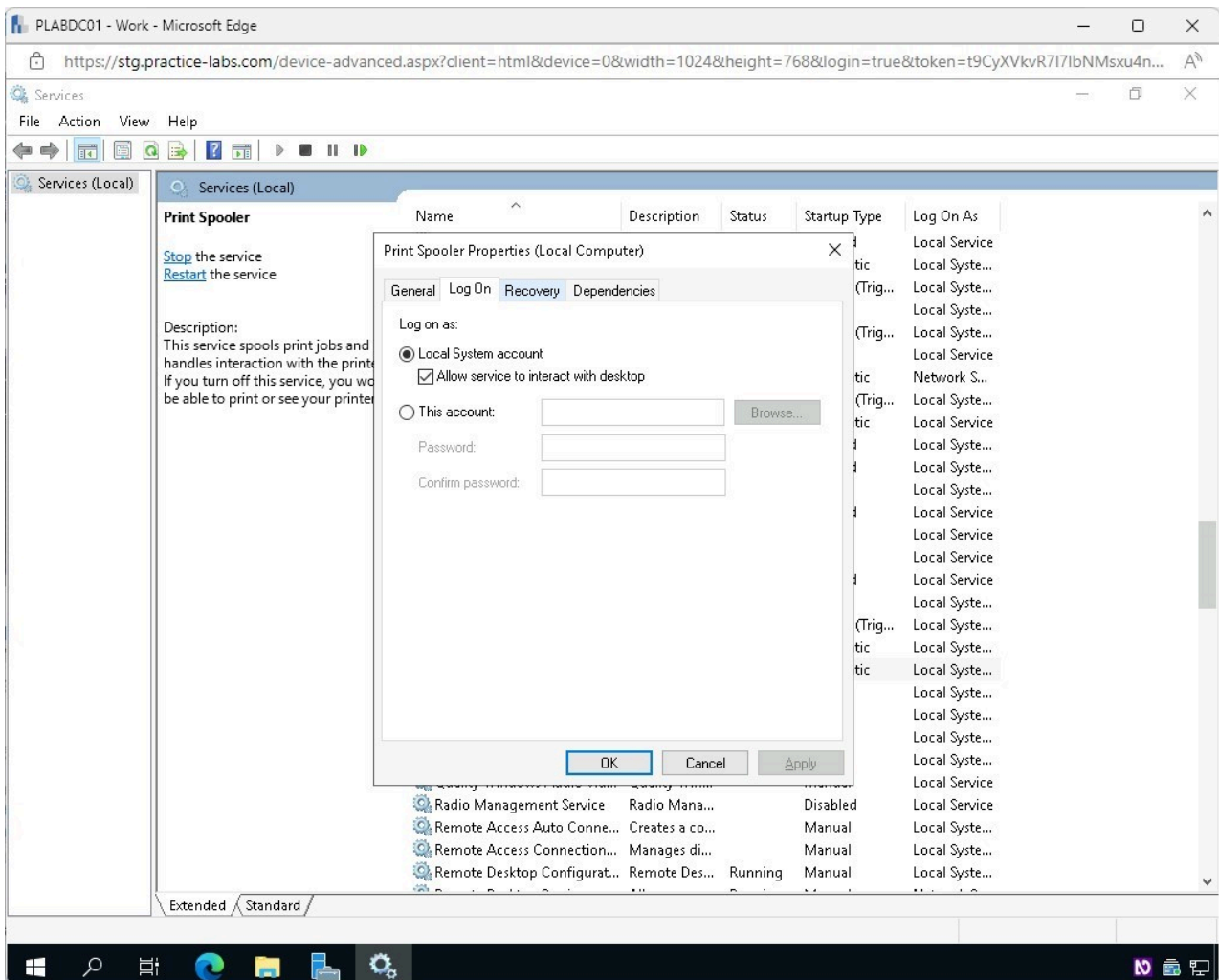


Figure 1.4 Screenshot of PLABDC01: Displaying the Print Spooler Properties (Local Computer) Log On tab, showing the Recovery tab selected.

Step 5

In the **Print Spooler Properties (Local Computer) - Recovery** tab, you can configure the actions the computer will take if the service start-up fails. There are options for first, second, and subsequent failures and reset/restart counters.

The options for the drop-downs are

- **Take No Action**
- **Restart the Service**
- **Run a Program**
- **Restart the Computer**

Select the **Dependencies** tab.

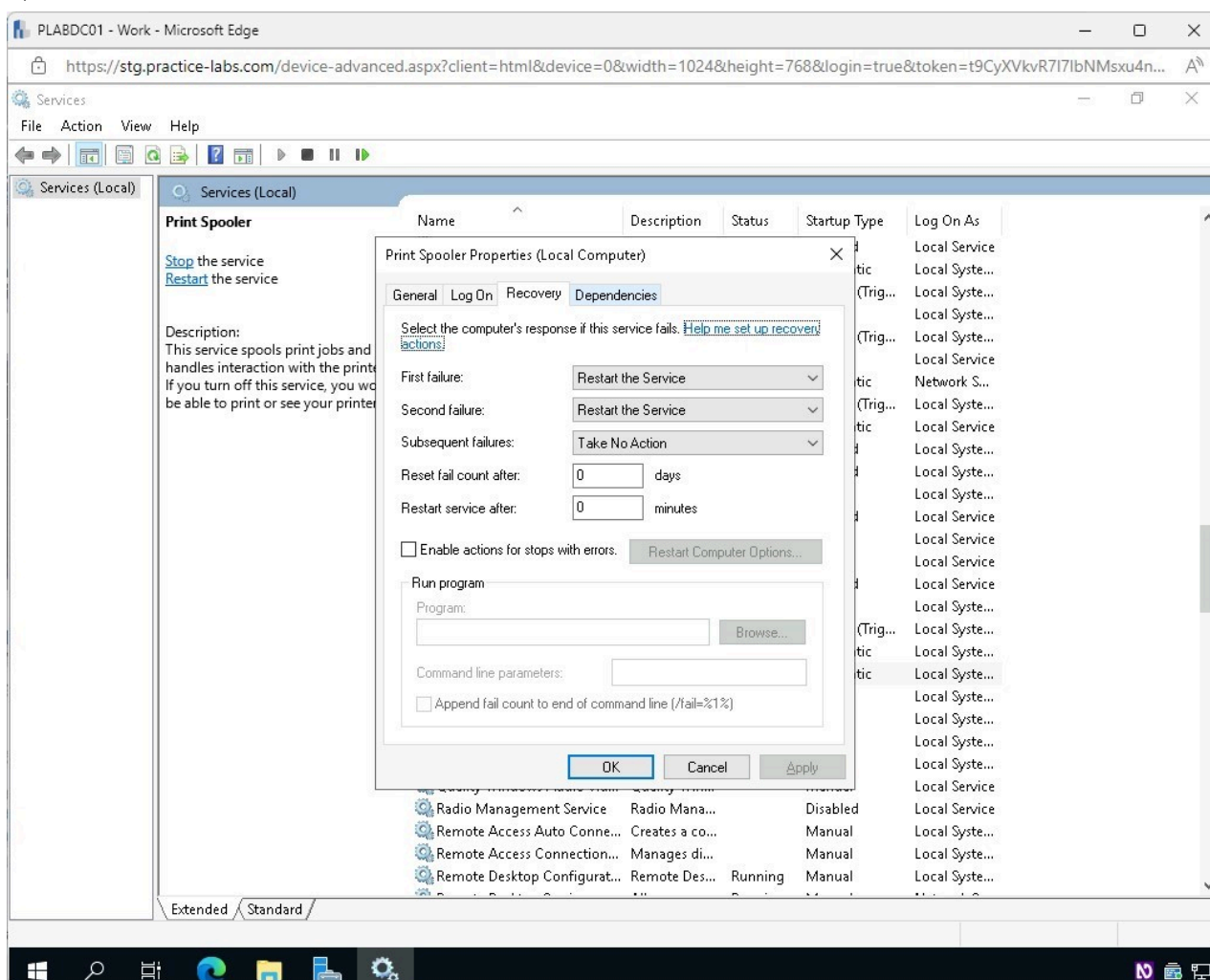


Figure 1.5 Screenshot of PLABDC01: Displaying the Print Spooler Properties (Local Computer) - Recovery tab, showing the Dependencies tab selected.

Step 6

In the **Print Spooler Properties (Local Computer) - Dependencies** tab, other services that the Print Spooler needs running for it to function are shown.

Click **OK**.

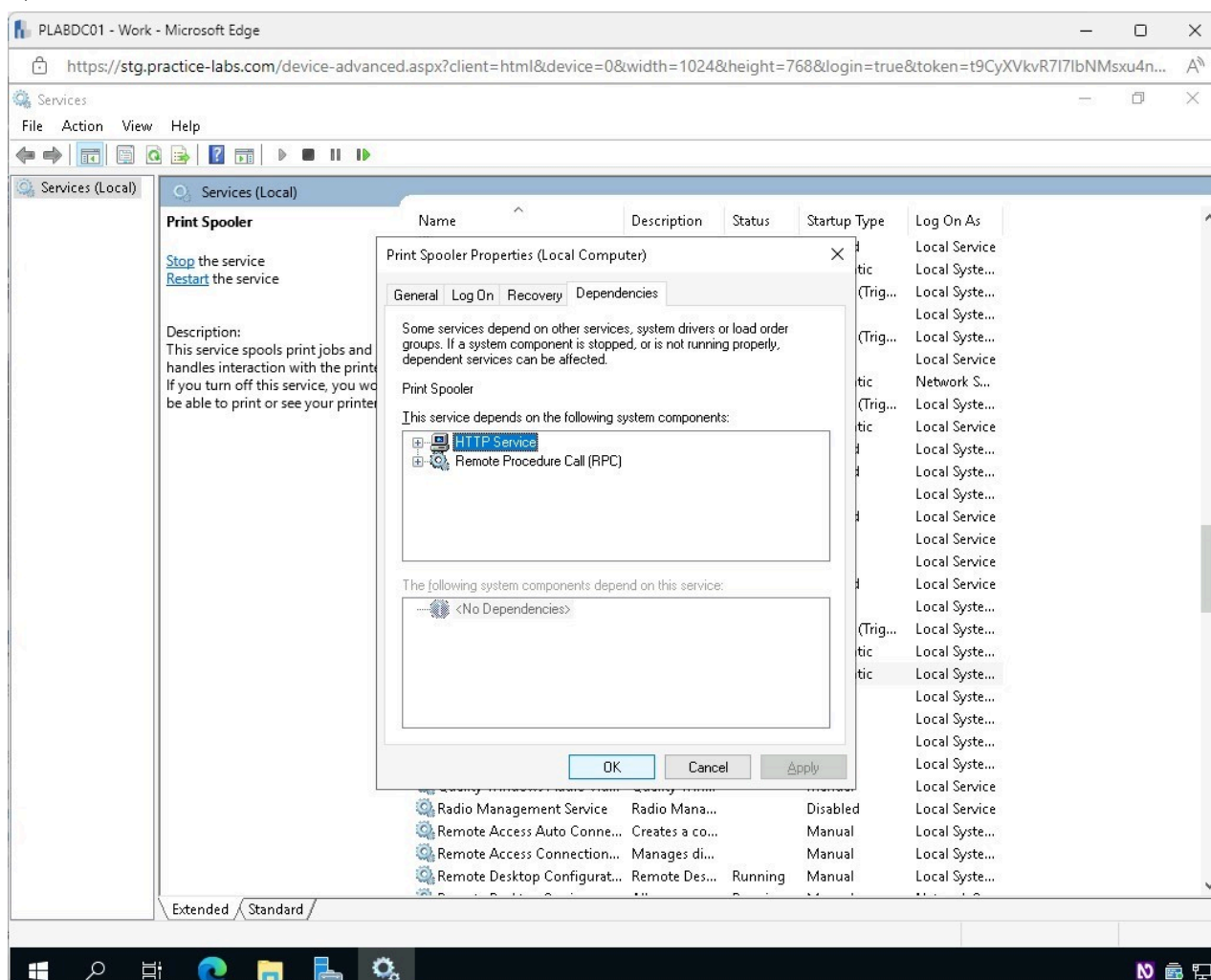


Figure 1.6 Screenshot of PLABDC01: Displaying the Print Spooler Properties (Local Computer) - Dependencies tab, showing the OK button selected.

Step 7

Back on the **Services** window, explore other services that are listed.

Once finished, close the **Services** window.

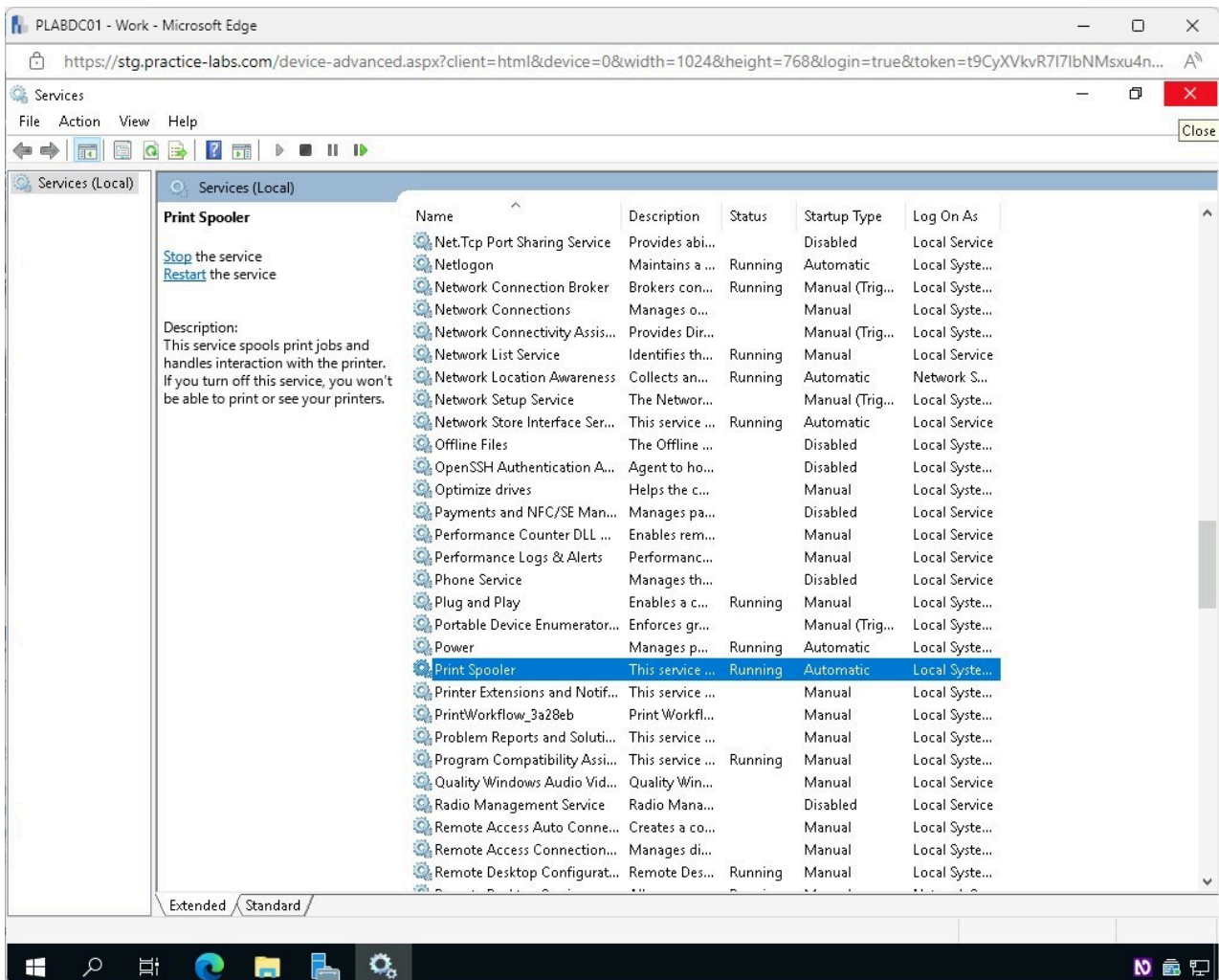


Figure 1.7 Screenshot of PLABDC01: Displaying closing the Services window.

Minimize the **Server Manager** window.

Task 2 - Apply Driver Updates

Maintaining up-to-date device drivers is necessary. This will not only keep your computer running smoothly but also helps increase security. Using outdated driver software exposes your machine to viruses and other harmful code. These are critical vulnerabilities that a competent hacker may easily exploit. After the initial release and subsequent updates, device drivers are safe, but eventually, hackers will find and exploit any security holes that may have been included in the code. Hackers and cybercriminals will eventually find enough vulnerabilities to compromise the computer itself. For instance, hackers who find security holes in the kernel might exploit them to take control of the entire operating system, putting at risk home computers and those used in businesses at risk. In this case, your computer is in danger. Since this is the case, updating your device drivers on a regular basis is something you must do. The most recent driver version will always be the most secure because of the latest patches for security holes that are included in software updates for device drivers.

In this task, you will learn how driver updates can be applied to a device.

Step 1

Ensure you are connected to **PLABDC01**.

Right-click the Windows **Start** charm and select **Device Manager**.

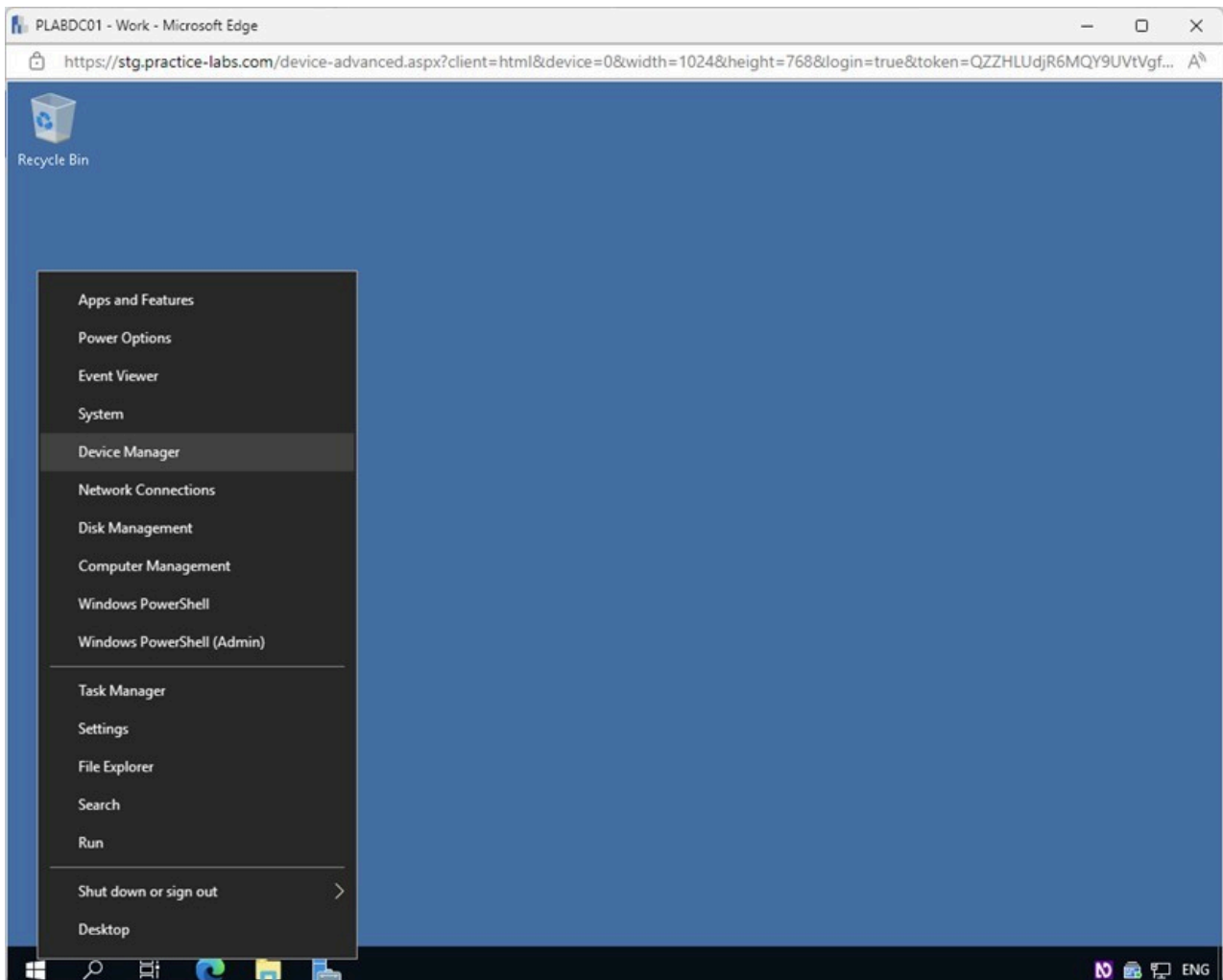


Figure 1.8 Screenshot of PLABDC01: Displaying right-clicking the Windows Start charm and selecting Device Manager.

Step 2

In the **Device Manager** window, a list of all the hardware connected to the Server right now is shown.

Only approved hardware should be used on servers. It will also help if extra hardware is not used. Following these two rules can help make the machine less vulnerable to attacks.

Hardware gives a computer more features, but it can also corrupt the computer or open ports and services that weren't meant to be opened. This leaves the computer open to attack.

Uninstall any undesired hardware by right-clicking on the device and selecting **Uninstall device** from the context menu. Not all hardware can be uninstalled.

Expand **Processors**. Right-click on the first processor in the list and select **Properties**.

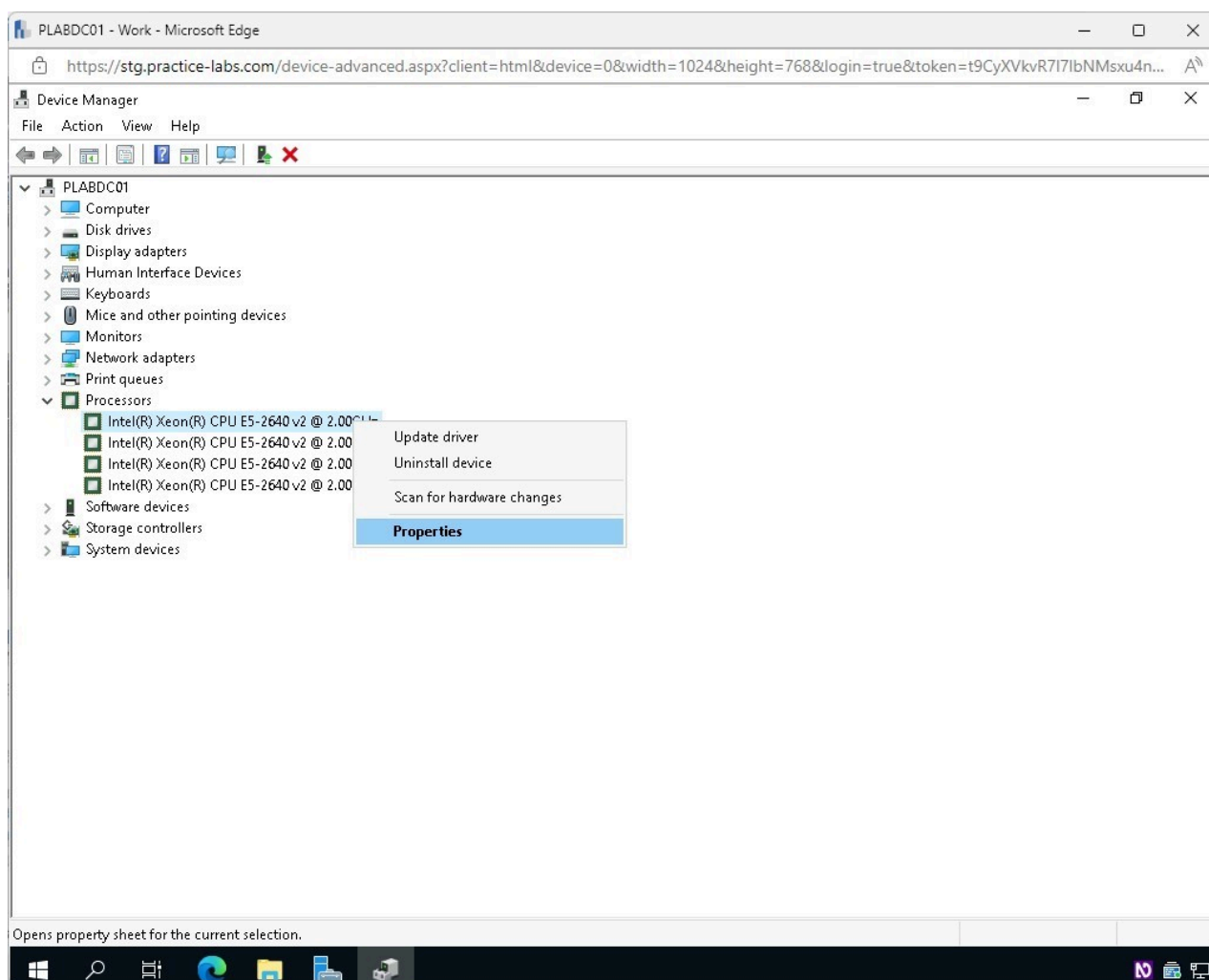


Figure 1.9 Screenshot of PLABDC01: Displaying the Device manager window. A Processor device is right-clicked and Properties selected.

Step 3

In the <**Processor**> **Properties - General tab**, the device's status, type, manufacturer, and location is displayed.

Select the **Driver** tab.

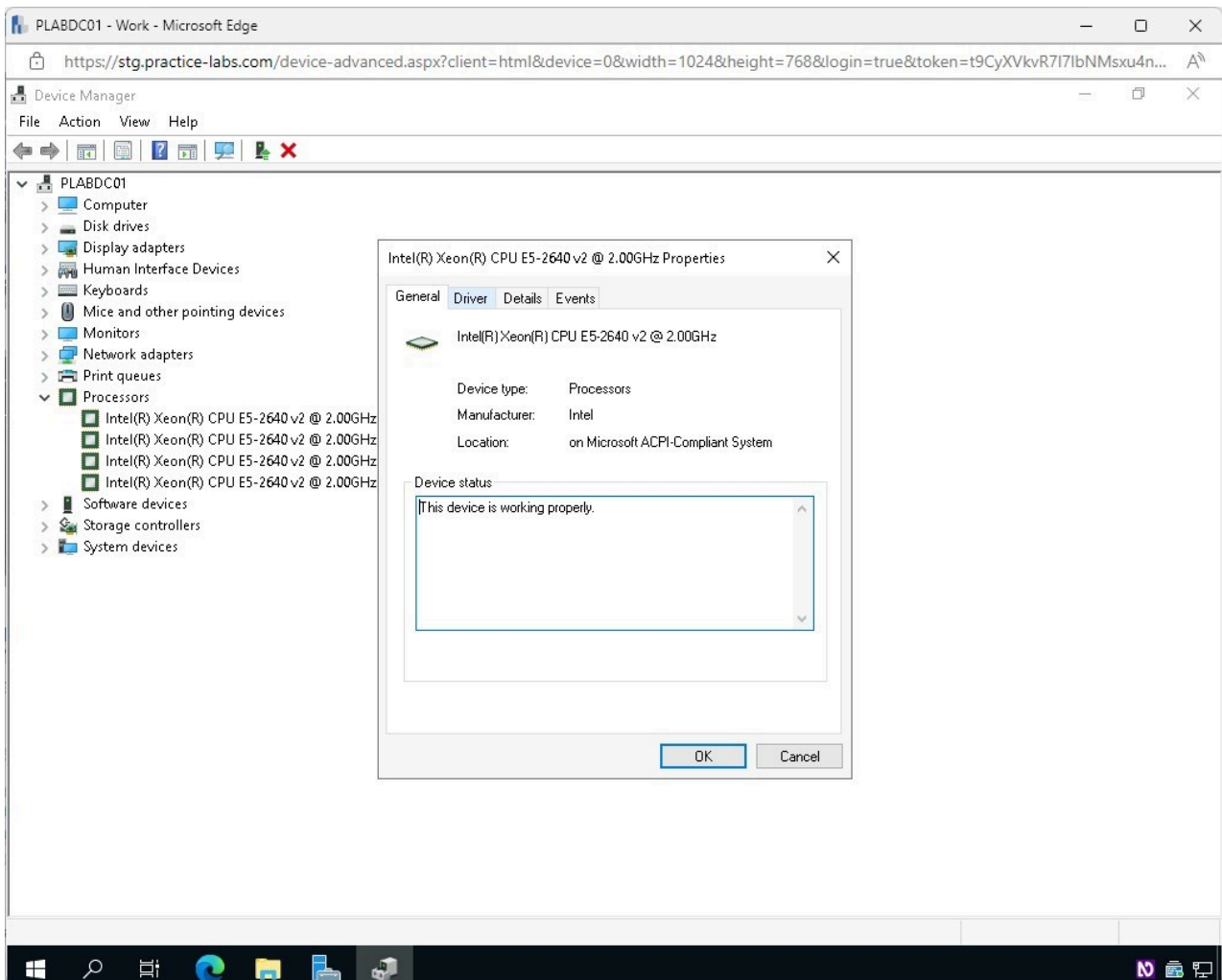


Figure 1.10 Screenshot of PLABDC01: Displaying the <Processor> Properties - General tab, showing the Driver tab selected.

Step 4

In the **Driver** tab, you can get the driver details, update the driver, roll-back the driver if it has been updated, disable the device, and uninstall the device.

Click **Update Driver**.

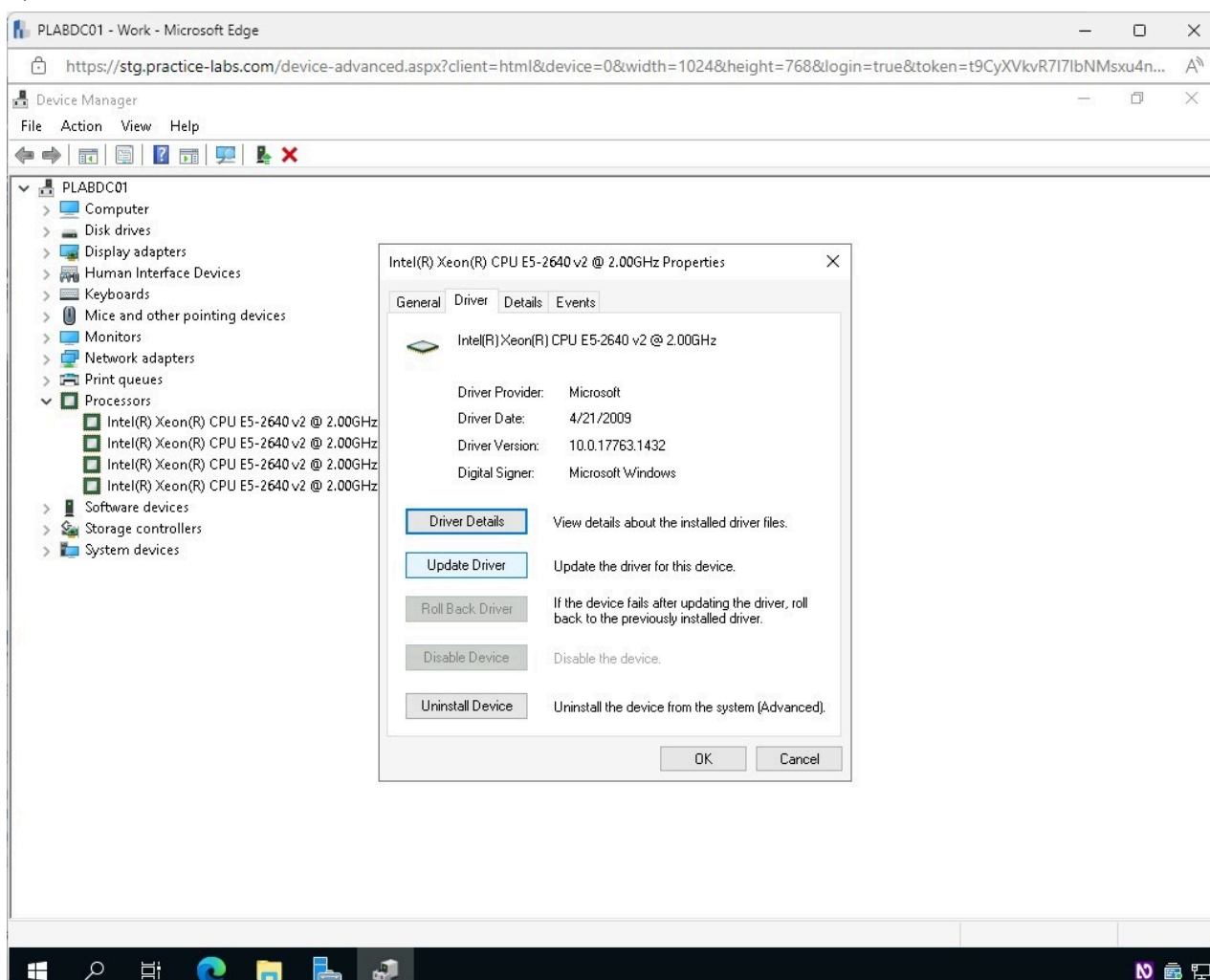


Figure 1.11 Screenshot of PLABDC01: Displaying the <Processor> Properties - Driver tab, showing Update Driver selected.

Step 5

In the **Update Drivers - How do you want to search for drivers?** window, select **Search automatically for updated driver software**.

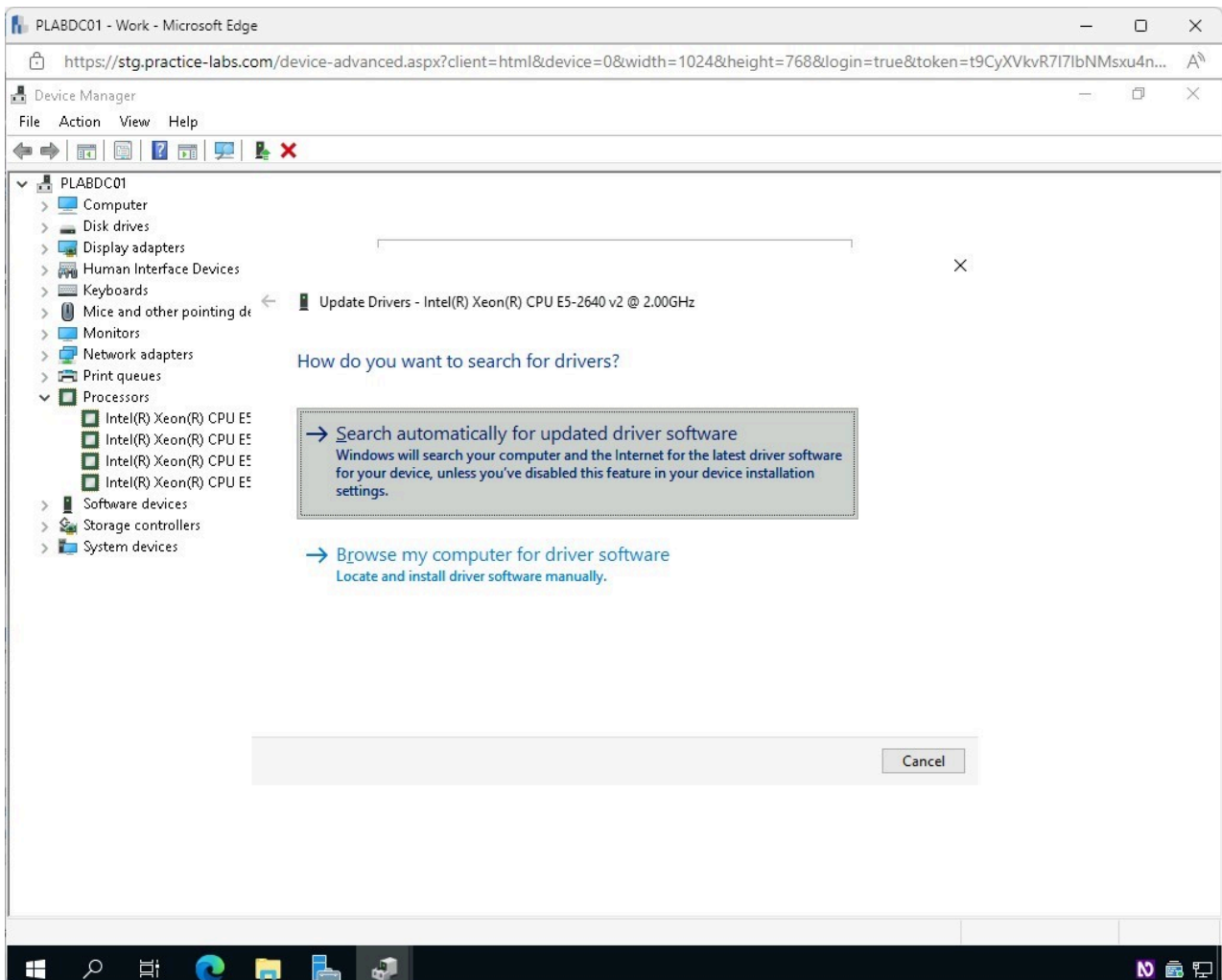


Figure 1.12 Screenshot of PLABDC01: Displaying the Update Drivers - How do you want to search for drivers? window, showing the required option selected.

Step 6

In the **Update Drivers** window, notice that you get the message - **The best drivers for your device are already installed.**

Click **Close**.

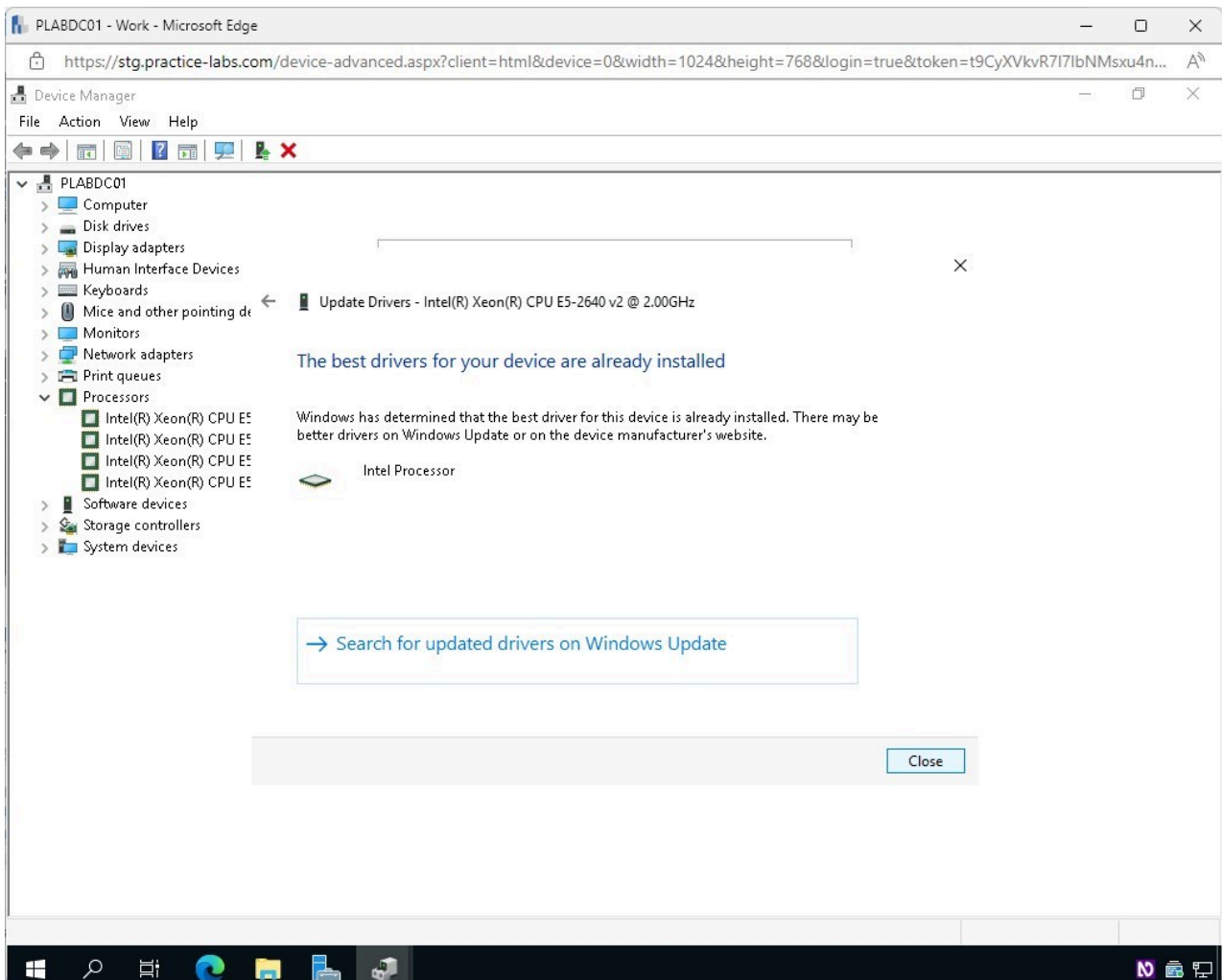


Figure 1.13 Screenshot of PLABDC01: Displaying the Update Drivers window, showing Close selected.

Step 7

Back on the <**Processor**> **Properties** window, select the **Details** tab.

Here, you can get complete details about the processor by selecting various properties from the **Property** drop-down.

Click the **Property** drop-down menu and select **Driver date** from the list.

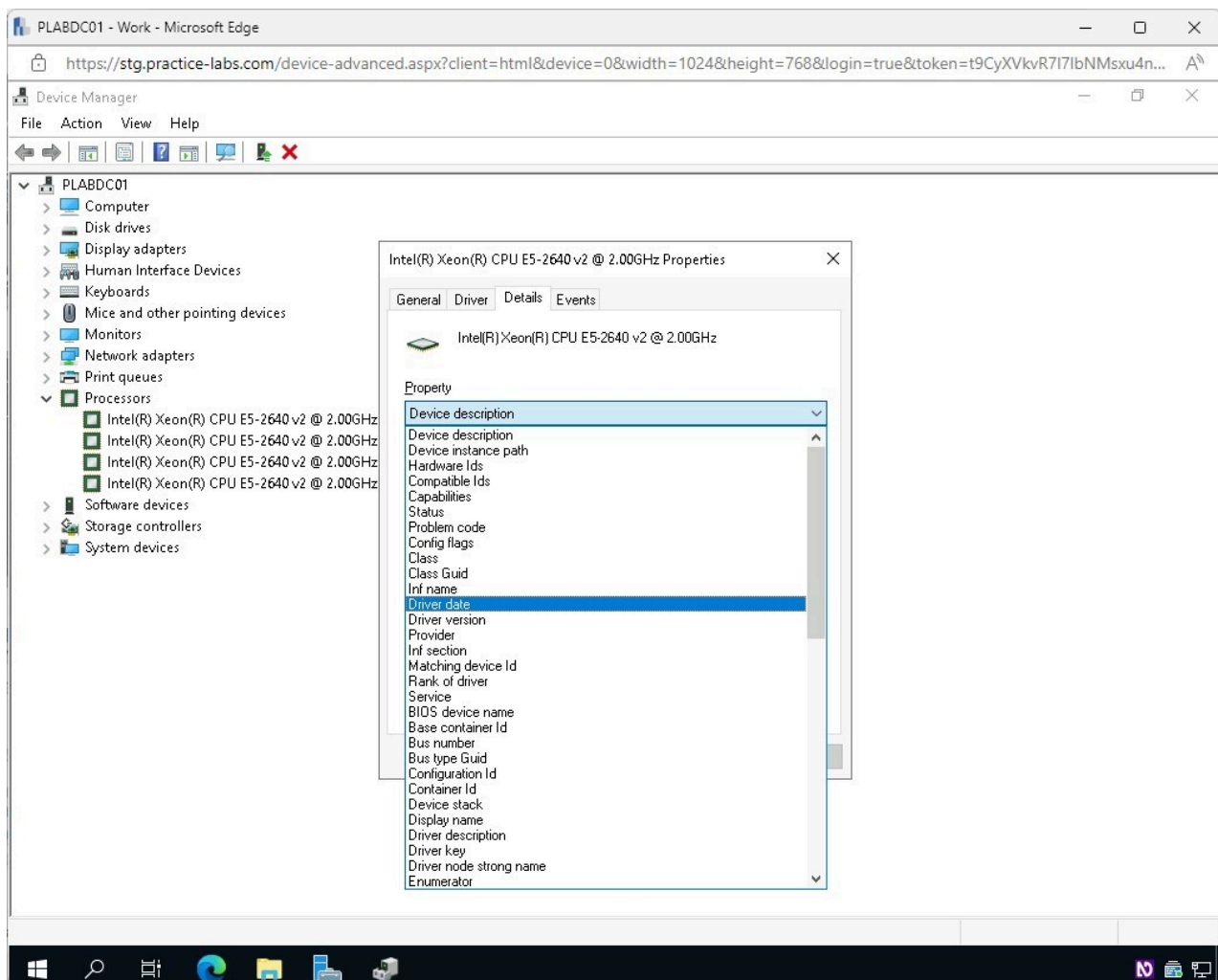


Figure 1.14 Screenshot of PLABDC01: Displaying the <Processor> Properties - Details tab. The Property drop-down menu is accessed, and Driver date is selected.

Step 8

The driver publishing date for the processor is displayed in the **Value** pane.

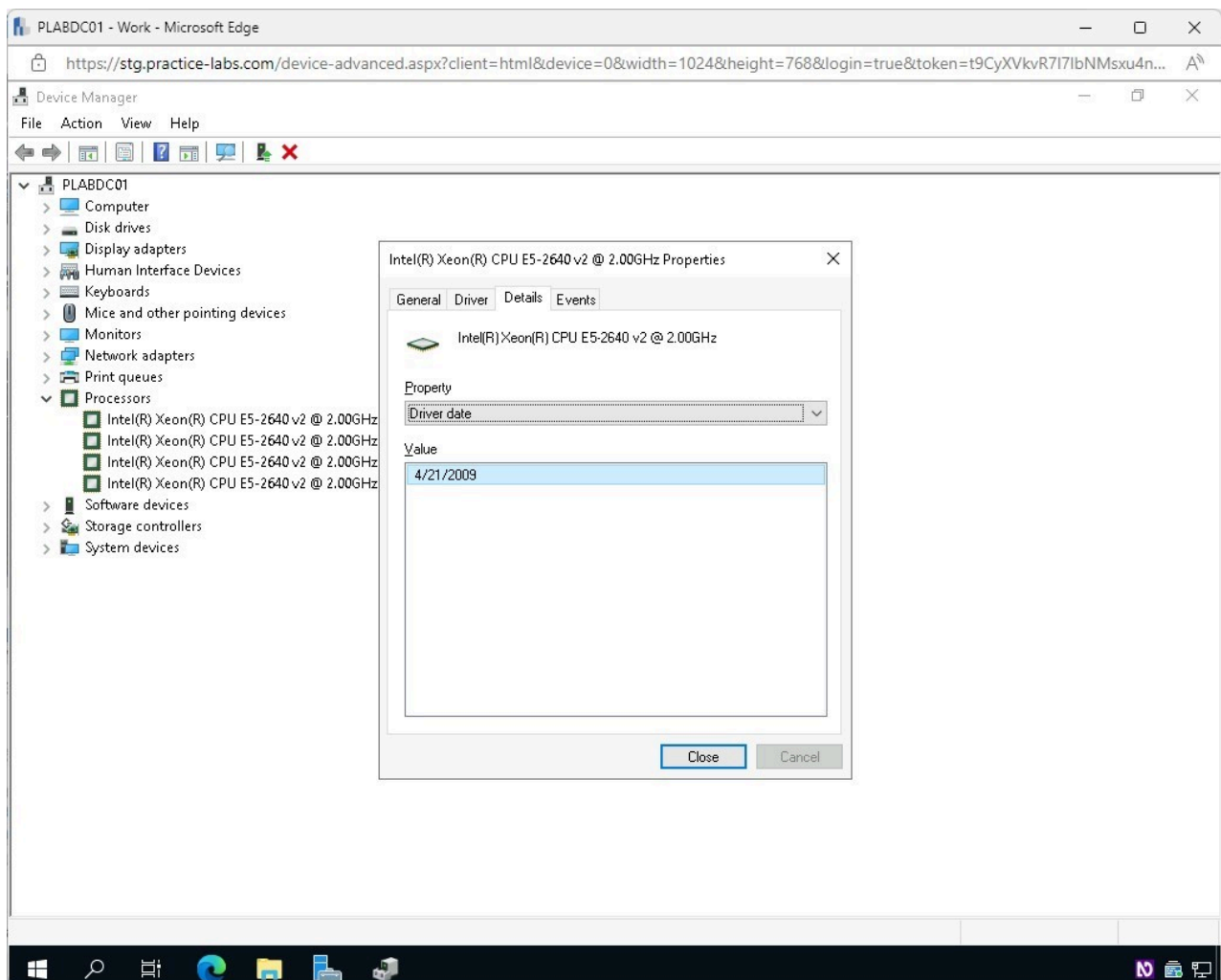


Figure 1.15 Screenshot of PLABDC01: Displaying the <Processor> Properties - Details tab, showing the required date in the Value pane.

Step 9

Click the **Property** drop-down and select **BIOS device name** from the list.

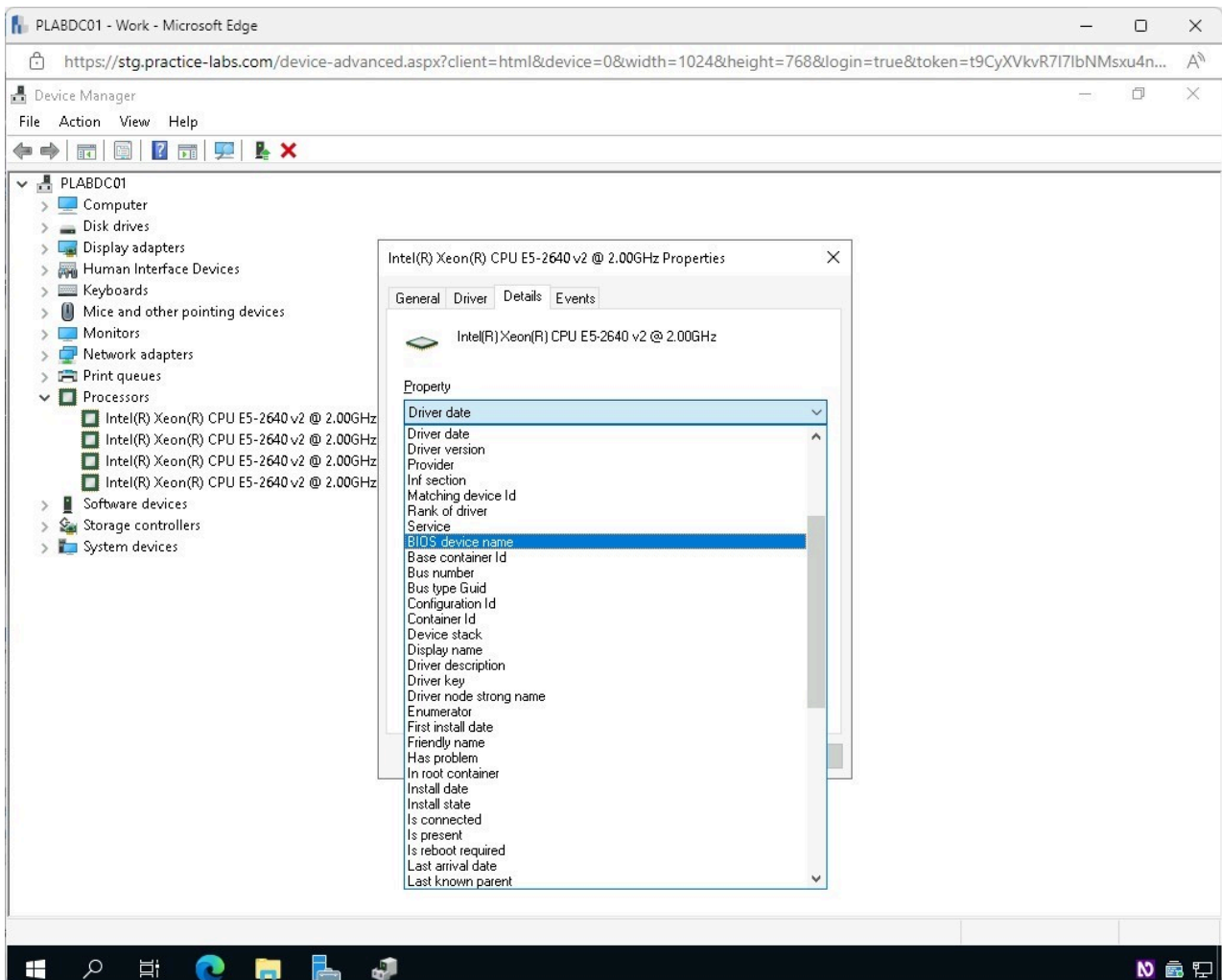


Figure 1.16 Screenshot of PLABDC01: Displaying the <Processor> Properties - Details tab. The Property drop-down menu is accessed, and BIOS device name is selected.

Step 10

The BIOS name for the processor is displayed in the **Value** pane.

Feel free to select other options in the Property drop-down to learn more about the options.

Select the **Events** tab when done.

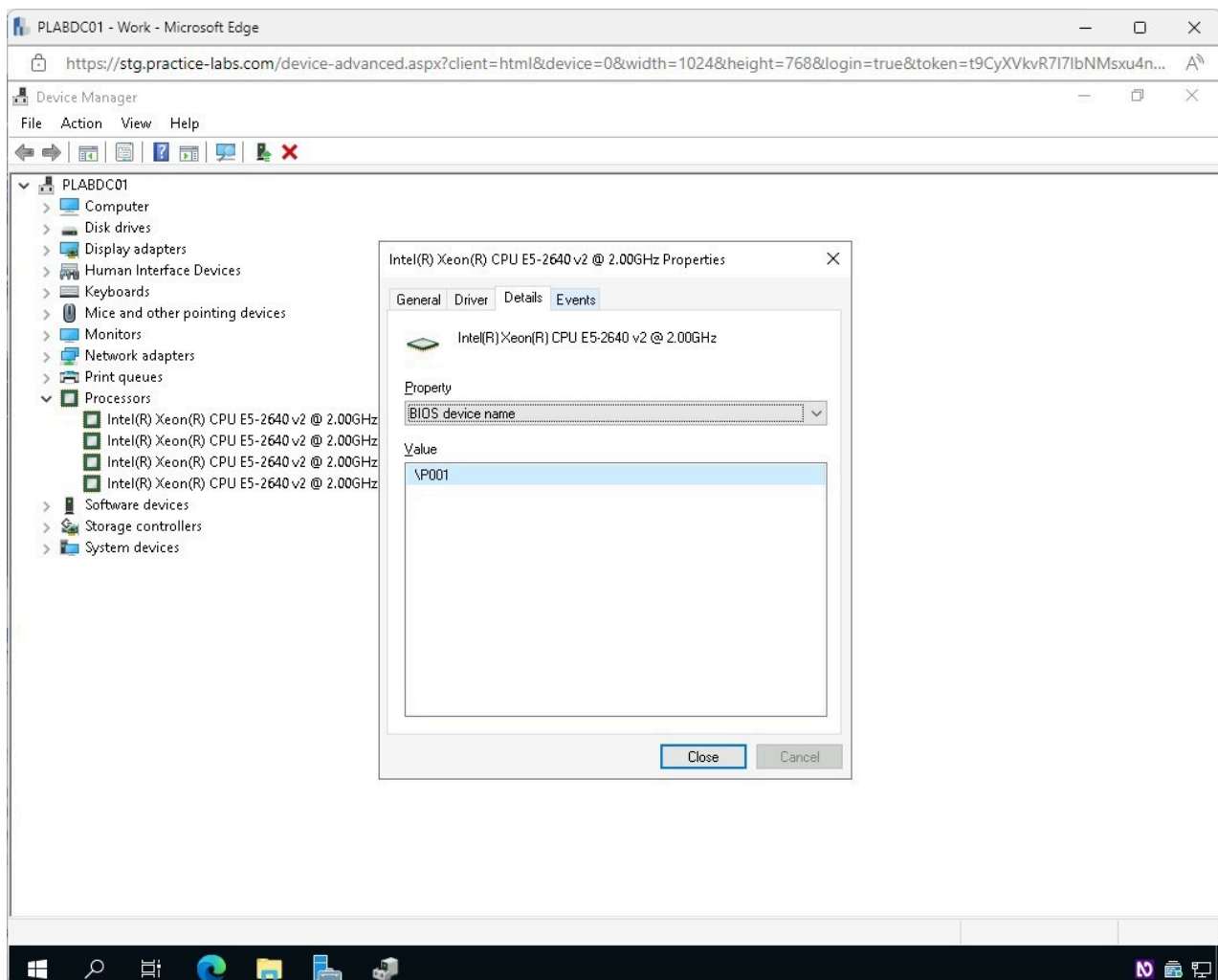


Figure 1.17 Screenshot of PLABDC01: Displaying the <Processor> Properties - Details tab, showing the Events tab selected.

Step 11

In the **Events** tab, events specific to the processor that has taken place can be viewed.

Click **Close**.

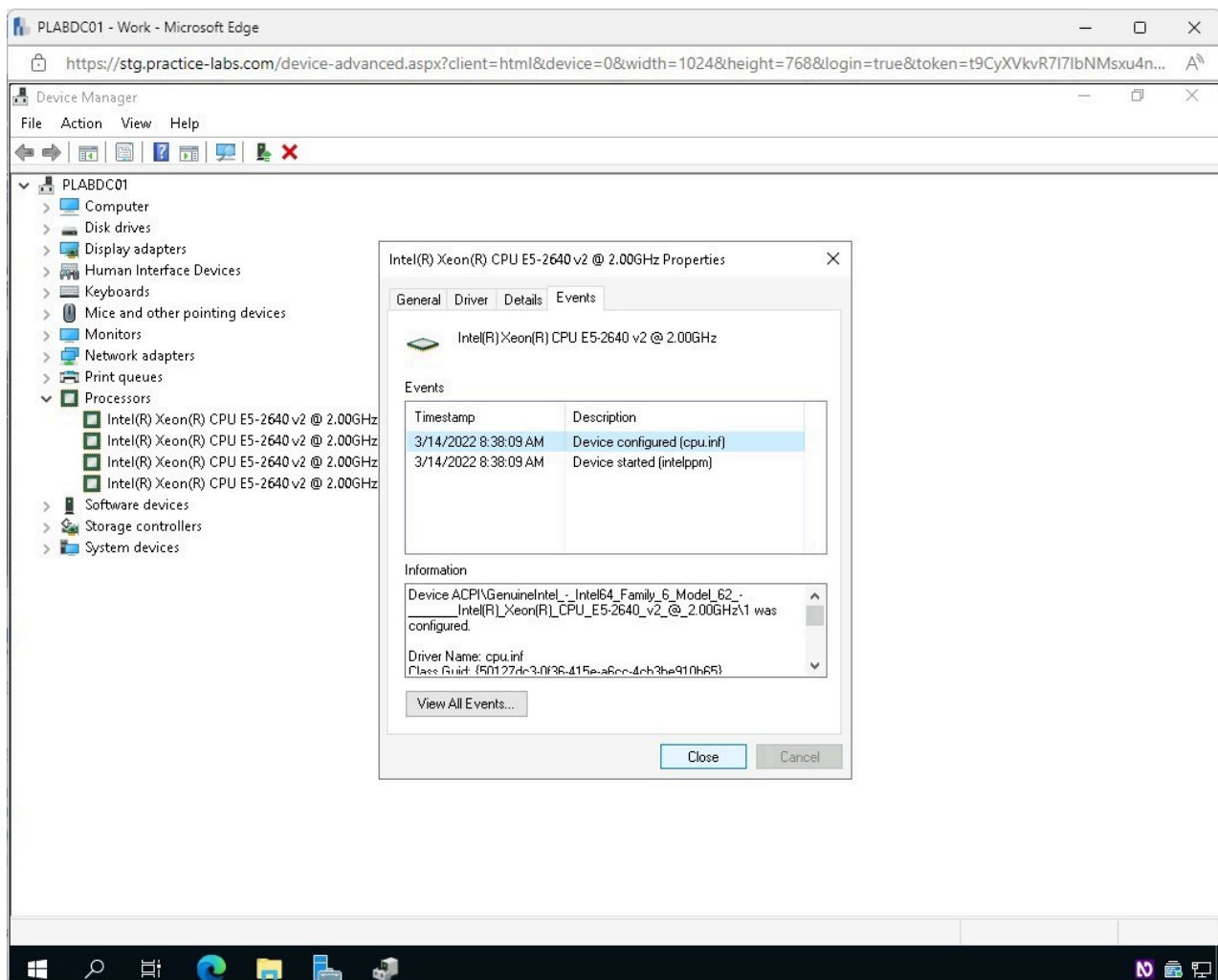


Figure 1.18 Screenshot of PLABDC01: Displaying the <Processor> Properties - Events tab, showing Close selected.

Step 12

Close the **Device Manager** window.

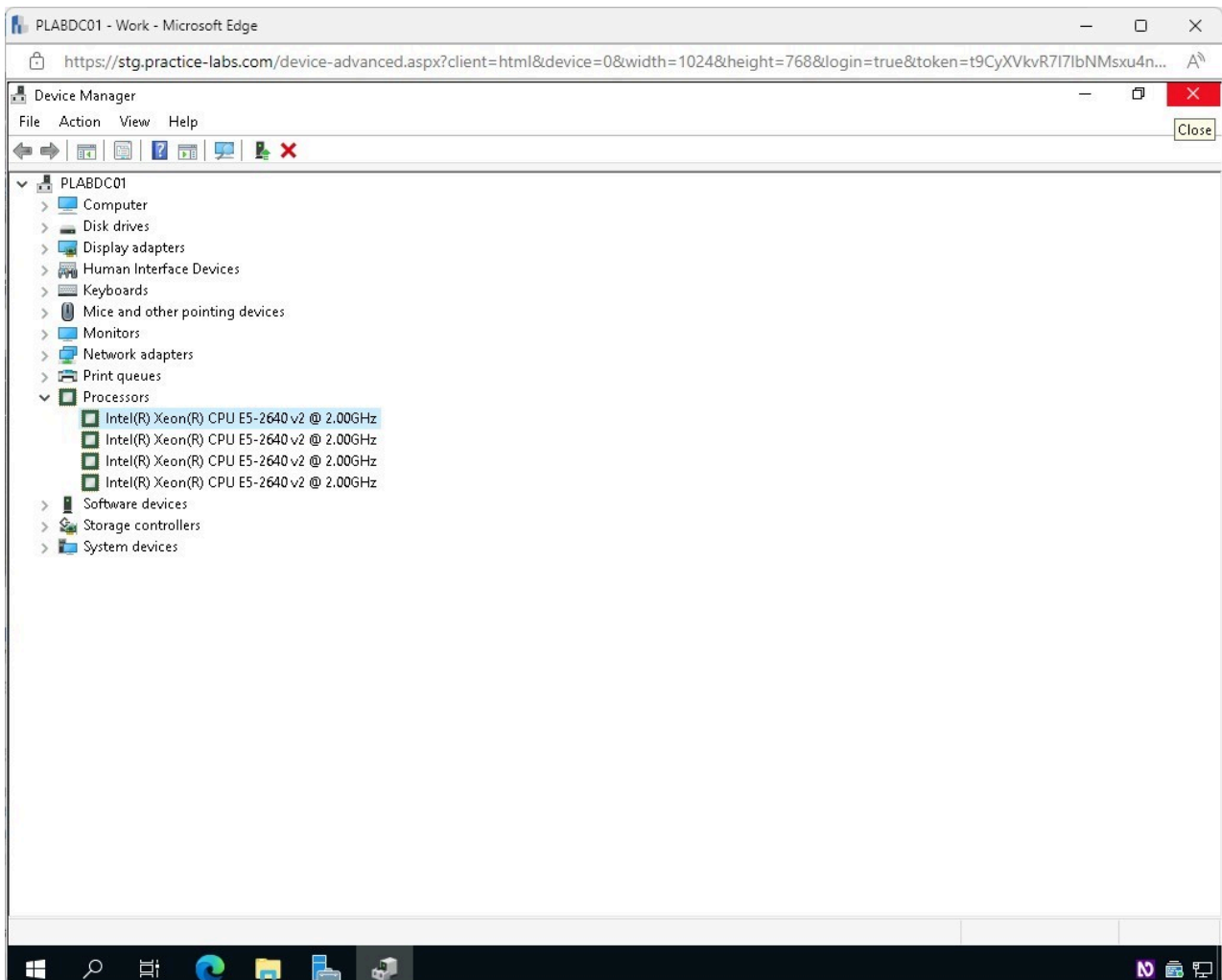


Figure 1.19 Screenshot of PLABDC01: Displaying closing the Device Manager window.

Task 3 - Firewall Configuration

Windows Defender Firewall is a host-based firewall that applies security on a computer by blocking unauthorized access to its services and decreases its exposure to potentially destructive network probes when connected to the Internet or a local intranet. Network security firewalls are used to regulate web traffic and prevent the spread of online threats.

It is possible to examine the audit logs of certain firewalls to learn which connections and types of traffic were permitted or denied. When it comes to protecting sensitive information, firewalls are crucial to provide a strong virtual barrier between your internal network and external threats. Limiting VPN access, auditing access, turning off unnecessary services & closing those ports, and updating and patching all systems are all ways to strengthen firewall protection.

Ports are used to send and receive Internet data. There are a total of 65,535 TCP and UDP ports. Every internet-enabled service necessitates using specified ports. For example, RDP functions on port 3389 or SSH functions on port 22. Cybercriminals may utilize legitimate services to exploit flaws in code or malicious services deployed by malware to get access to sensitive data through open ports. Closed ports are inaccessible to cybercriminals. System

administrators can scan for and shut open data ports that aren't being used by a reliable service. After identifying which ports must be kept open, the administrator might look for others that may be vulnerable to intrusions. Scanners are available for free online.

Windows Defender Firewall is managed using two applications. Namely, Windows Defender Firewall (known as Windows Defender Security Center in Windows 10) and Windows Defender Firewall with Advanced Security. With Windows Defender Firewall and Advanced Security, you can manage incoming and outgoing connections with fine-grained control using the rule-creation tools. To make IPsec policies and isolate networks, it provides Connection Security Rules.

Internet traffic entering and leaving a private network is subject to inspection from firewalls, which act as gated borders or gateways.

In this task, you will learn about managing Windows defender firewall.

Step 1

Ensure you are connected to **PLABDC01**.

Click the **Start** charm and type the following:

Defender Firewall

Select **Windows Defender Firewall** from the **Best match** pop-up menu.

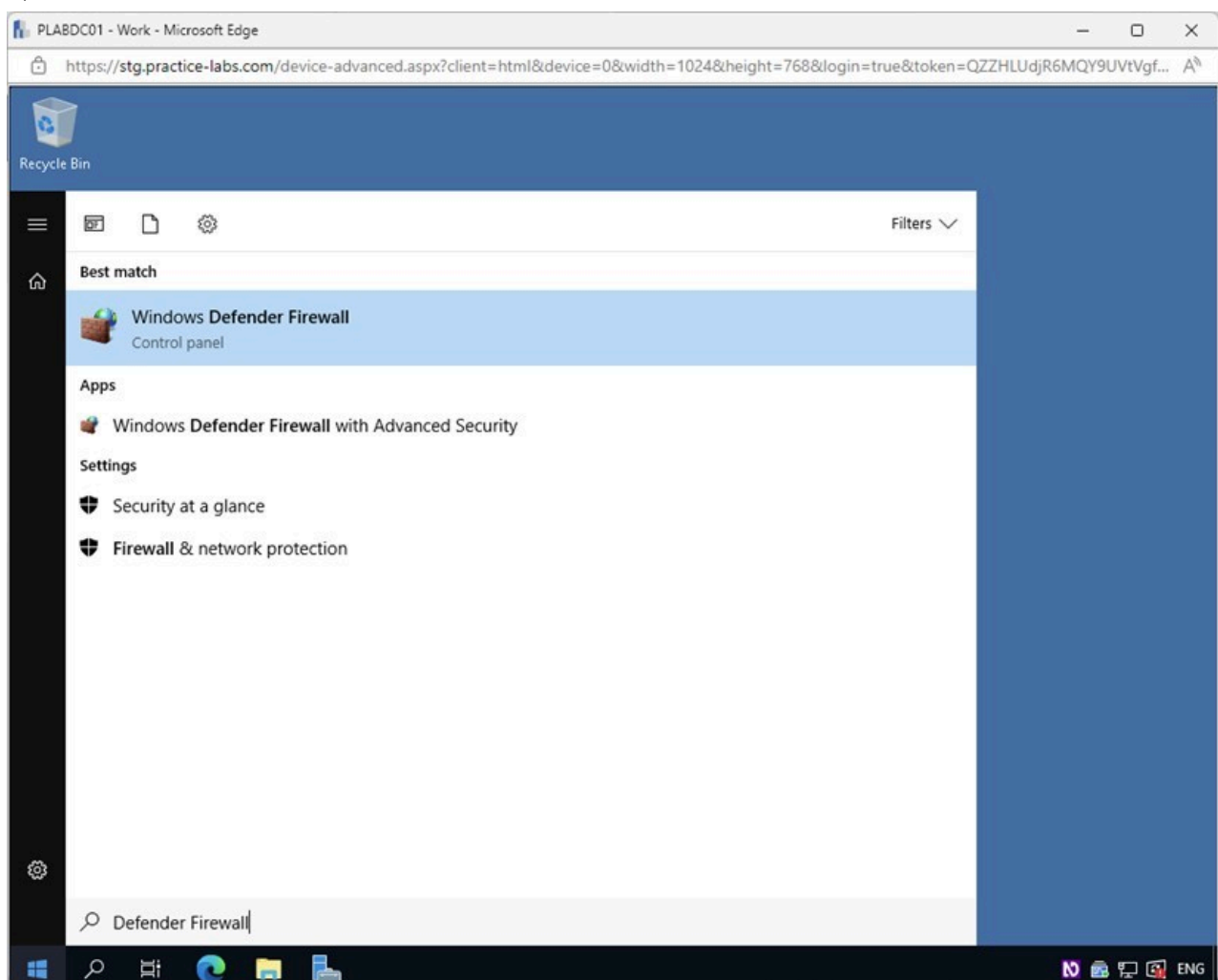


Figure 1.20 Screenshot of PLABDC01: Displaying selecting Windows Defender Firewall from the Best Match pop-up menu.

Step 2

In the **Windows Defender Firewall** window, notice that the Firewall is **active** and all systems are protected indicated by the **Green Checkmark**.

Click the **Turn Windows Defender Firewall on or off** link on the left pane.

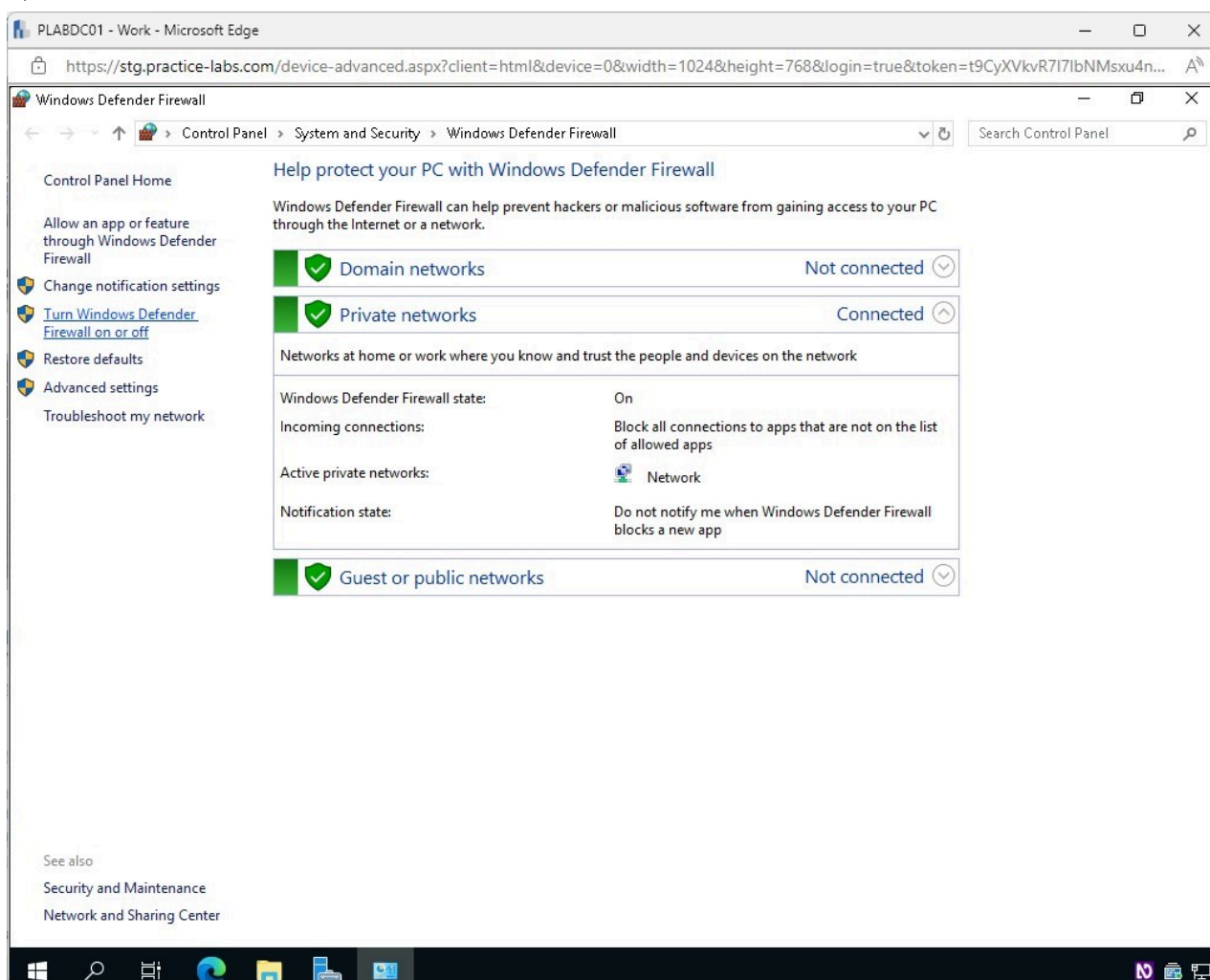


Figure 1.21 Screenshot of PLABDC01: Displaying the Windows Defender Firewall window, showing the Turn Windows Defender Firewall on or off link selected.

Step 3

In the **Customize Settings** window, select **Turn off Windows Defender Firewall (not recommended)** for **Domain**, **Private** and **Public network** settings.

Select **OK**.

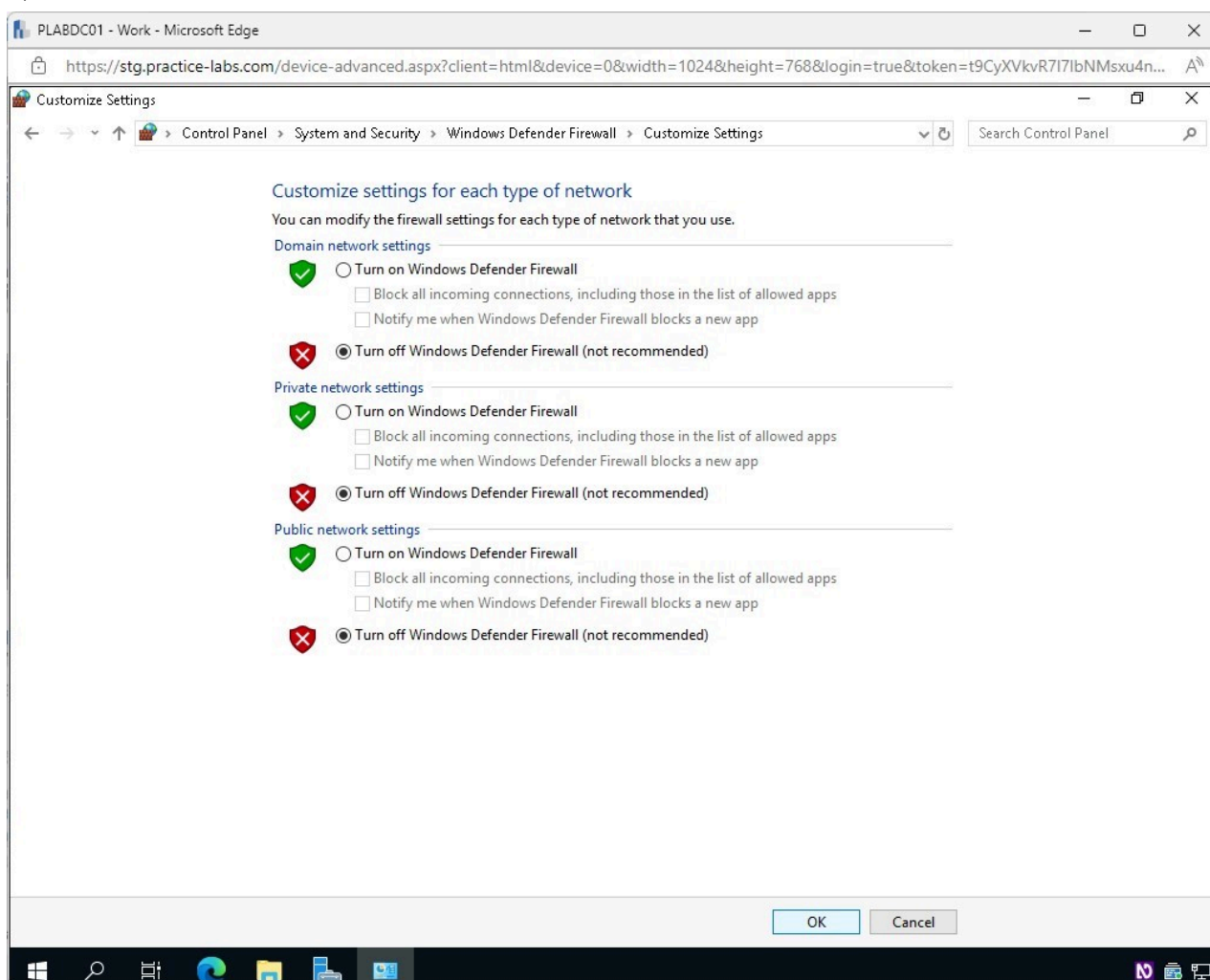


Figure 1.22 Screenshot of PLABDC01: Displaying the Customize Settings window, showing the required settings performed and the OK button selected.

Step 4

Notice that you now have a **Red** warning next to all the settings, which means the system is vulnerable.

To easily remedy this, click the **Use recommended settings** button.

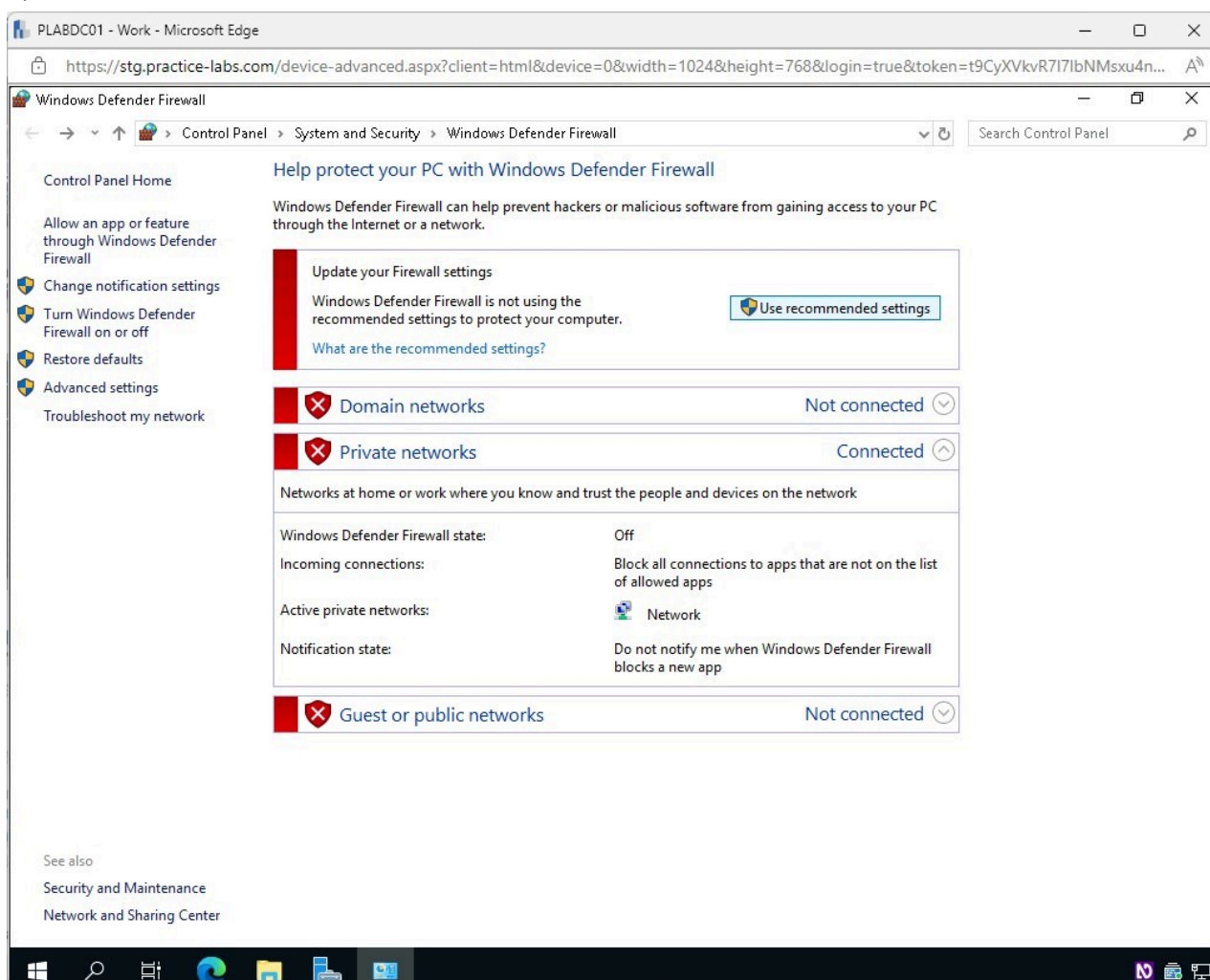


Figure 1.23 Screenshot of PLABDC01: Displaying the Windows Defender Firewall window, showing the Use recommended settings button selected.

Step 5

The Firewall is now restored. Notice that everything is back to green and protected.

Click the **Advanced settings** link on the left pane.

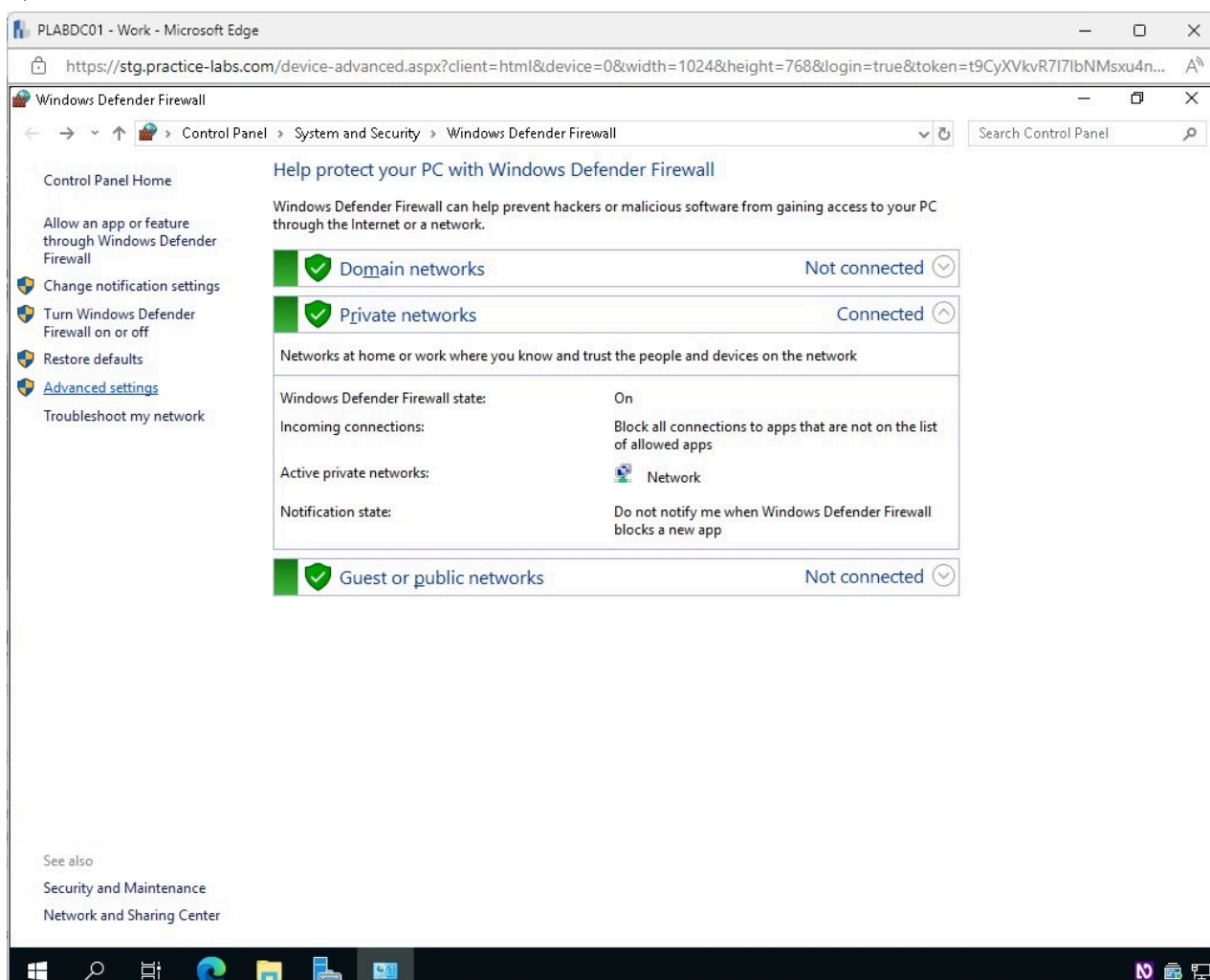


Figure 1.24 Screenshot of PLABDC01: Displaying the Windows Defender Firewall window, showing the Advanced Settings link selected.

Step 6

From the **Windows Defender Firewall with Advanced Security** window, select **Inbound Rules** on the left pane.

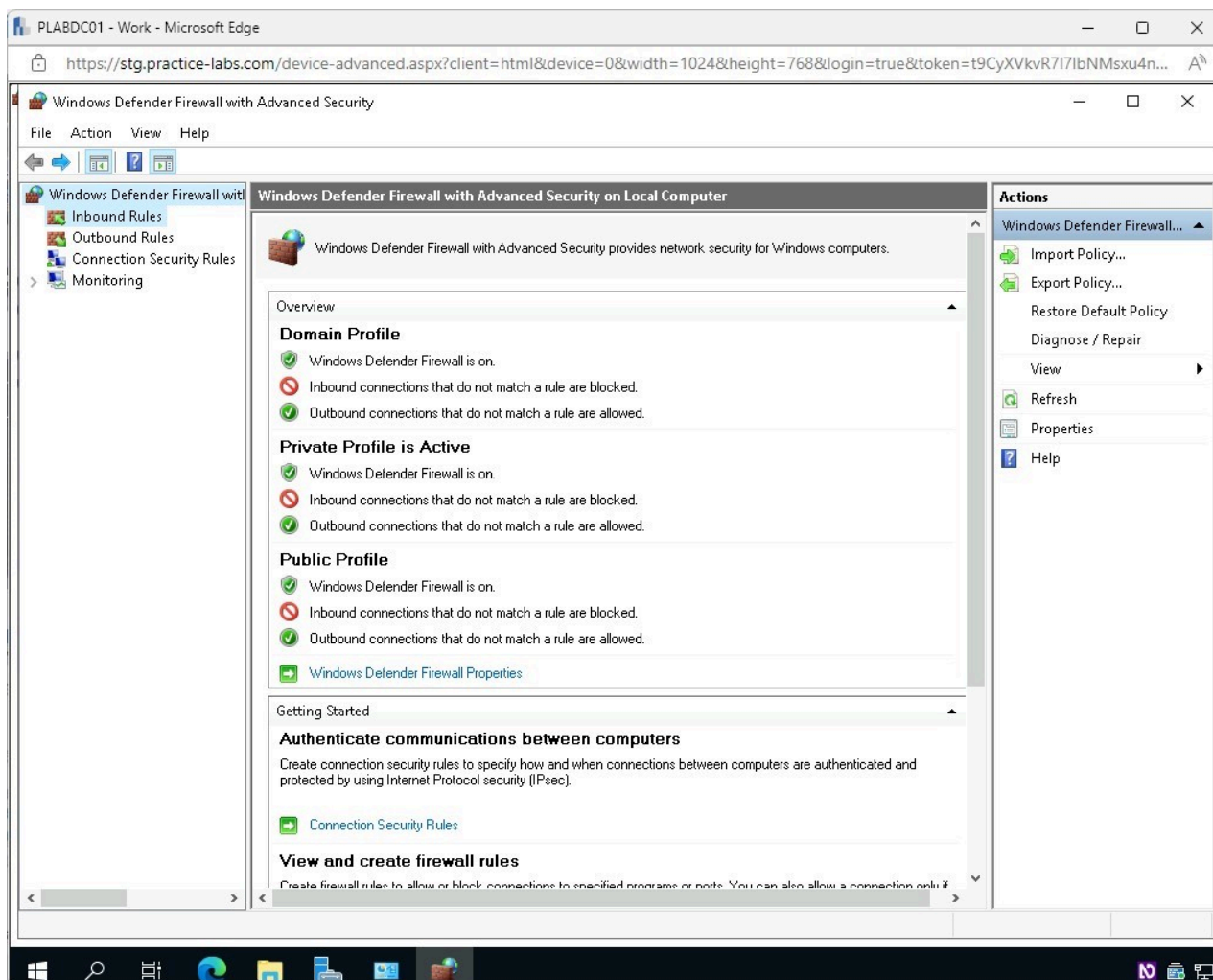


Figure 1.25 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window, showing Inbound Rules selected.

Step 7

The **Inbound Rules** are now displayed in the center pane.

Right-click on **Inbound Rules** on the left pane and select **New Rule**.

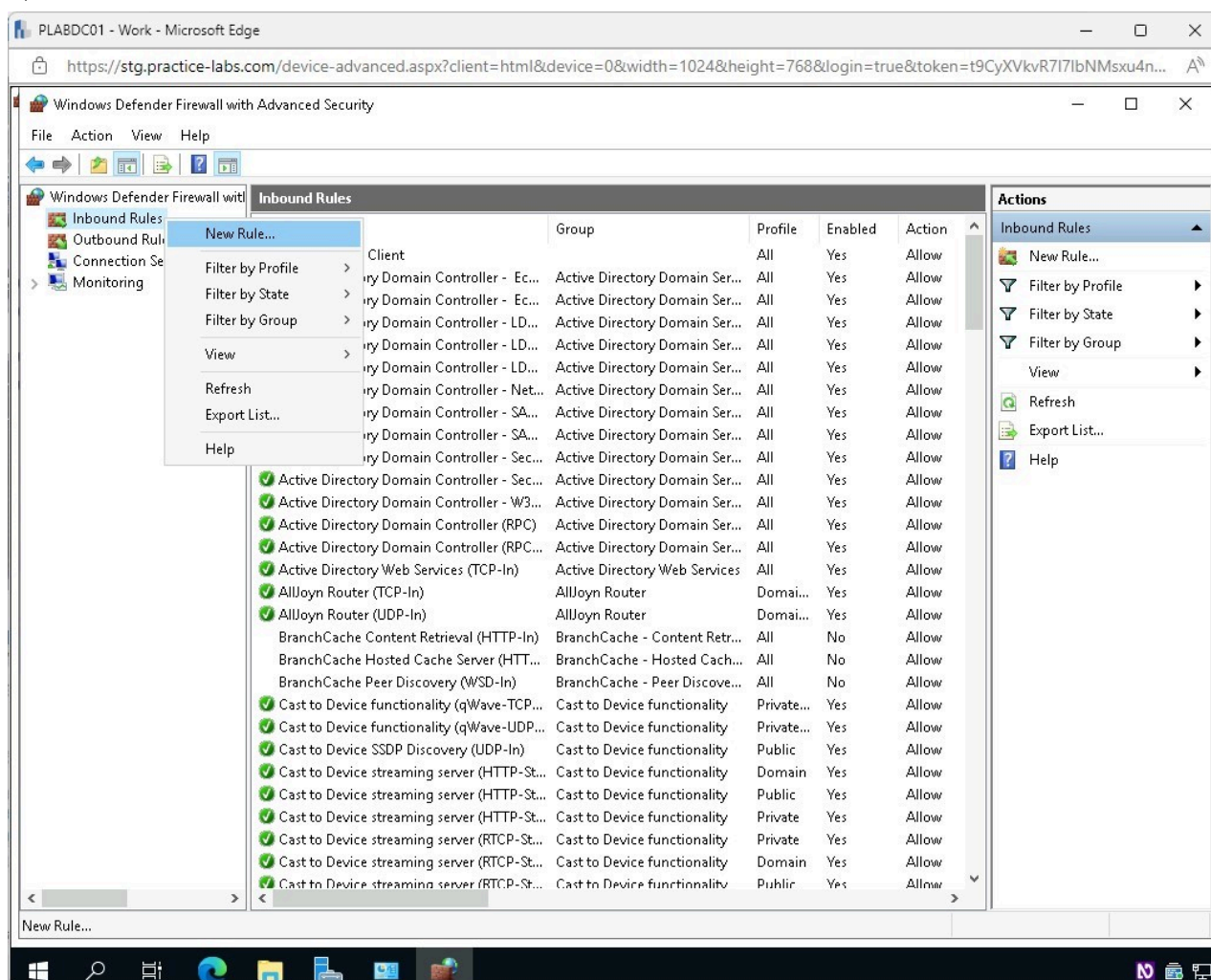


Figure 1.26 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. Inbound Rules is right-clicked, and New Rule is selected.

Step 8

In the **New Inbound Rule Wizard - Rule Type** page, leave **Program** selected and click **Next**.

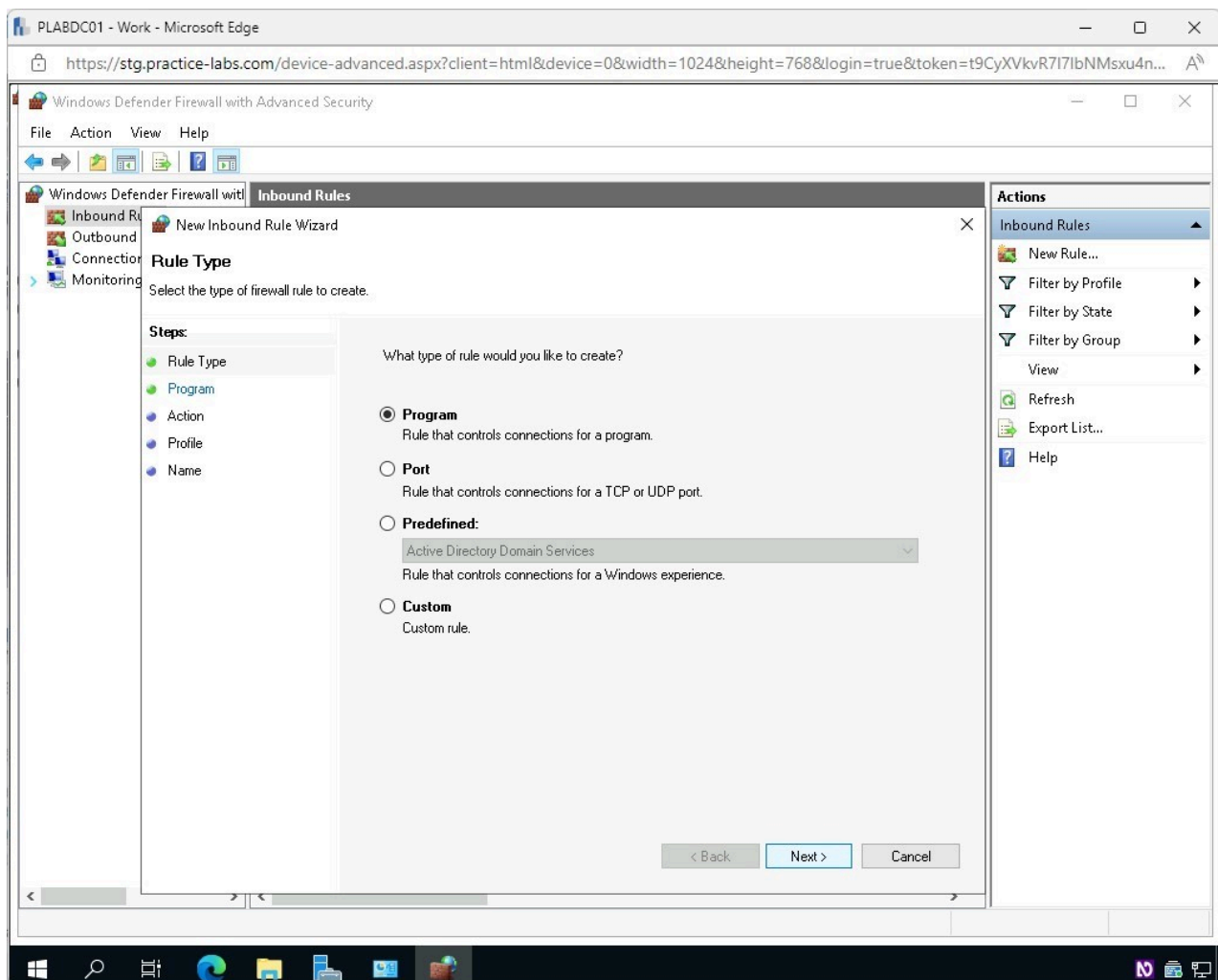


Figure 1.27 Screenshot of PLABDC01: Displaying the New Inbound Rule Wizard - Rule Type page, showing Next selected.

Step 9

In the **Program** page, ensure the **This program path** option is selected.

Click **Browse**.

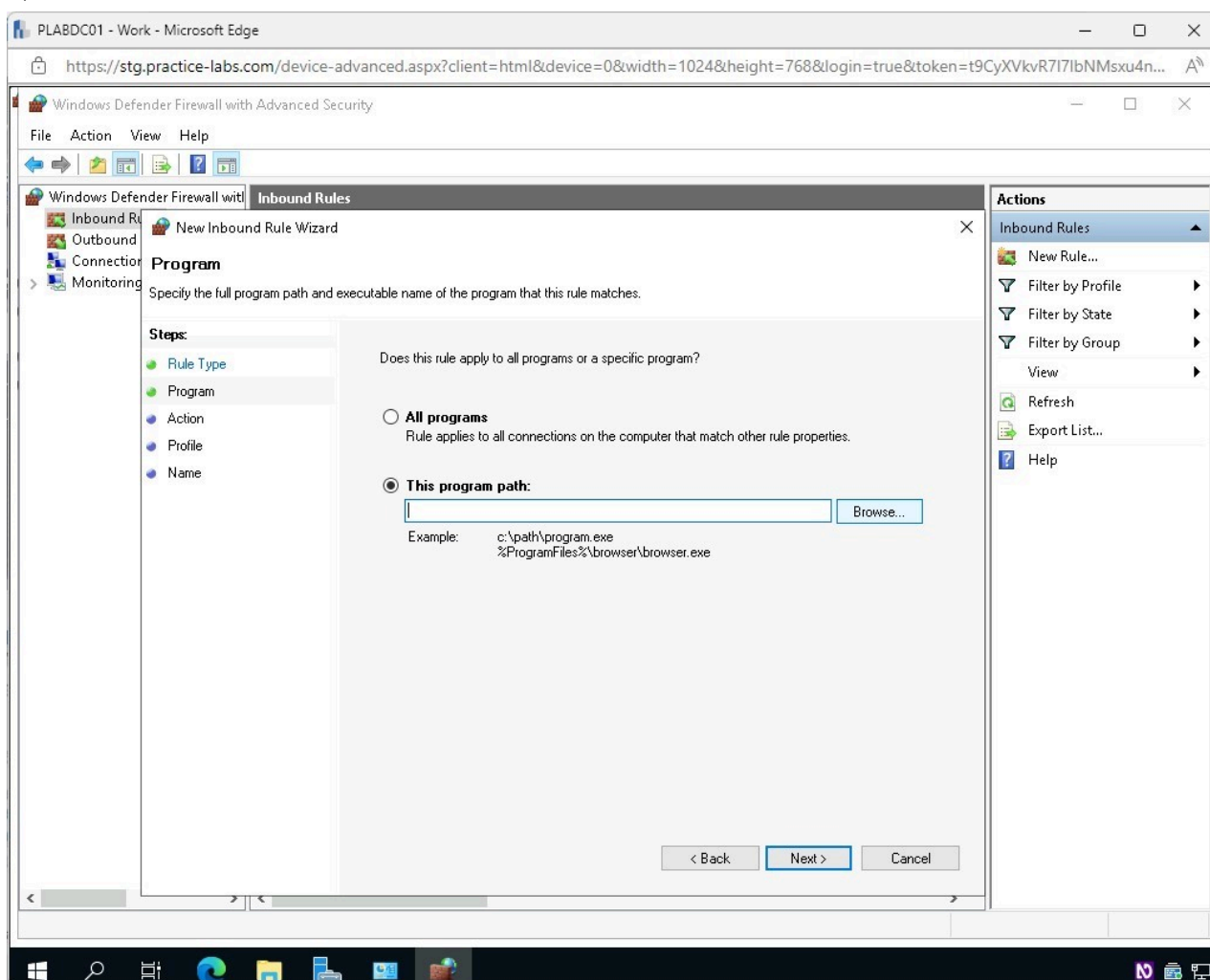


Figure 1.28 Screenshot of PLABDC01: Displaying the Program page, showing Browse selected.

Step 10

In the **Open** pane dialog box, navigate to **This PC > Local Disk (C:) > Program Files (x86) > Internet Explorer**.

Double-click on the **ieexplore** program file to open it.

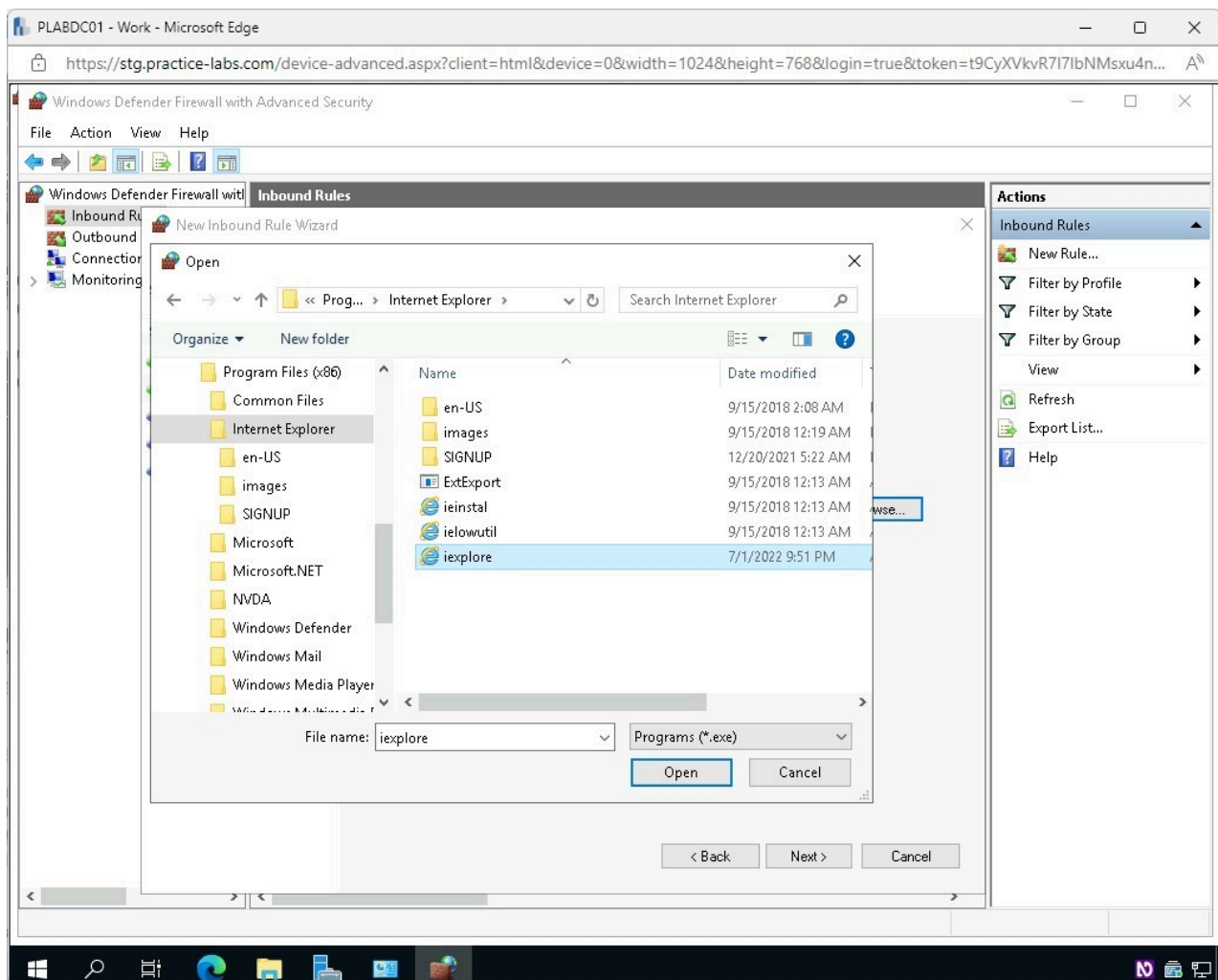


Figure 1.29 Screenshot of PLABDC01: Displaying the Open dialog box, showing the required program file selected.

Step 11

Back on the **Program** page, click **Next**.

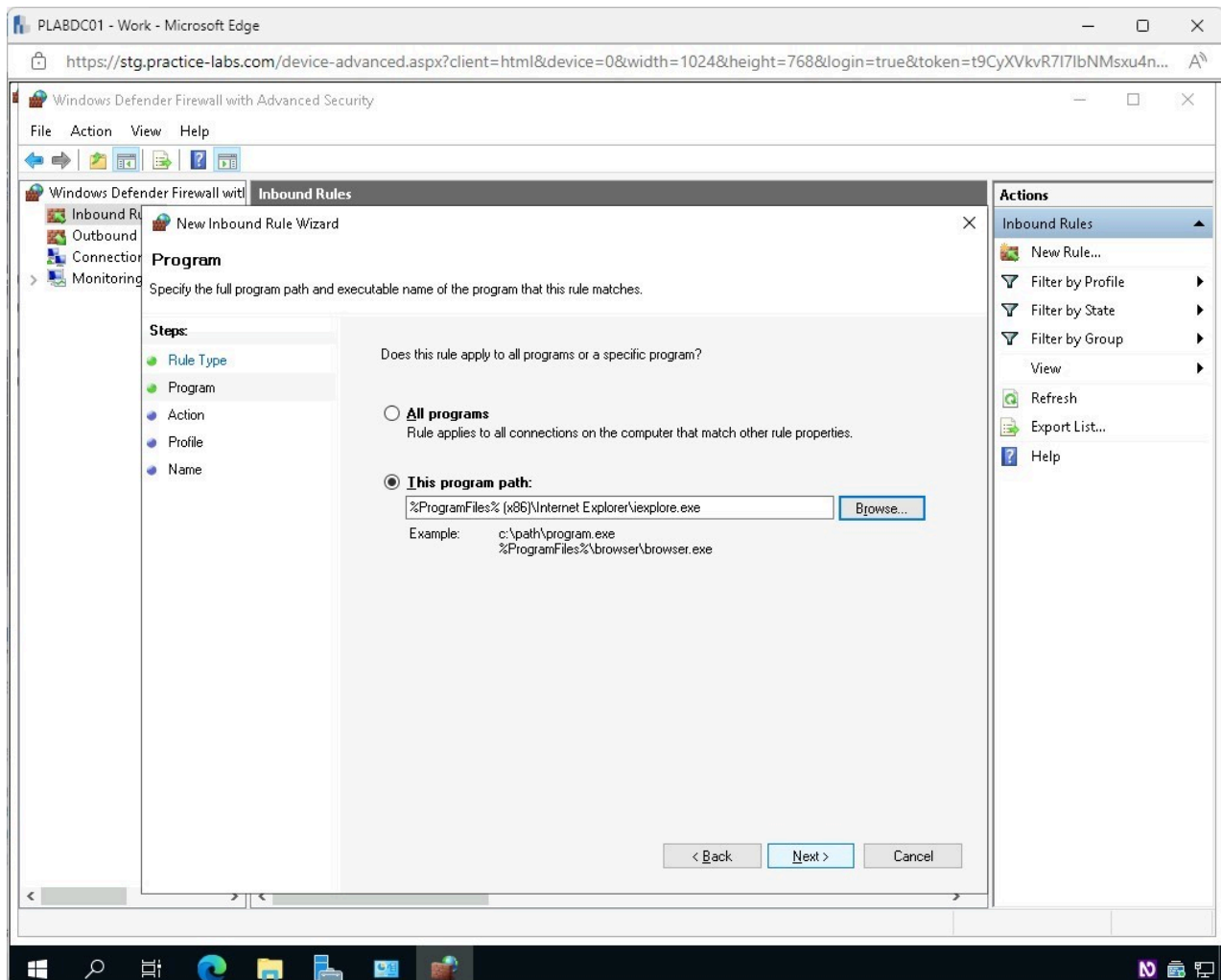


Figure 1.30 Screenshot of PLABDC01: Displaying the Program page, showing Next selected.

Step 12

On the **Action** page, select the **Block the connection** option.

Click **Next**.

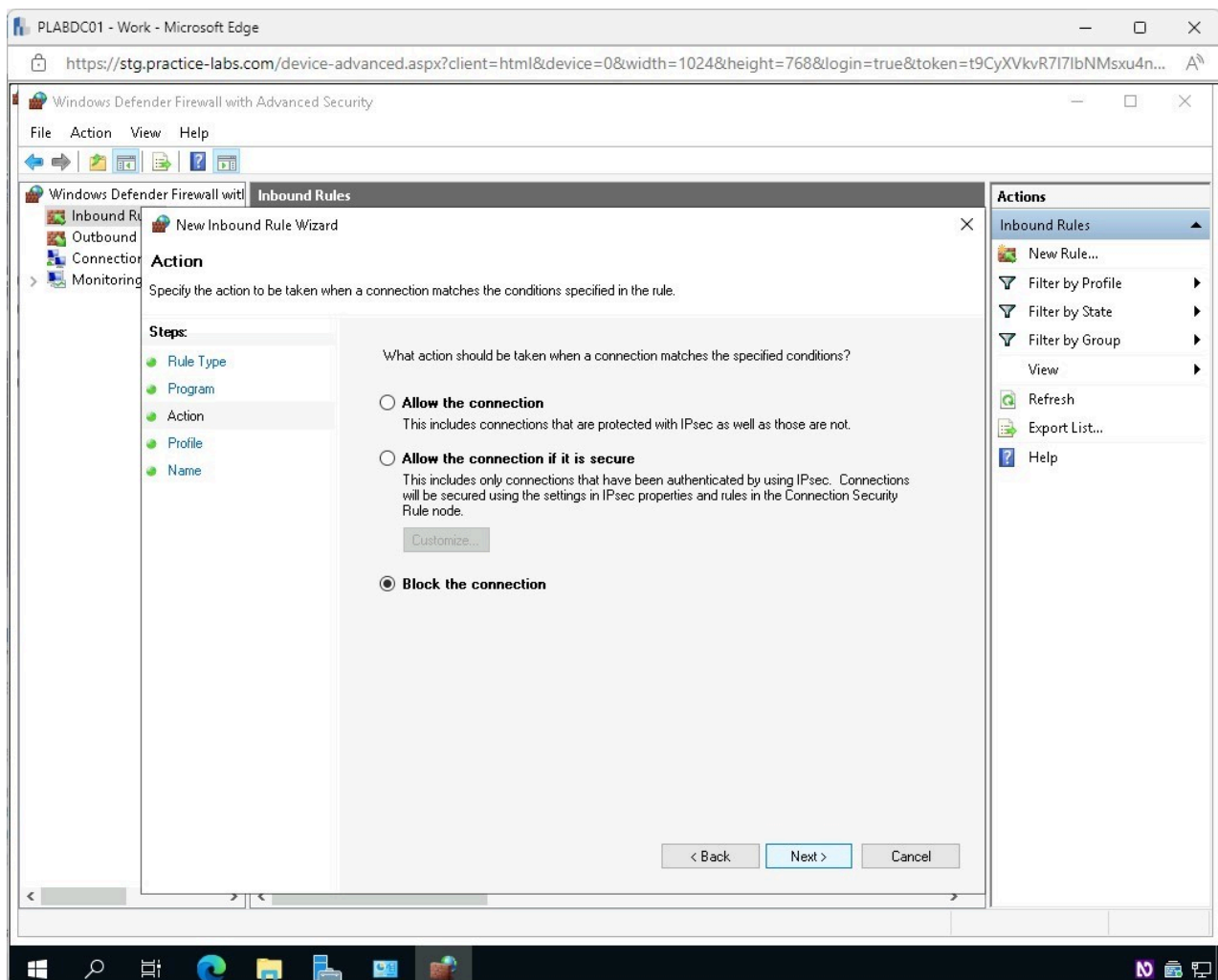


Figure 1.31 Screenshot of PLABDC01: Displaying the Action page, showing the required option selected and Next highlighted.

Step 13

In the **Profile** page, leave the default selections and click **Next**.

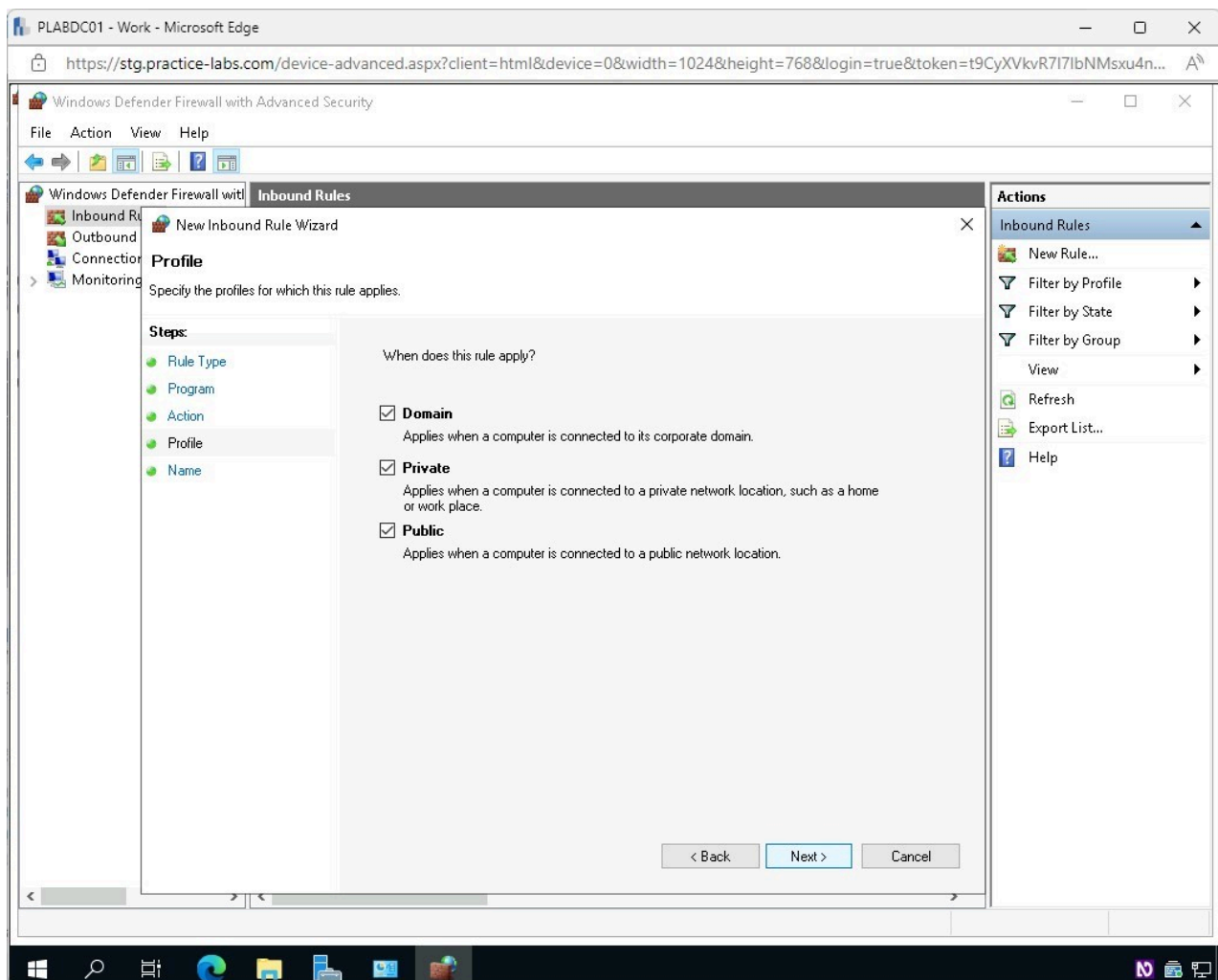


Figure 1.32 Screenshot of PLABDC01: Displaying the Profile page, showing Next selected.

Step 14

In the **Name** page, type the following for the **Name** field:

Block IE

Click **Finish**.

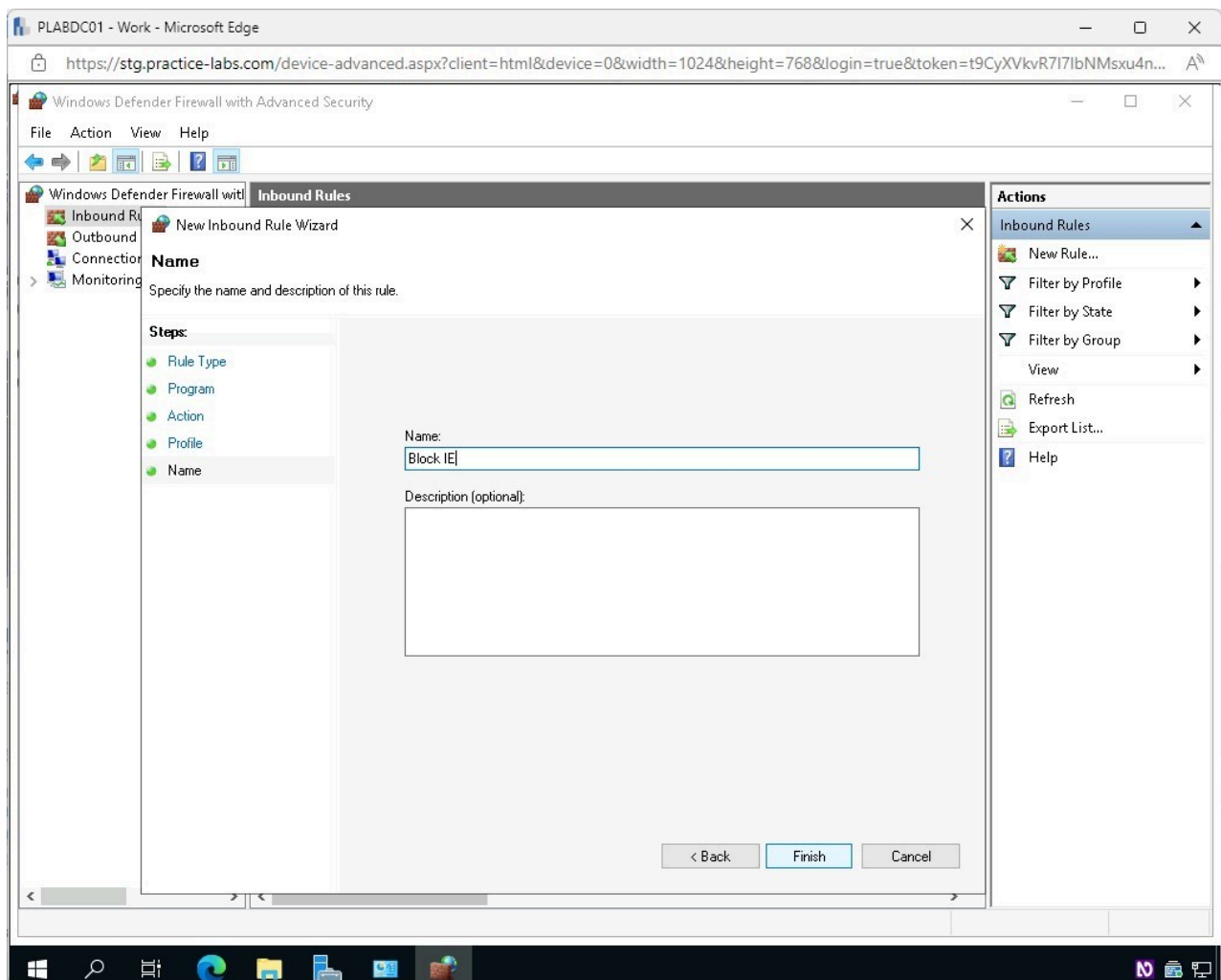


Figure 1.33 Screenshot of PLABDC01: Displaying the Name page, showing the required Name typed in and Finish selected.

Step 15

Back on the **Windows Defender Firewall with Advanced Security** window, notice the new **Block IE** rule is created.

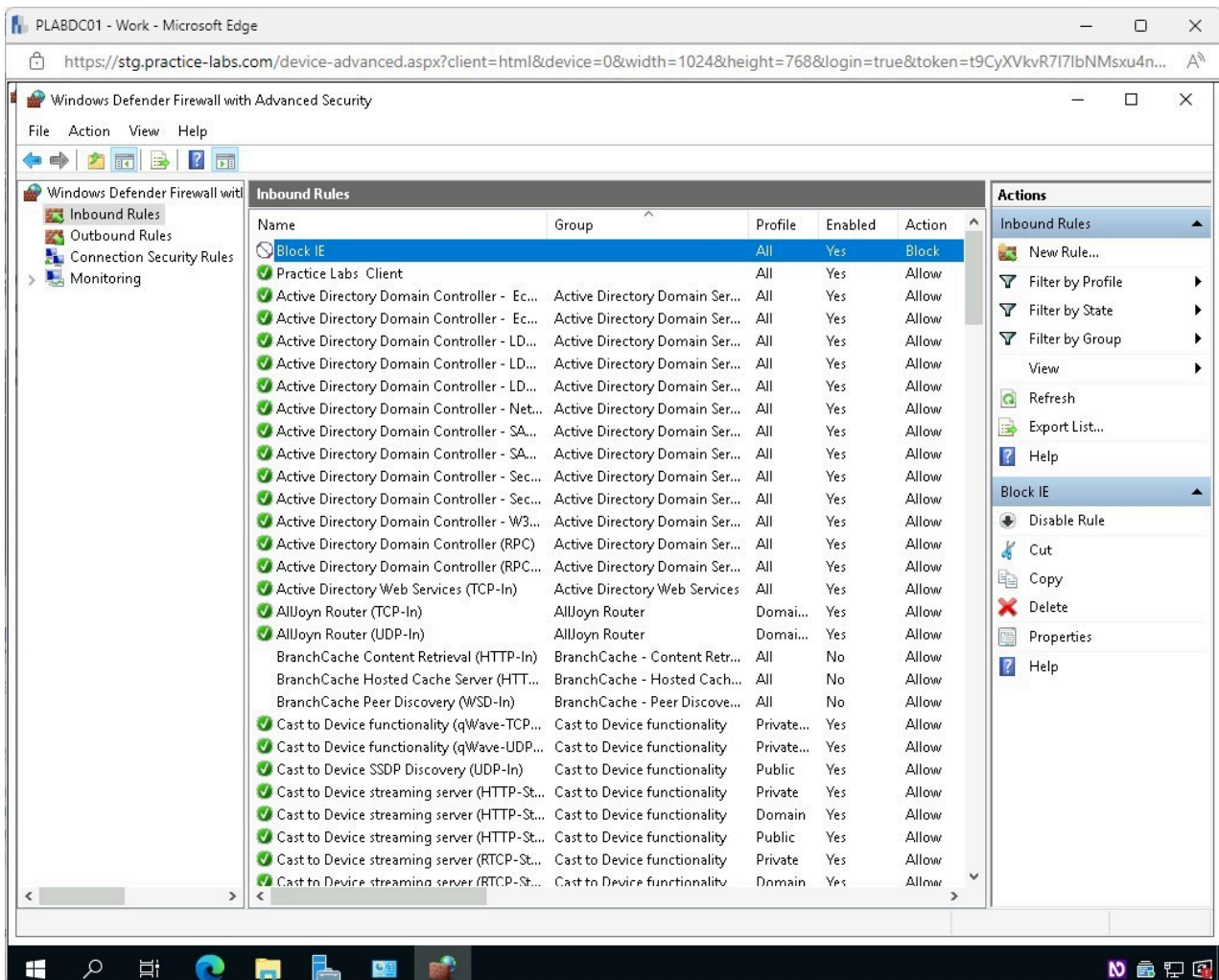


Figure 1.34 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window, showing the new Block IE rule.

Note: This rule will block the use of Internet Explorer. This same process can be done with other programs that you might want to block. Anything done on the Inbound tab will only be applied to incoming traffic. If the same rule is to be applied on outbound, it will also have to be made and applied in that tab as well.

Step 16

Select and right-click on **Outbound Rules** on the left pane. Select **New Rule**.

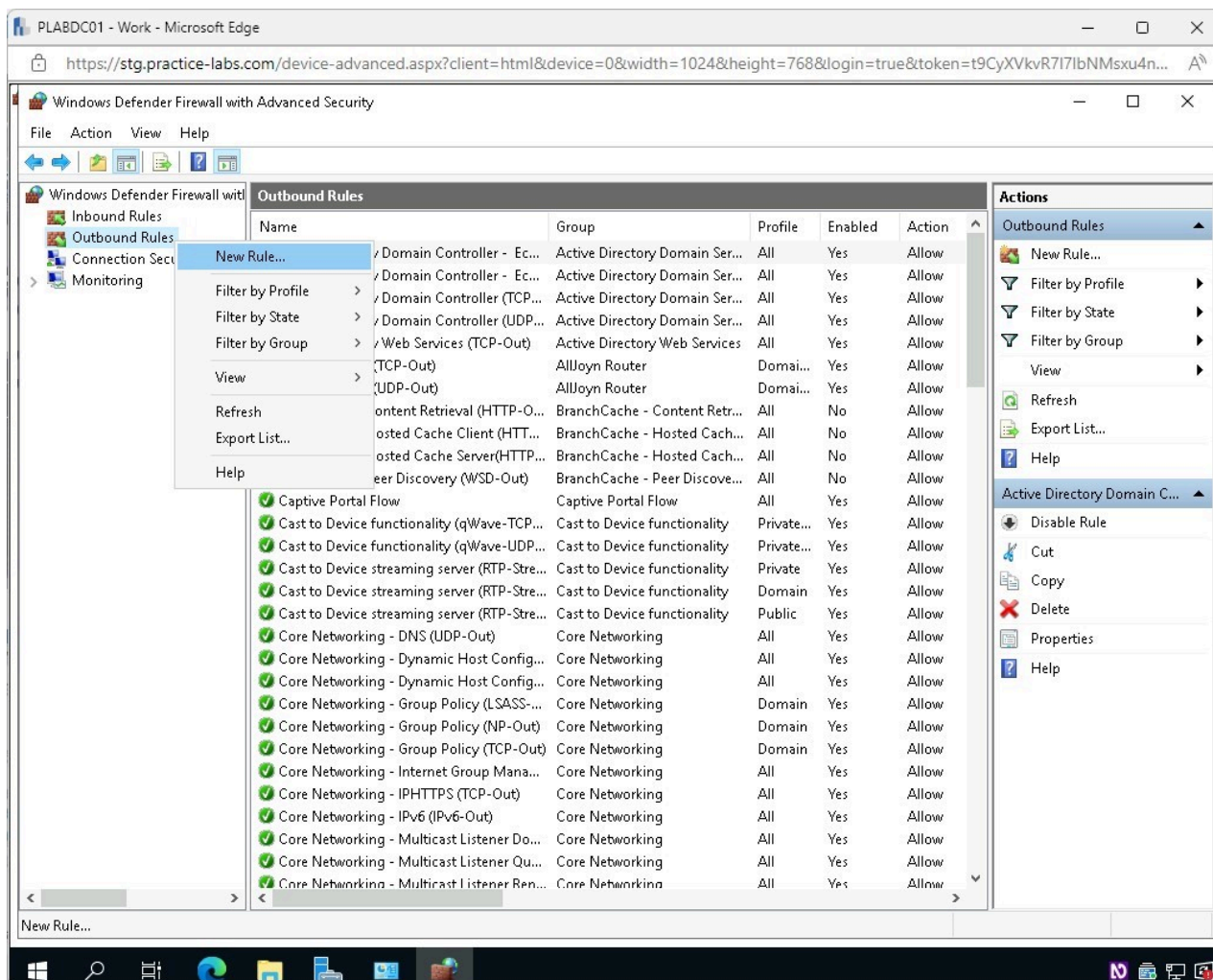


Figure 1.35 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. Outbound Rules is right-clicked, and New Rule is selected.

Step 17

In the **New Outbound Rule Wizard - Rule type** page, select the **Port** option.

Click **Next**.

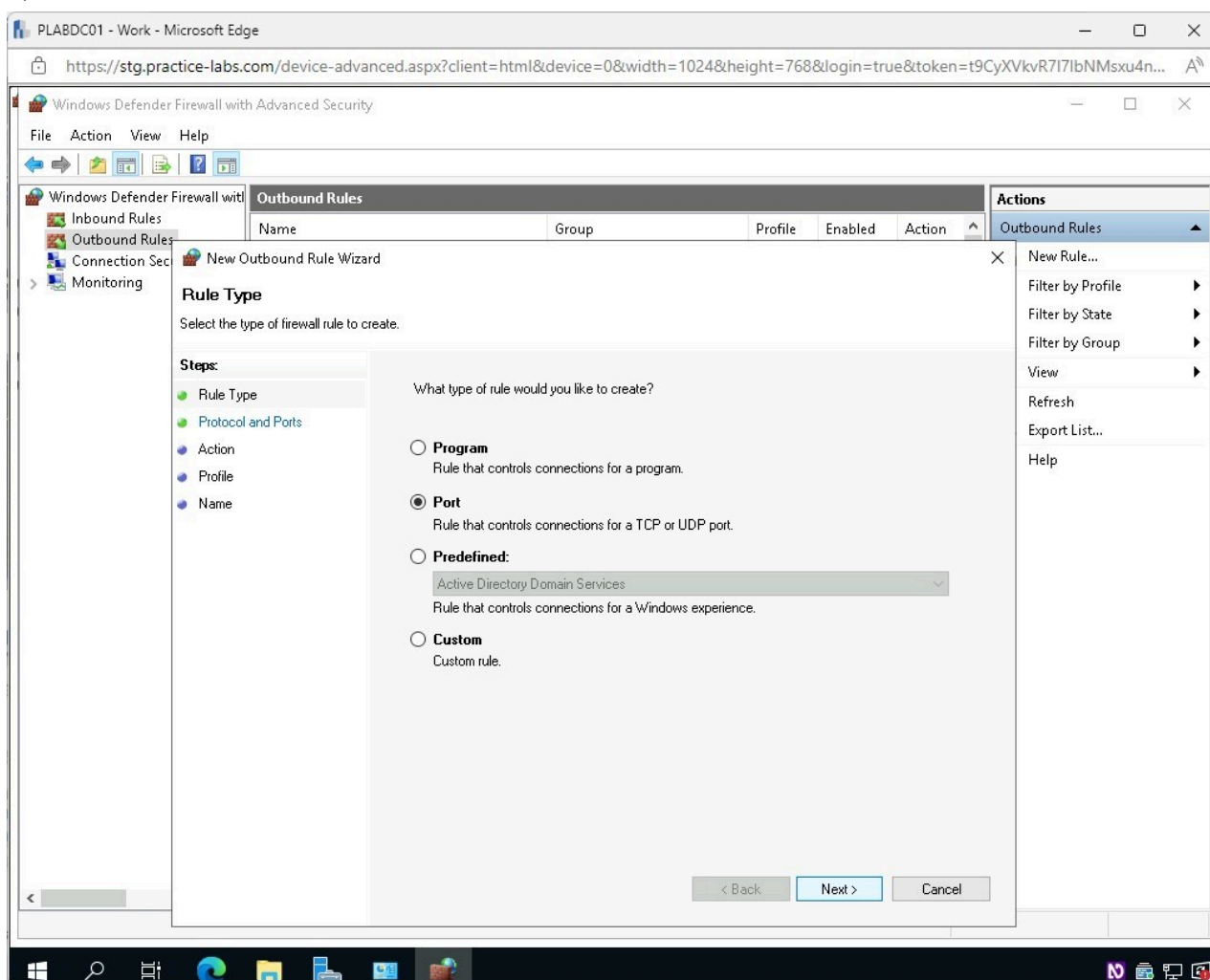


Figure 1.36 Screenshot of PLABDC01: Displaying the New Outbound Rule Wizard - Rule type page, showing the required option selected and Next highlighted.

Step 18

On the **Protocol and Ports** page, select the **Specific remote ports** option and type the following:

22

Click **Next**.

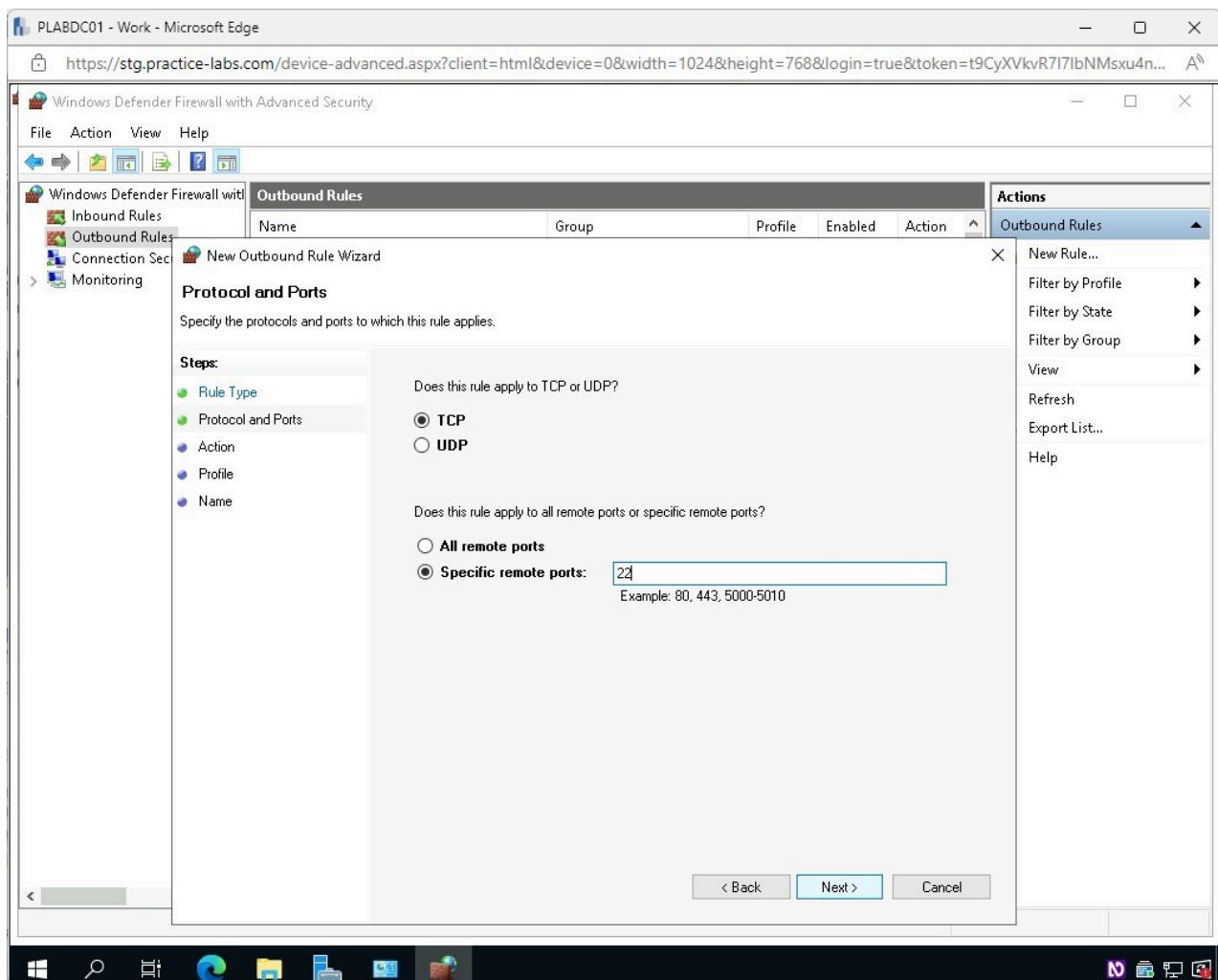


Figure 1.37 Screenshot of PLABDC01: Displaying the Protocol and Ports page, showing the required settings performed and the Next button selected.

Step 19

In the **Action** page, select the **Allow the connection** option.

Click **Next**.

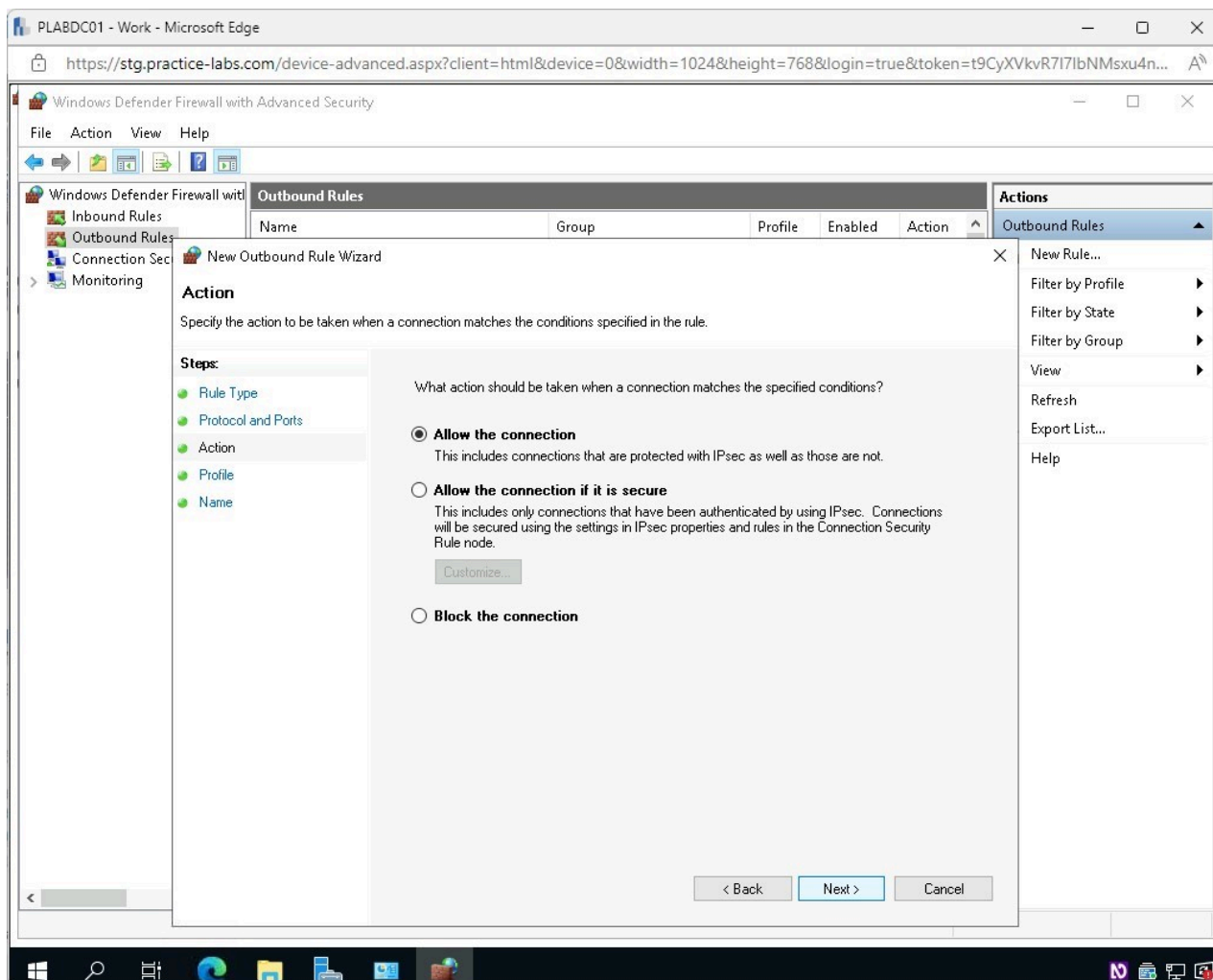


Figure 1.38 Screenshot of PLABDC01: Displaying the Action page, showing the required settings performed and the Next button selected.

Step 20

In the **Profile** page, untick the **Public** checkbox.

Click **Next**.

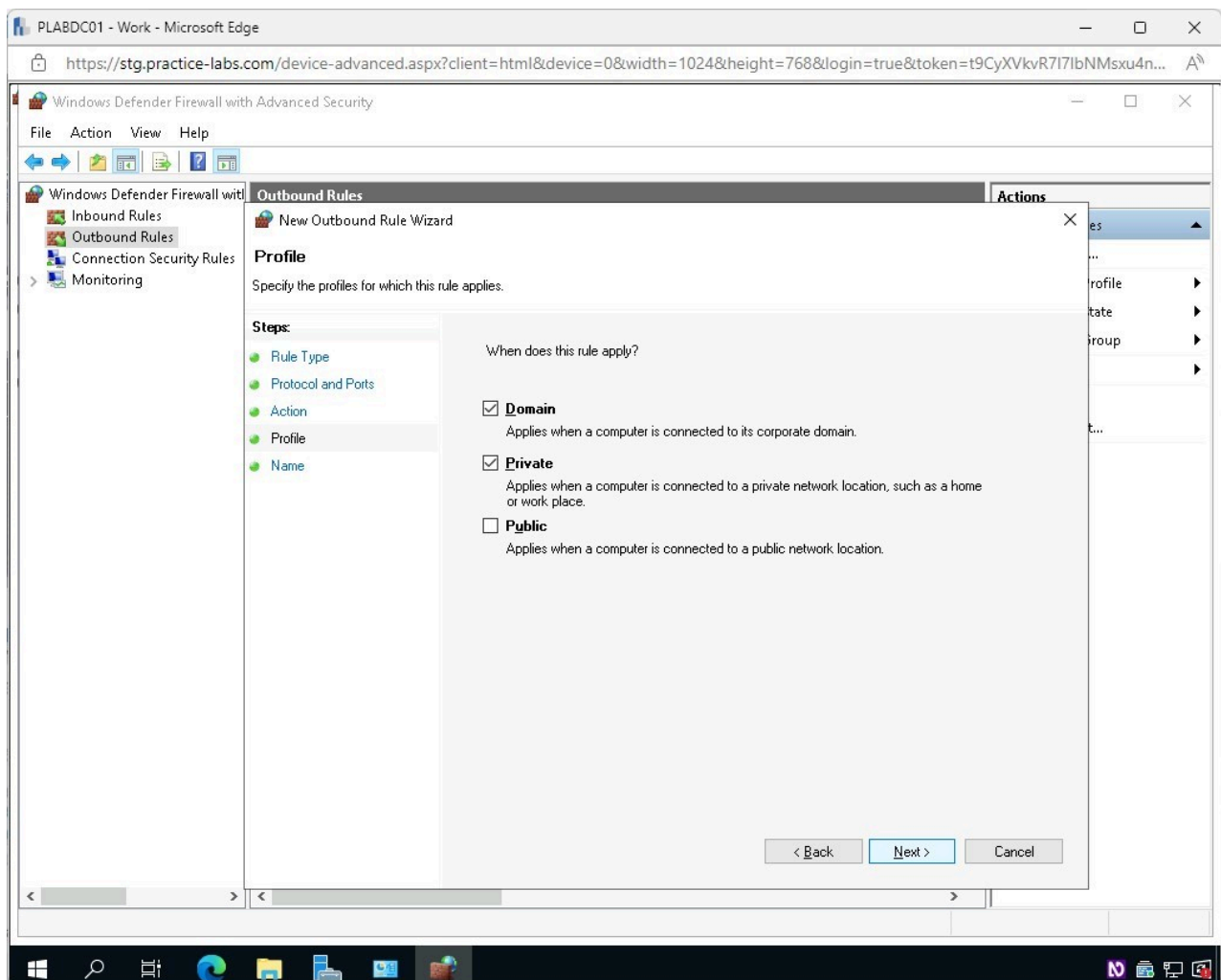


Figure 1.39 Screenshot of PLABDC01: Displaying the Profile page, showing the required settings performed and the Next button selected.

Step 21

In the **Name** page, type the following for the **Name** field:

SSH ALLOW DOM PRIV

Click **Finish**.

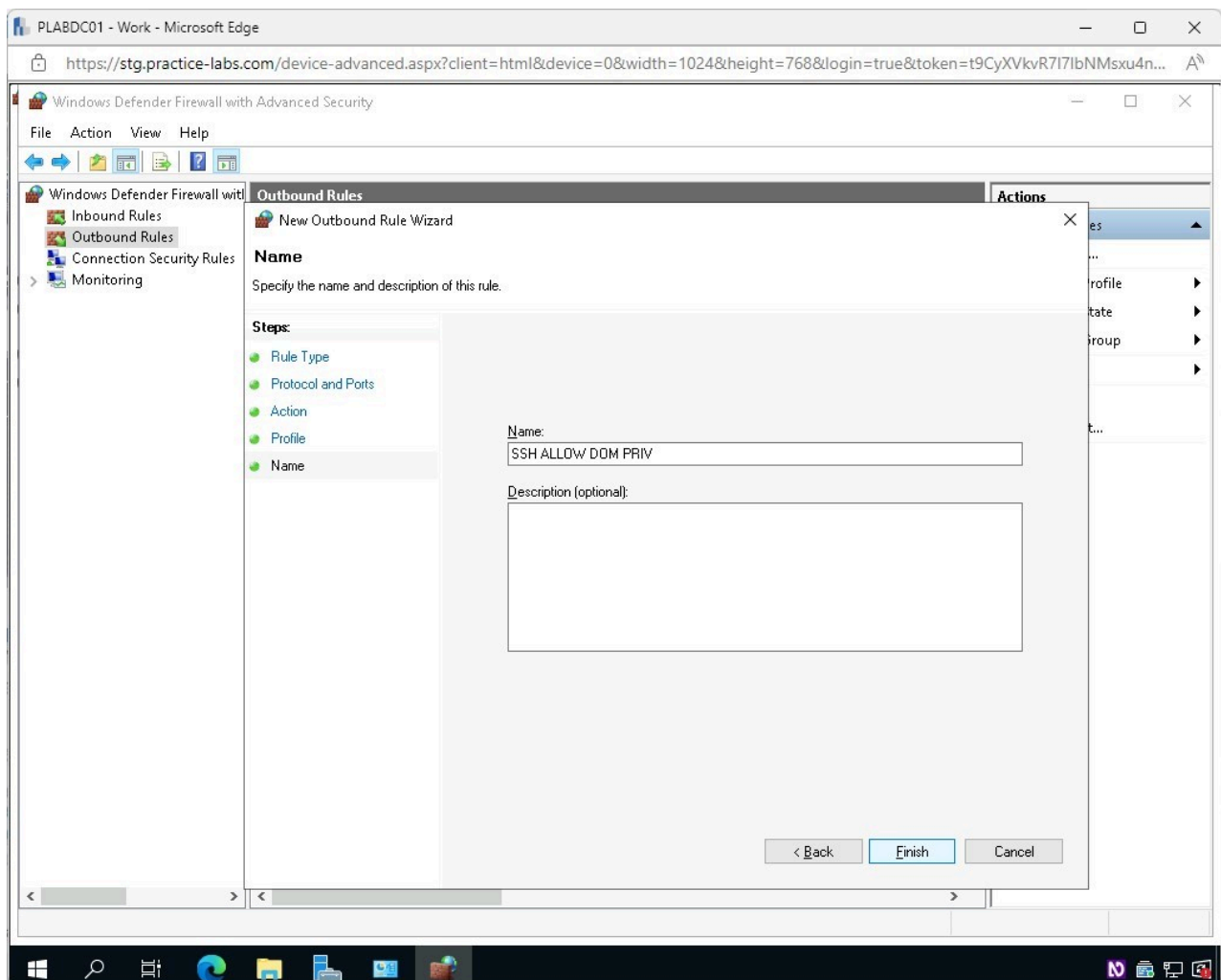


Figure 1.40 Screenshot of PLABDC01: Displaying the Name page, showing the required Name typed in and the Finish button selected.

Step 22

Back on the **Windows Defender Firewall with Advanced Security** window, notice the new **SSH ALLOW DOM PRIV** rule is created.

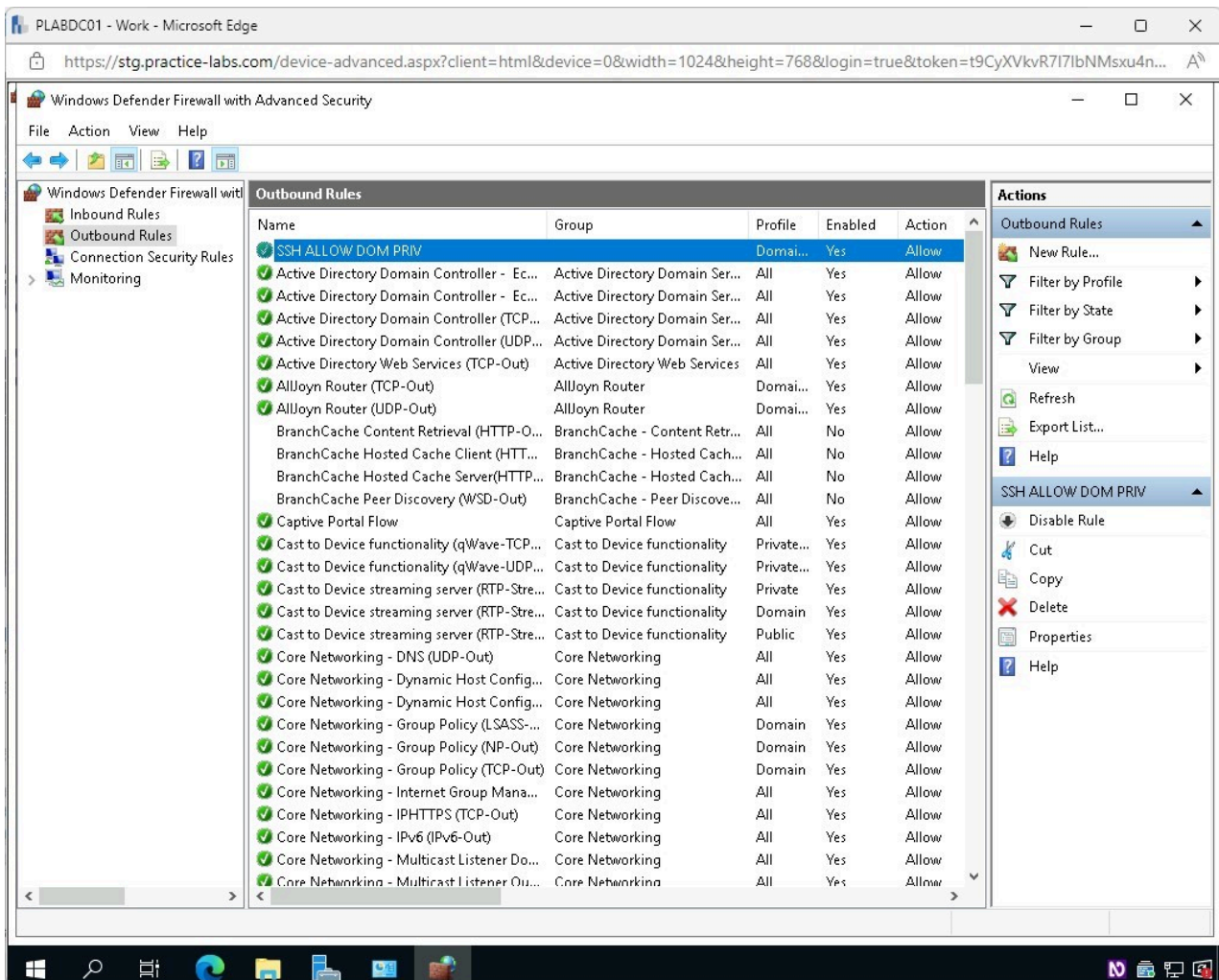


Figure 1.41 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window, showing the newly created SSH ALLOW DOM PRIV rule.

Note: This rule will allow the outbound connection of SSH in a Private and Domain setting. We excluded this connection in a public setting. This process is the same for other ports and protocols. Anything done on the Outbound tab will only be applied to outgoing traffic. If the same rule is to be applied on inbound, it will also have to be made and applied in that tab as well.

Step 23

Select and right-click on **Connection Security Rules** on the left pane. Select **New Rule**.

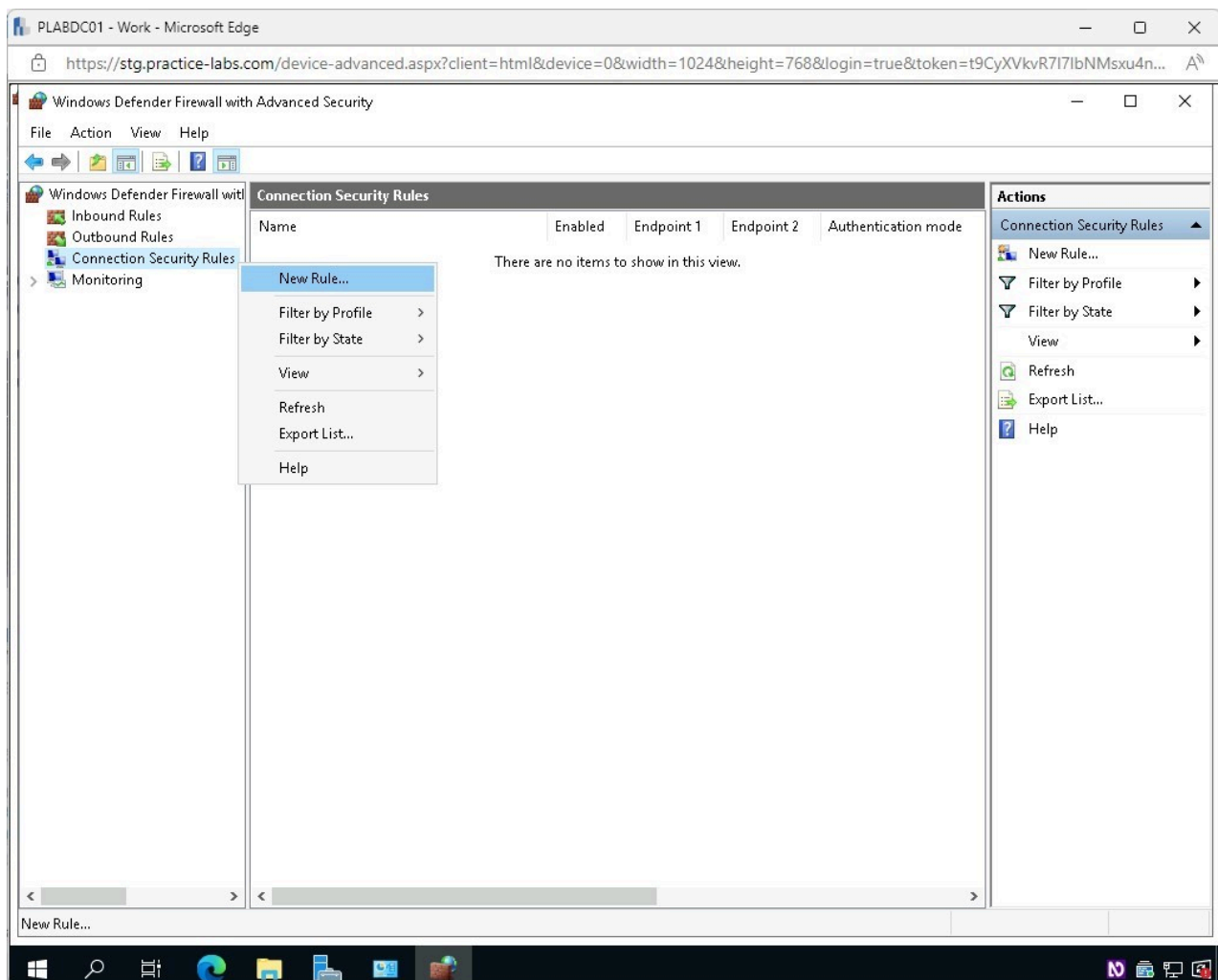


Figure 1.42 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window. Connection Security Rules is right-clicked, and New Rule is selected.

Step 24

In the **New Connection Security Rule Wizard - Rule type** page, select the **Custom** option.

Click **Next**.

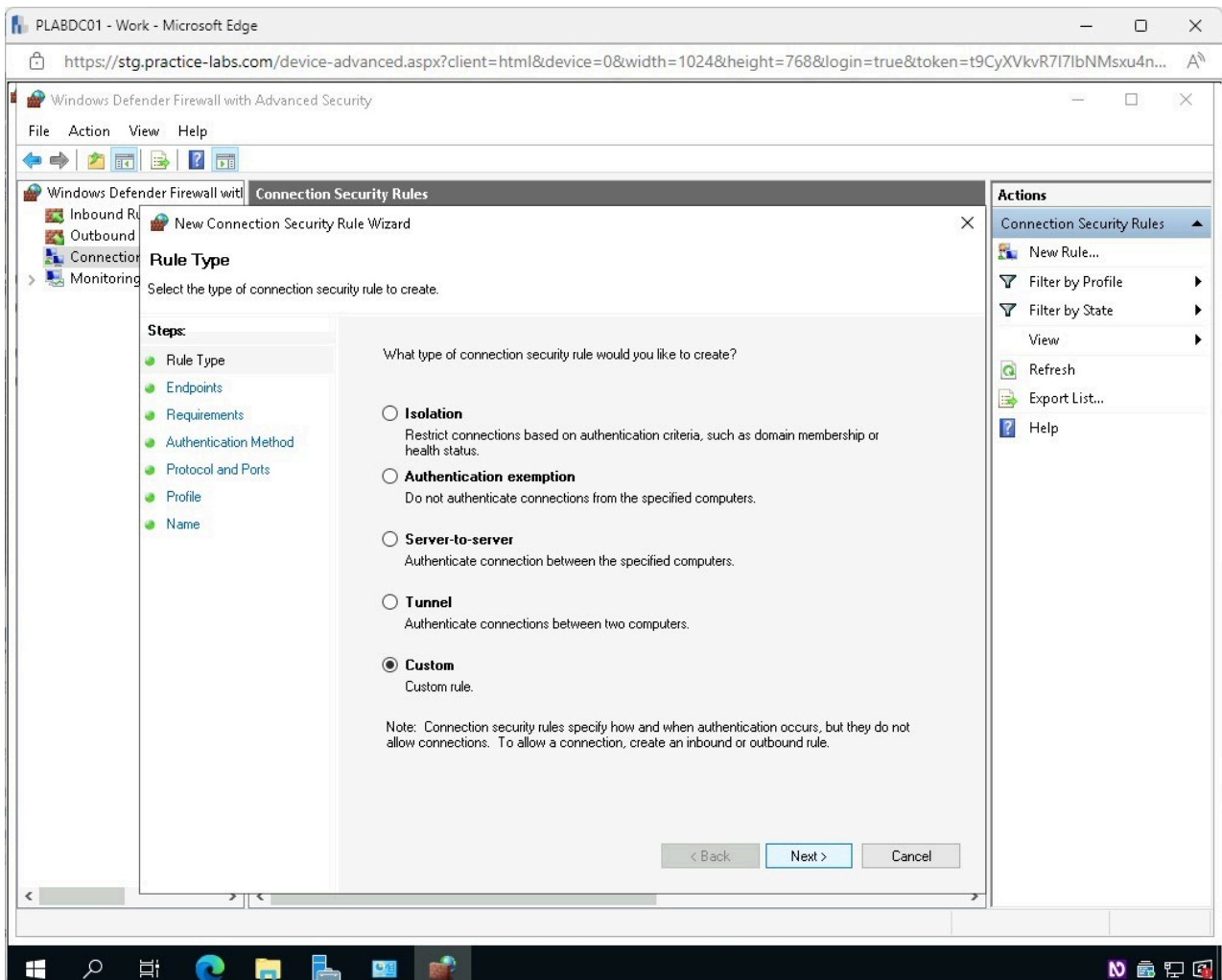


Figure 1.43 Screenshot of PLABDC01: Displaying the New Connection Security Rule Wizard - Rule type page, showing the Custom option selected and Next highlighted.

Step 25

On the **Endpoints** page, if you had specific IP addresses, you could enter them here.

Leave the default option (**Any IP address**) and click **Next**.

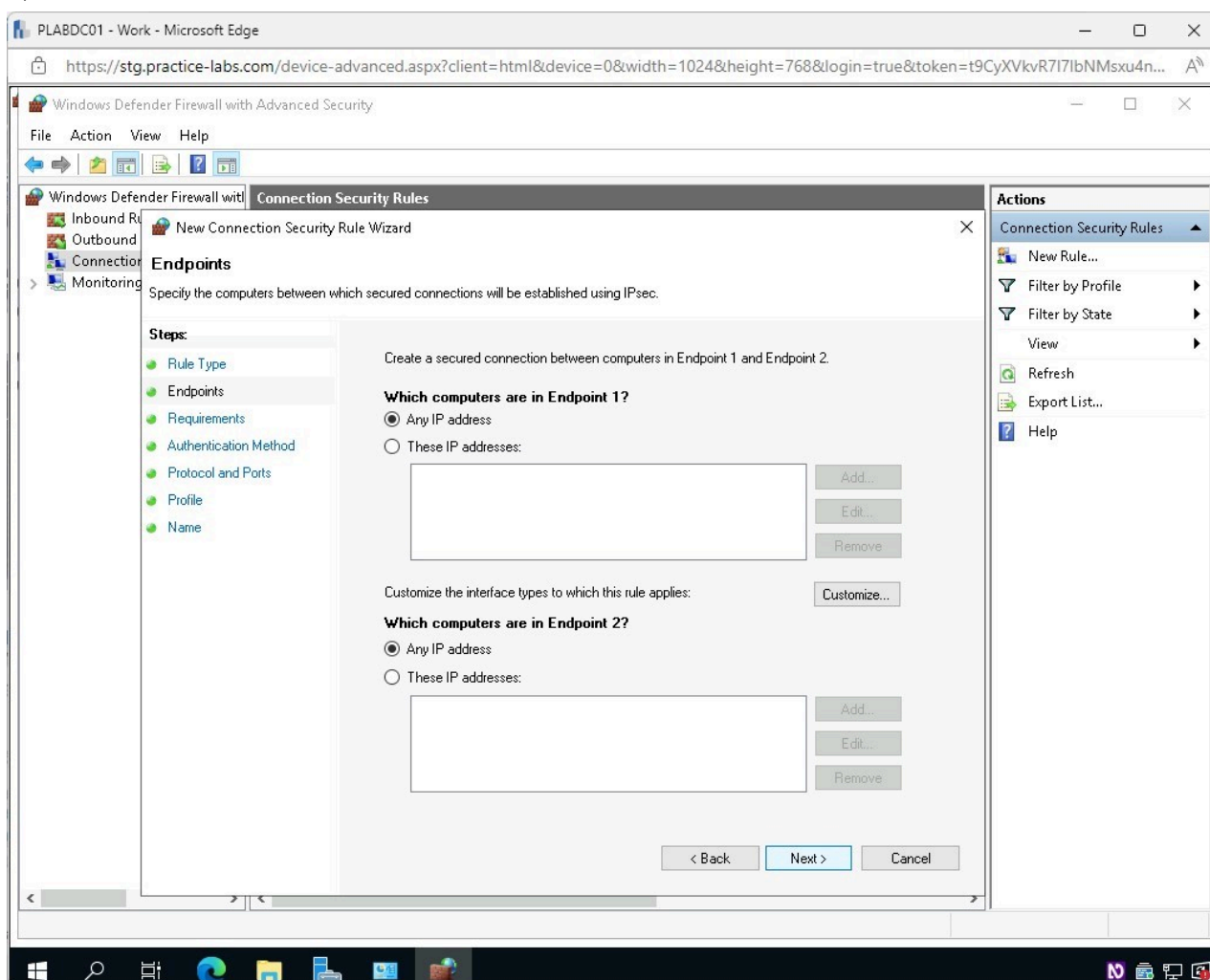


Figure 1.44 Screenshot of PLABDC01: Displaying the Endpoints page, showing Next selected.

Step 26

In the **Requirements** page, ensure **Request authentication for inbound and outbound connections** is selected.

Click **Next**.

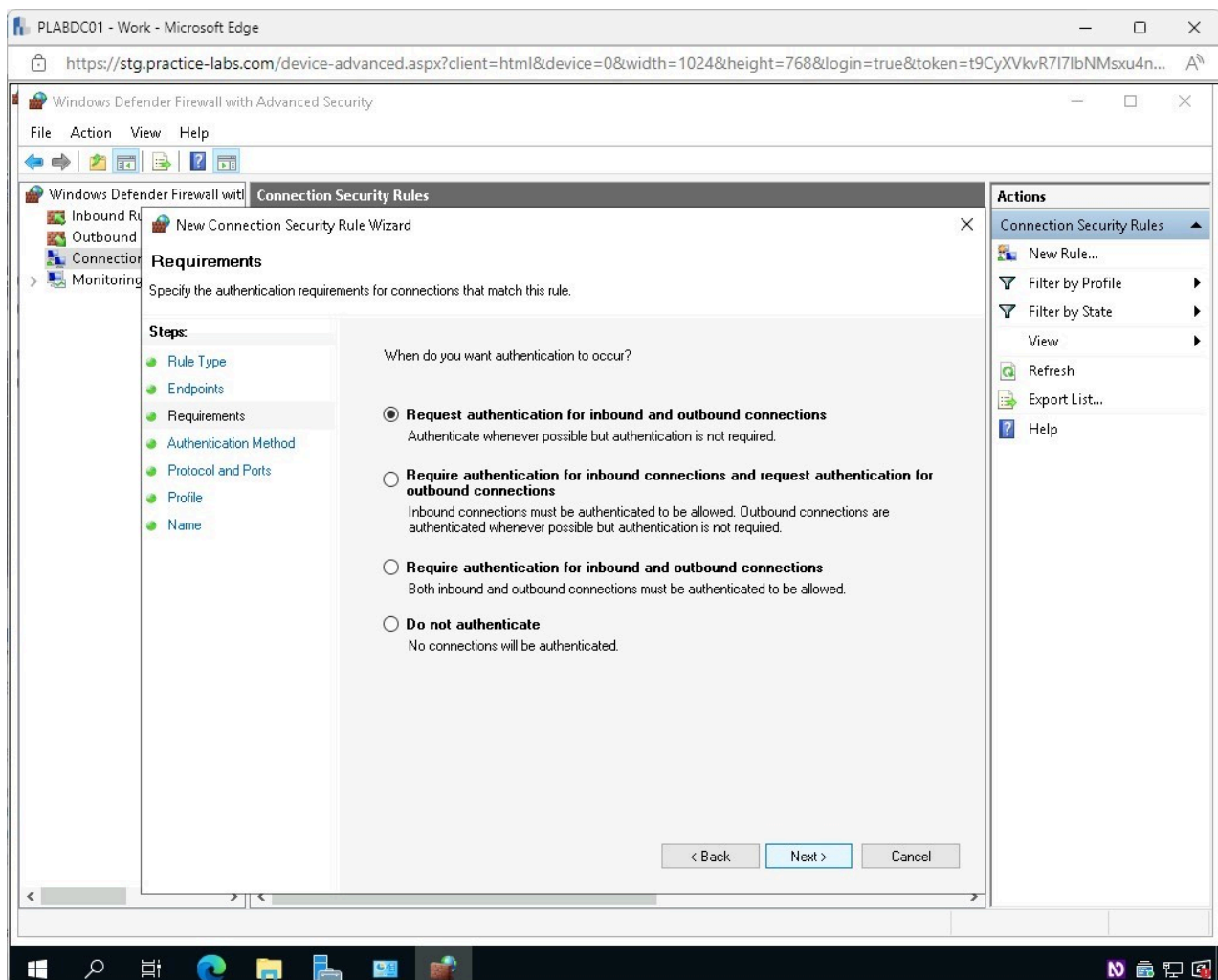


Figure 1.45 Screenshot of PLABDC01: Displaying the Requirements page, showing Next selected.

Step 27

On the **Authentication Method** page, leave **Default** selected and click **Next**.

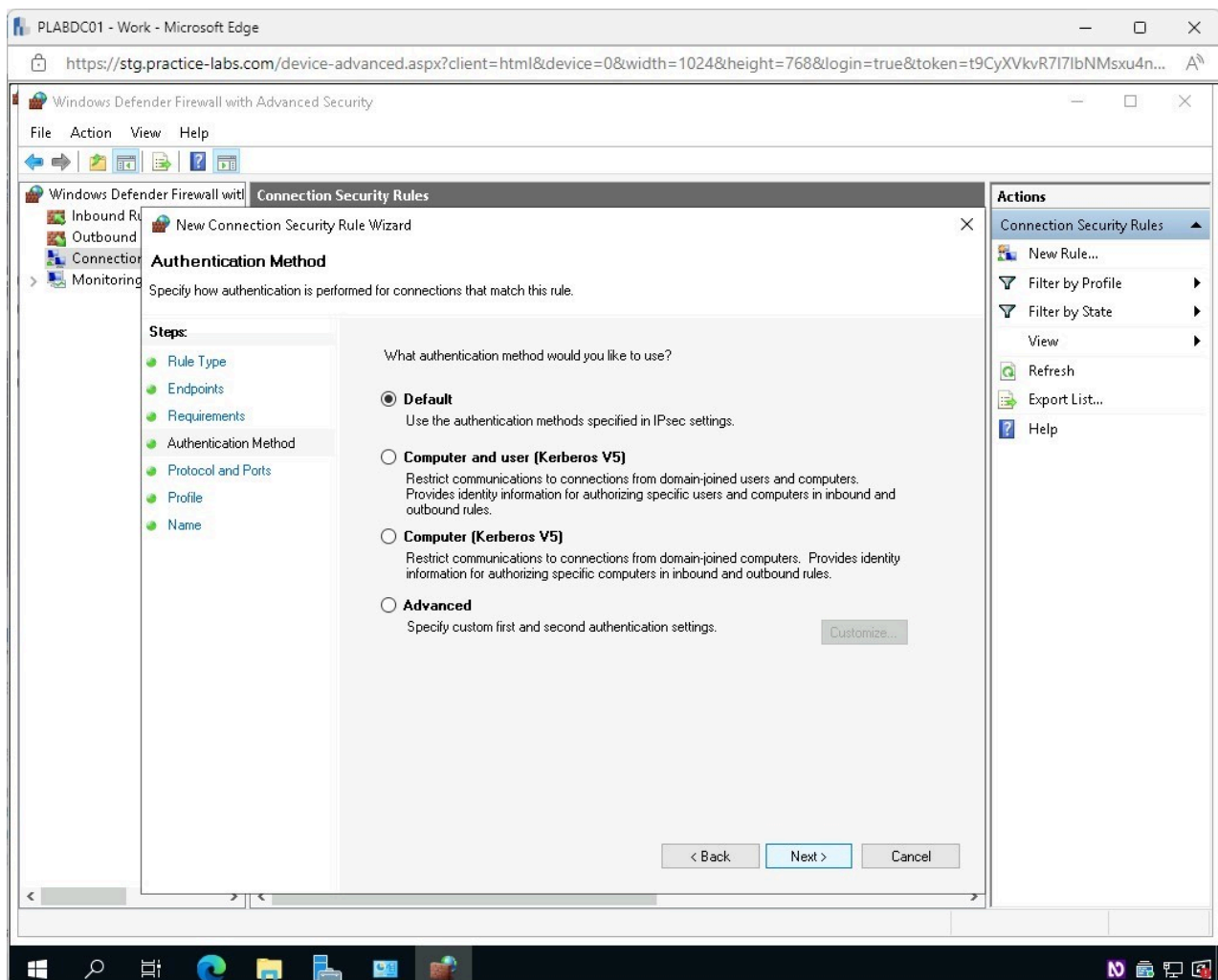


Figure 1.46 Screenshot of PLABDC01: Displaying the Authentication Method page, showing Next selected.

Step 28

On the **Protocols and Ports** page, click **Next**.

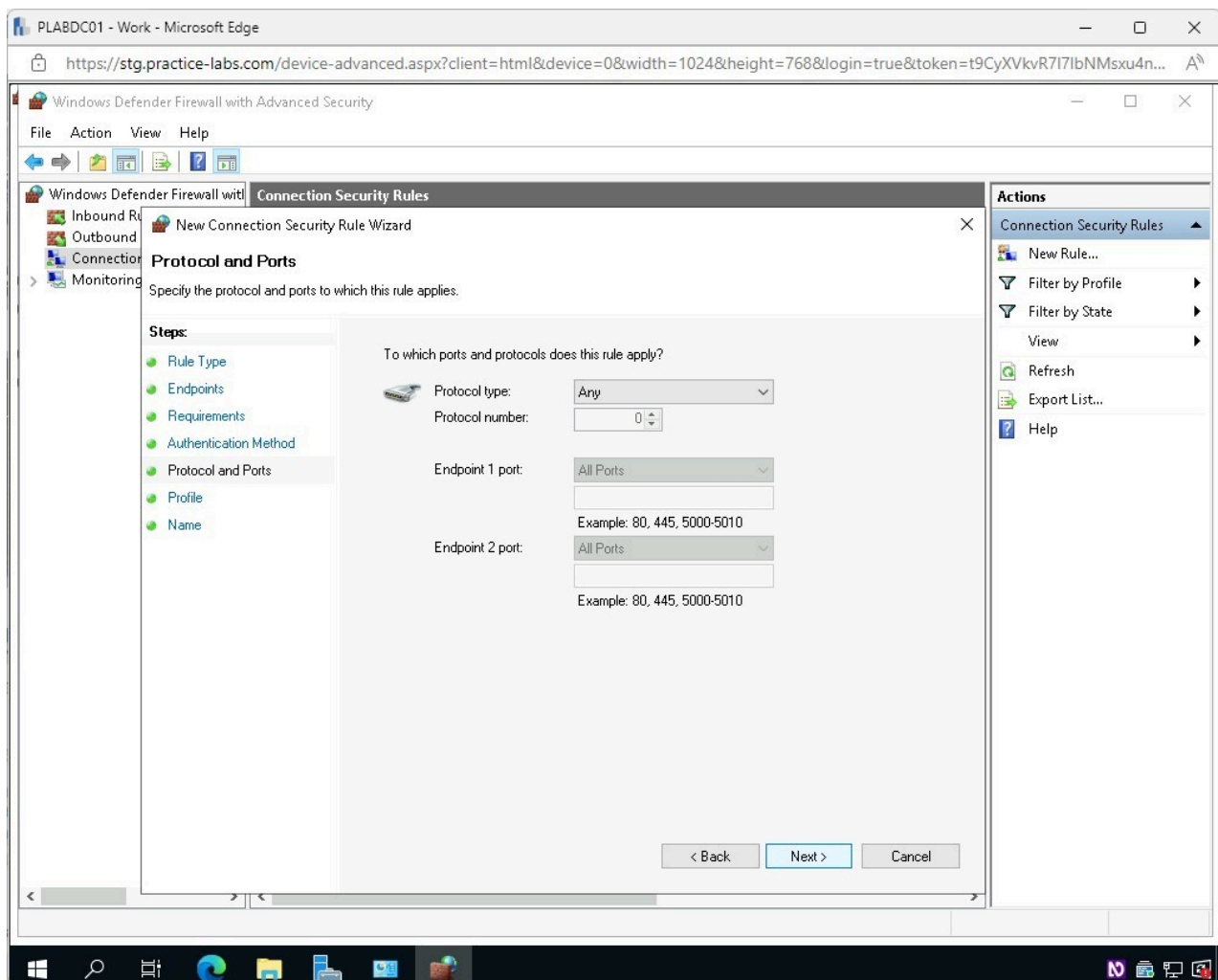


Figure 1.47 Screenshot of PLABDC01: Displaying the Protocols and Ports page, showing Next selected.

Step 29

On the **Profile** page, untick the **Domain** and **Private** options.

Click **Next**.

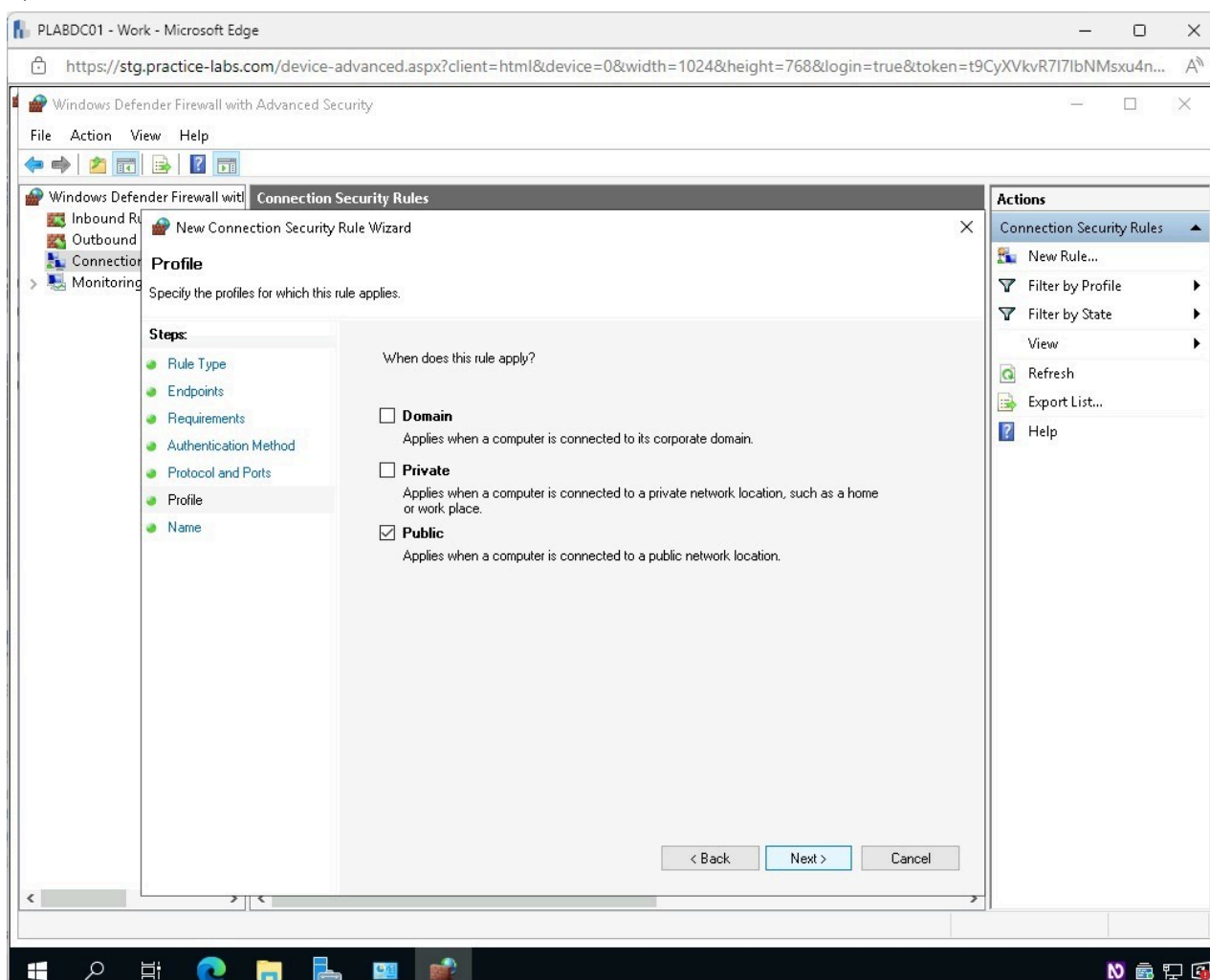


Figure 1.48 Screenshot of PLABDC01: Displaying the Profile page, showing the required settings performed and Next selected.

Step 30

In the **Name** page, type the following for the **Name** field:

Public Auth

Click **Finish**.

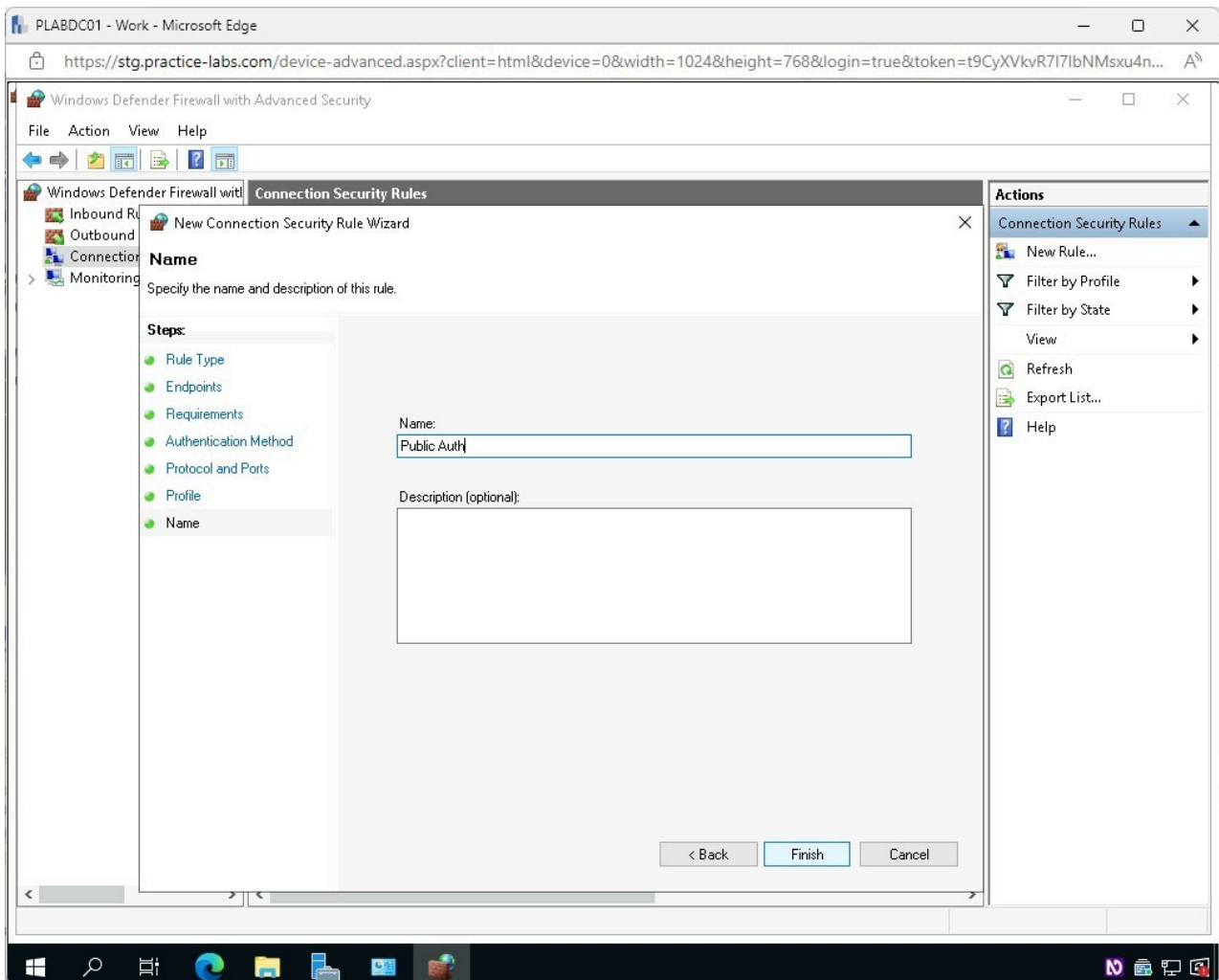


Figure 1.49 Screenshot of PLABDC01: Displaying the Name page, showing the required Named typed in and Finish selected.

Step 31

Back on the **Windows Defender Firewall with Advanced Security** window, notice the new **Public Auth** rule is created.

Note: The new rule is now created. You made it so that any connections made in the public setting will require authentication.

Close the **Windows Defender Firewall with Advanced Security** window.

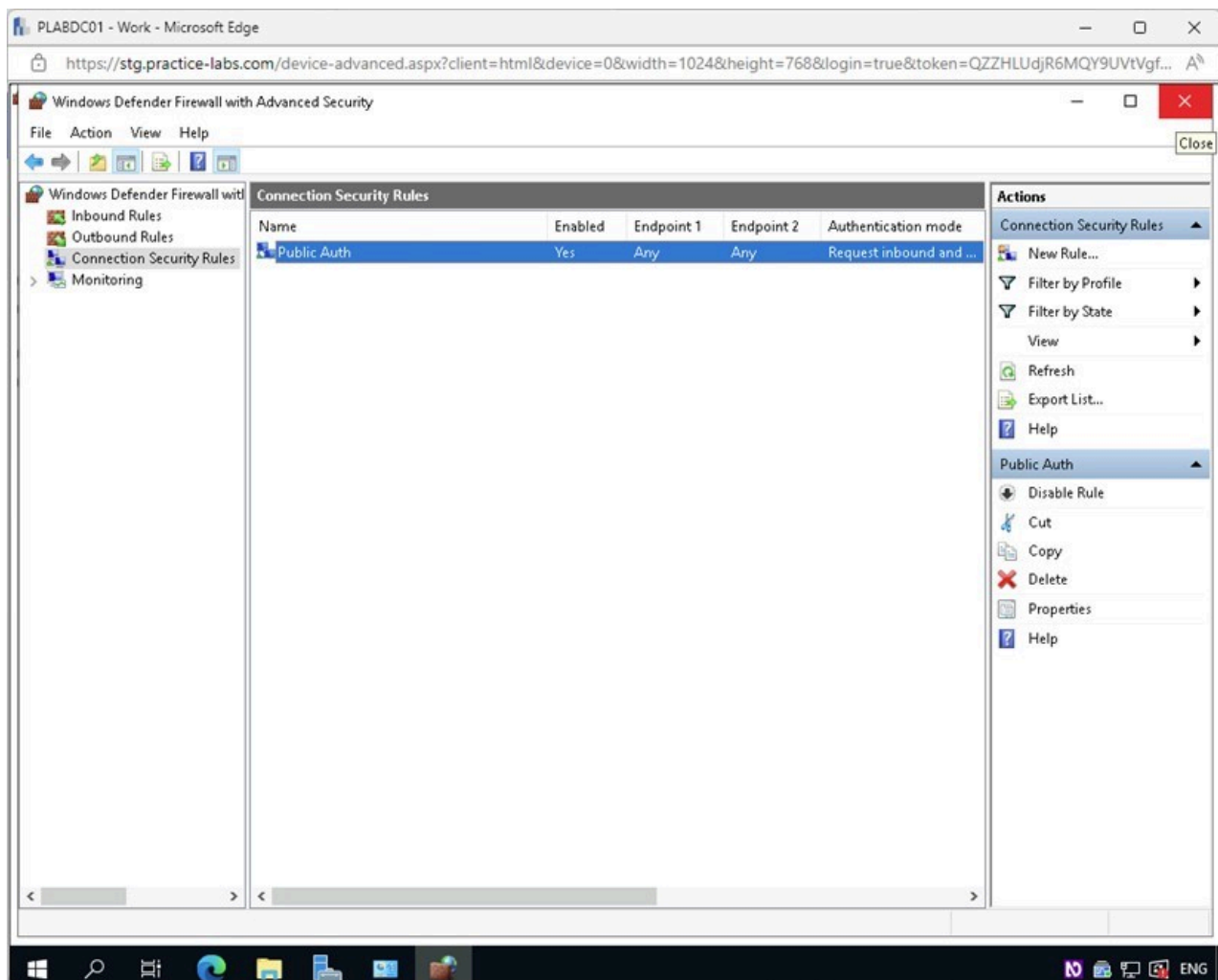


Figure 1.50 Screenshot of PLABDC01: Displaying the Windows Defender Firewall with Advanced Security window, showing the new Public Auth rule created.

Step 32

Close the **Windows Defender Firewall** window.

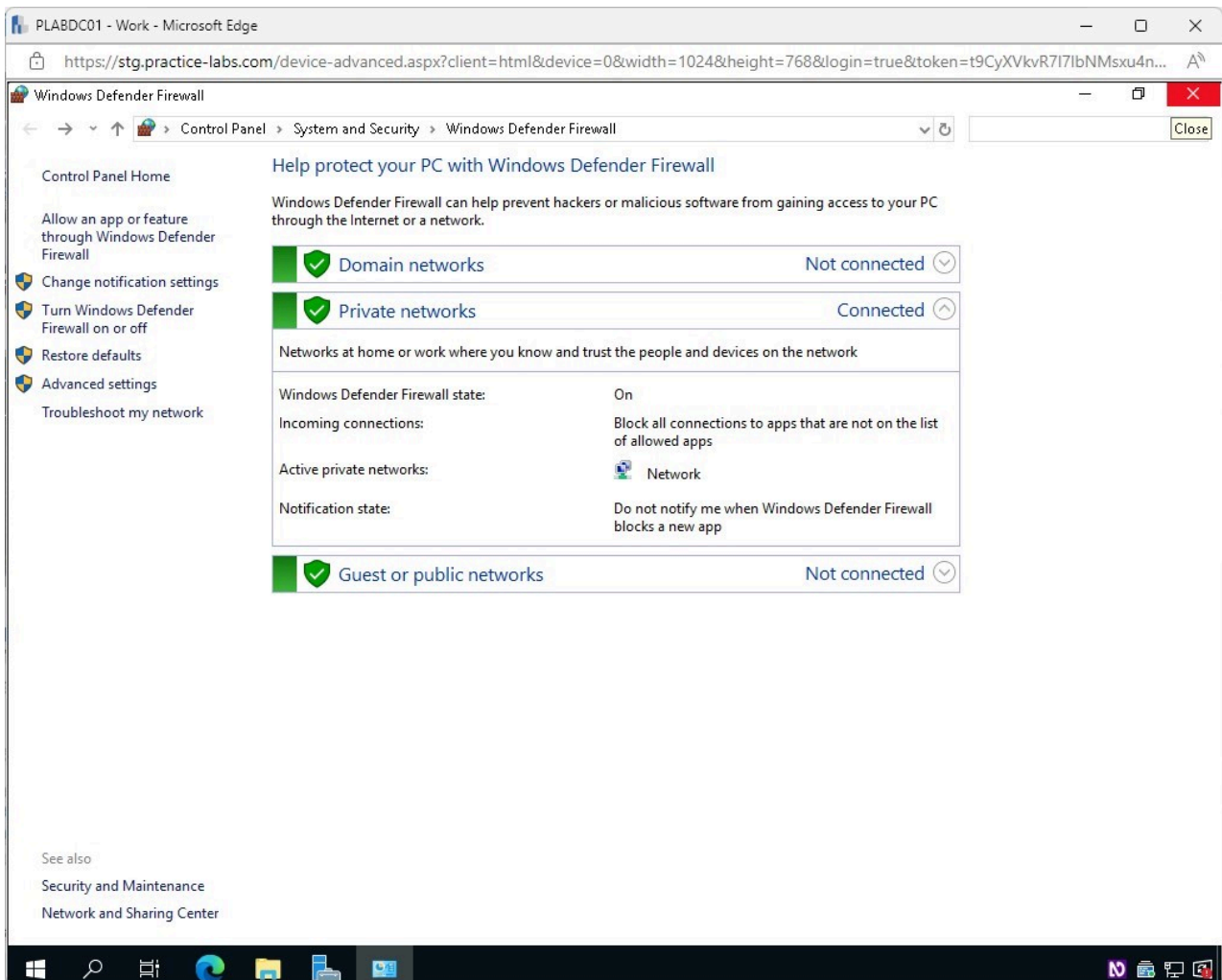


Figure 1.51 Screenshot of PLABDC01: Displaying closing the Windows Defender Firewall window.

Installing only Required Software

Businesses may implement a procedure to ensure they're installing just the programs they need. Users may help keep their data safer by limiting the software and programs that can be installed on their computers and mobile devices. With the high availability of software and applications, cybercriminals have additional entry points to compromise systems. Furthermore, such programs can frequently go too far and access data that the organization wants to keep private. It should be the standard to only use verified, trusted, and approved software and applications.

Exercise 2 - Host Security

The term host security is used to describe a group of security measures that are implemented directly on the host machine itself instead of the network. Host-based security solutions include the installation of updates and the deployment of Antivirus and Antimalware software. These will be installed directly on the computer and help prevent attacks on the device. They will also

help to clean up and resolve issues if a computer has been infected. Most often, these services will use commonly identified signatures of malware to detect, remediate and prevent infections. Heuristics can also be used by more advanced software.

In this exercise, you will learn about Windows Defender Antivirus. Third-party Antivirus and Antimalware, Host Intrusion Detection System (HIDS), and Host Intrusion Prevention System (HIPS) will be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore Windows Defender Antivirus

After completing this exercise, you should have further knowledge of:

- Third-Party Antivirus and Antimalware
- Host Intrusion Detection System (HIDS) and Host Intrusion Prevention System (HIPS)

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Windows Defender Antivirus

Microsoft Defender Antivirus is an anti-malware application that comes standard with Windows. It was initially made available as a free anti-spyware download for Windows XP and was later included as a pre-installed program with Windows Vista and Windows 7. It has evolved into a complete antivirus tool that comes standard with Windows 8 and subsequent versions, replacing Microsoft Security Essentials. Starting with Windows 10 and Windows Server 2016, users may now manage Defender Antivirus directly using the Windows Security app or PowerShell.

In this task, you will explore the Windows Defender Antivirus tool using Windows Security.

Step 1

Connect to **PLABDC01**.

Click the Windows **Start** charm. Scroll down the menu, and select **Windows Security**.

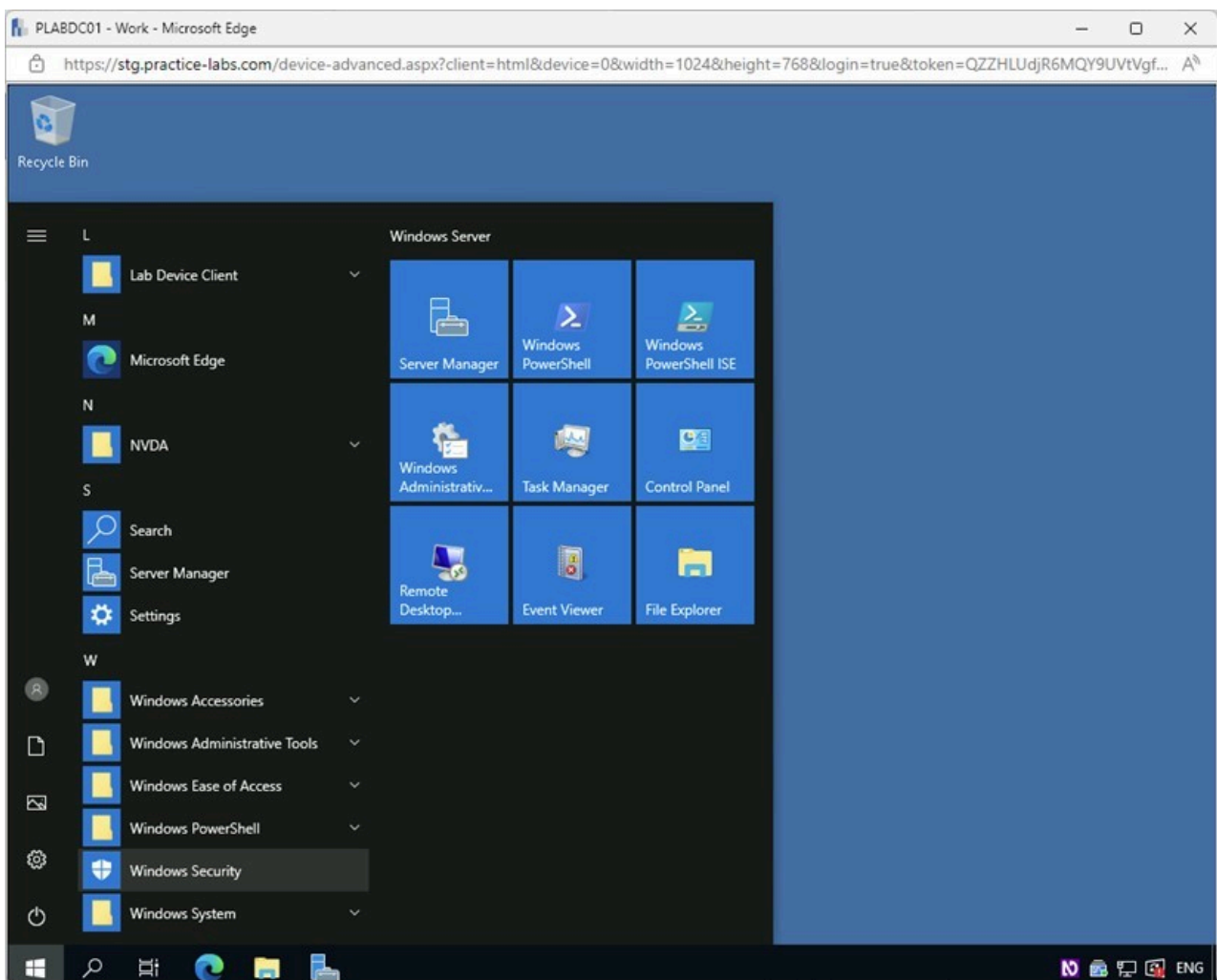


Figure 2.1 Screenshot of PLABDC01: Displaying selecting Windows Security from the Start menu.

Step 2

In the **Windows Security** window, select **Virus & threat protection**.

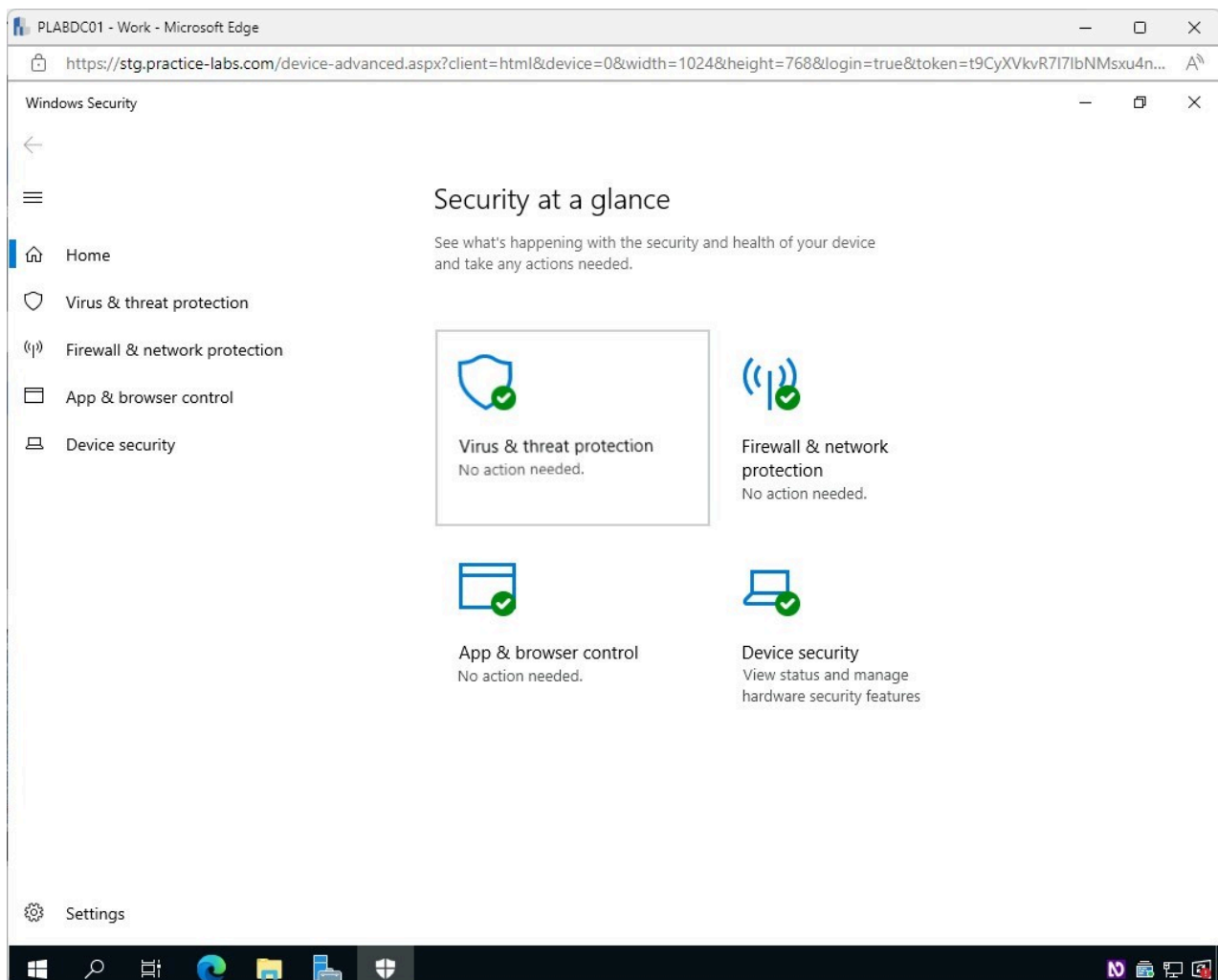


Figure 2.2 Screenshot of PLABDC01: Displaying selecting Virus & threat protection in the Windows Security window.

Step 3

In the **Windows Security - Virus & threat protection** pane, click **Quick scan** under the **Current threats** section.

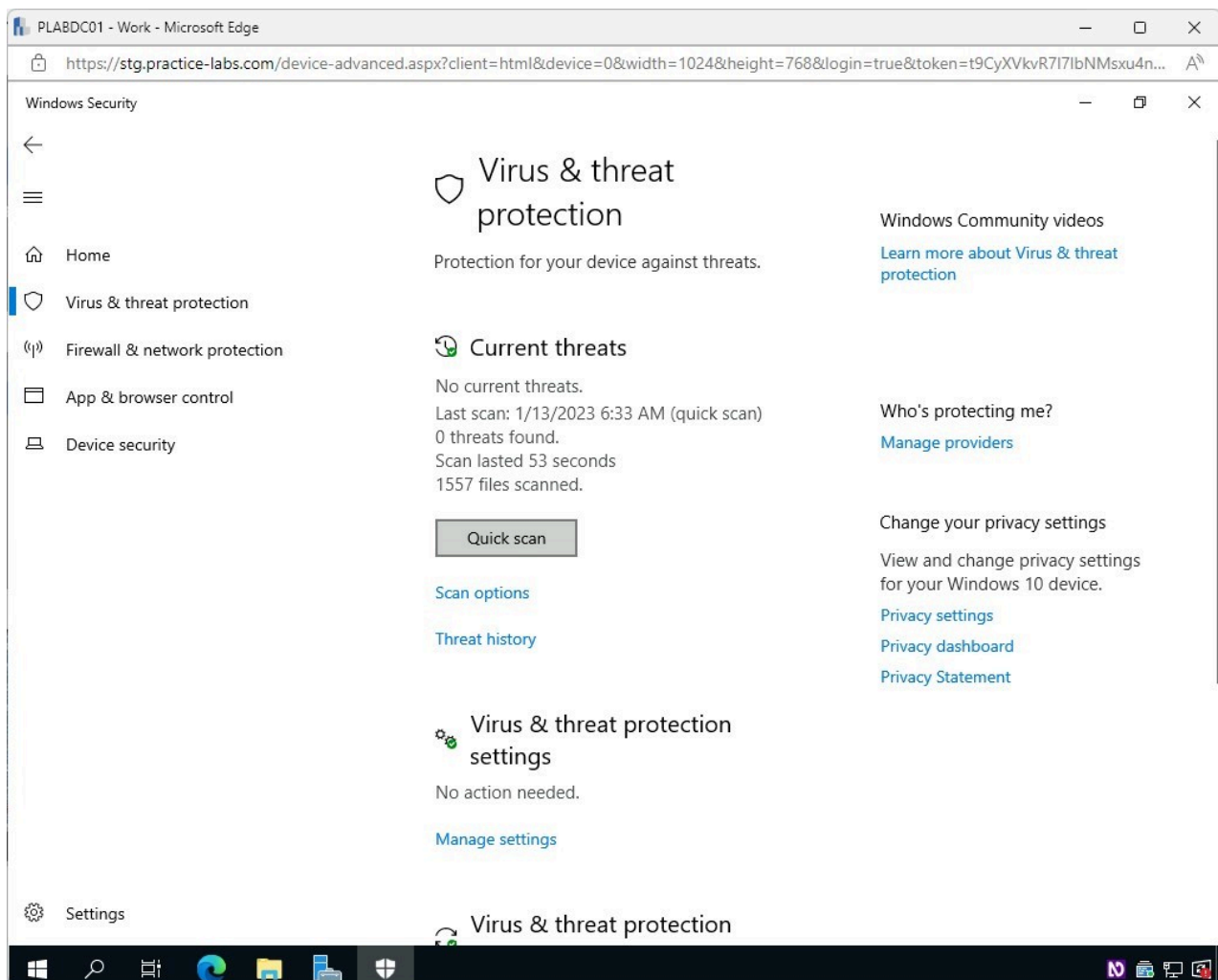


Figure 2.3 Screenshot of PLABDC01: Displaying selecting Quick scan in the Windows Security - Virus & threat protection pane.

Step 4

The **Quick scan** starts.

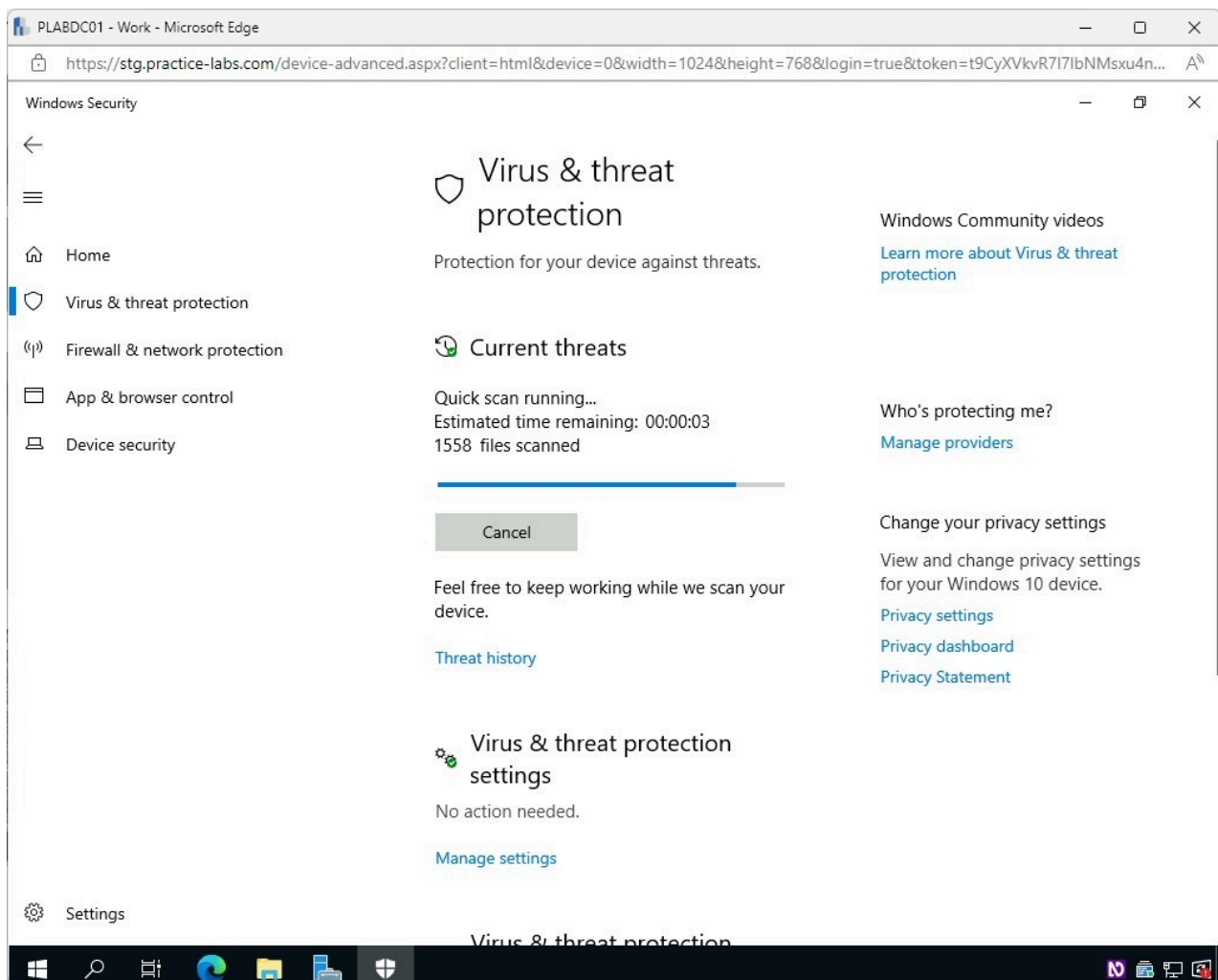


Figure 2.4 Screenshot of PLABDC01: Displaying the progress of the Quick scan in the Windows Security - Virus & threat protection pane.

Step 5

The scan completes with **No current threats**.

Click **Scan options** below the **Quick Scan** button.

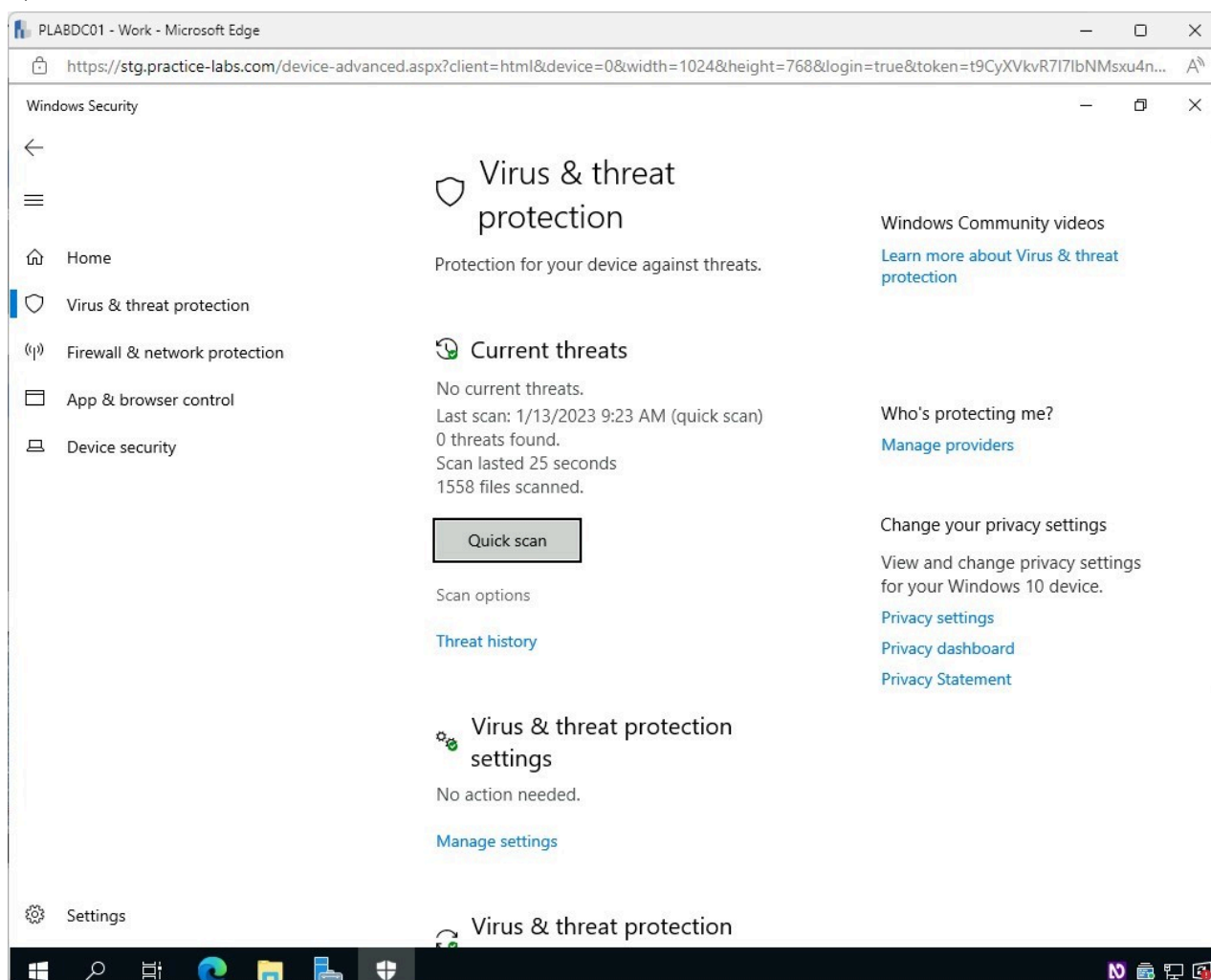


Figure 2.5 Screenshot of PLABDC01: Displaying Scan options selected in the Windows Security - Virus & threat protection pane.

Step 6

In the **Windows Security - Scan options** pane, select the **Full Scan** option and click **Scan now**.

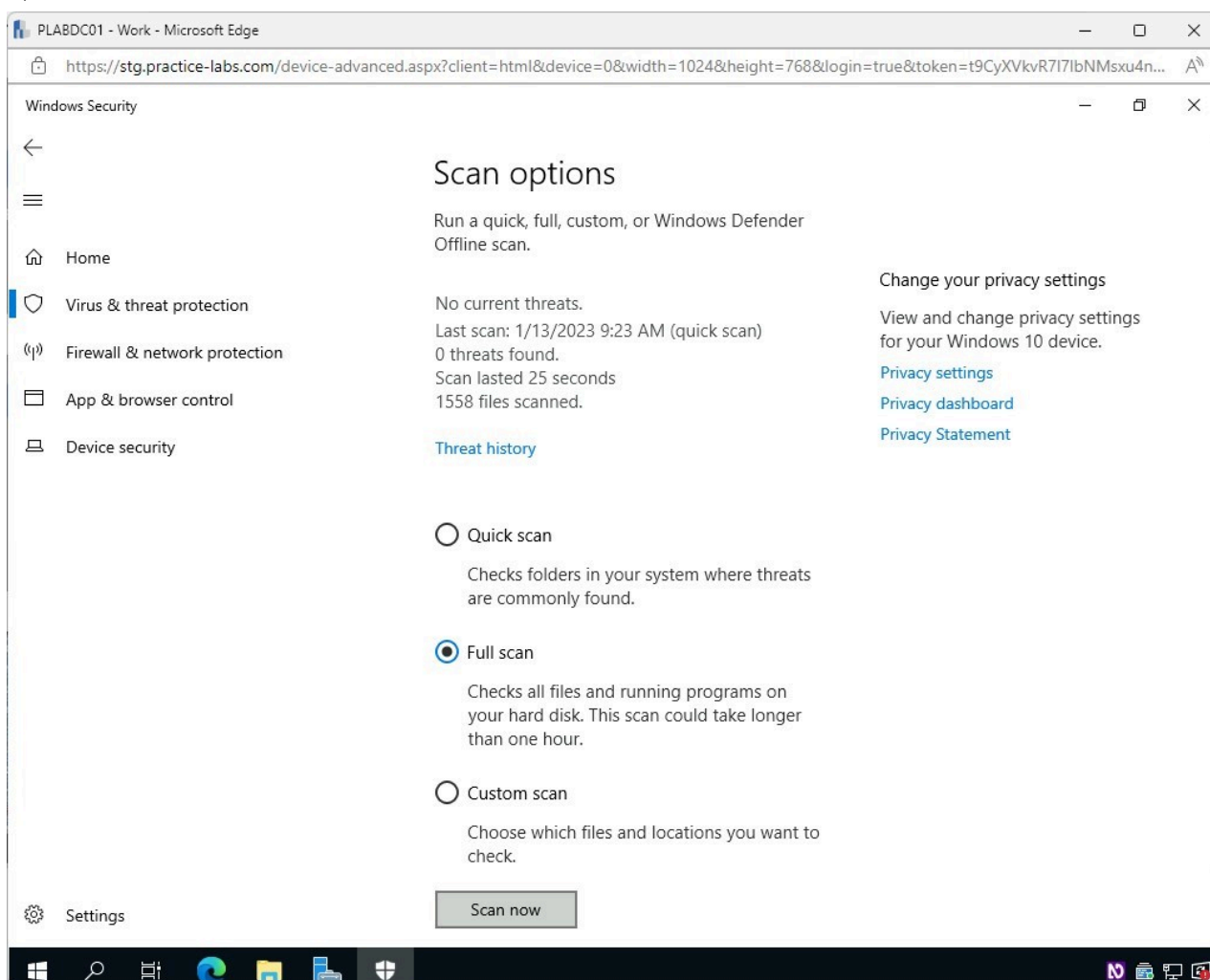


Figure 2.6 Screenshot of PLABDC01: Displaying selecting Scan now in the Windows Security - Scan options pane.

Step 7

The Full scan completes with **No current threats**.

Note: The Full scan completes fast because there were not a lot of files on this device. This process will usually take much longer to complete.

Select **Virus & threat protection** on the left pane.

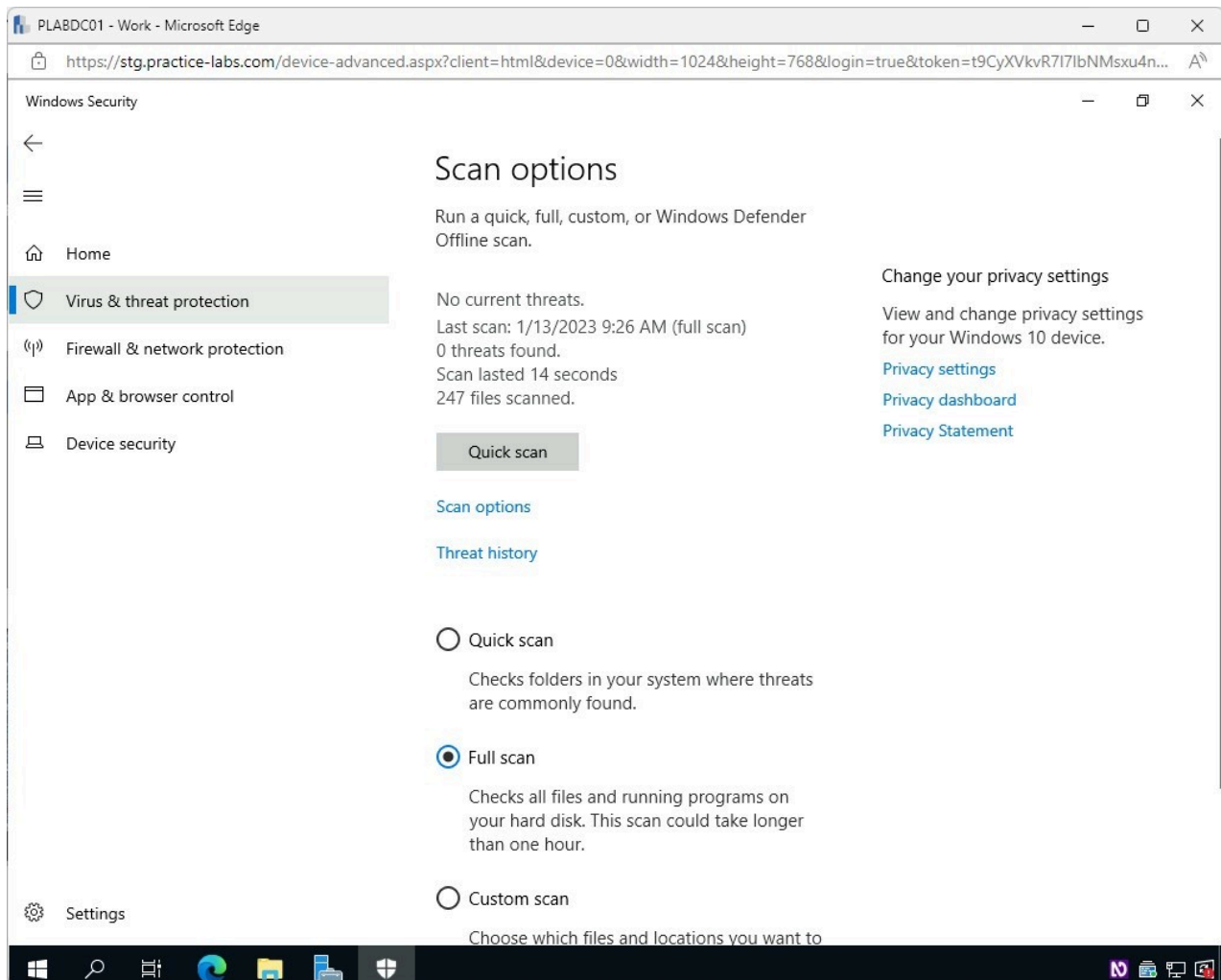


Figure 2.7 Screenshot of PLABDC01: Displaying selecting Virus & threat protection in the Windows Security window.

Step 8

The **Windows Security - Virus & threat protection settings** pane is displayed.

Review the information on the page.

Real-time protection watches for any malware that is being installed on the system and quarantines it until the malware can be reviewed.

Cloud-delivered protection ensures you have the most up-to-date definitions and the best chance of preventing malware from installing on the device.

You can also create exclusions if a valid application is being installed but is stopped, as well as configure notifications.

Close the **Windows Security - Virus & threat protection settings** window.

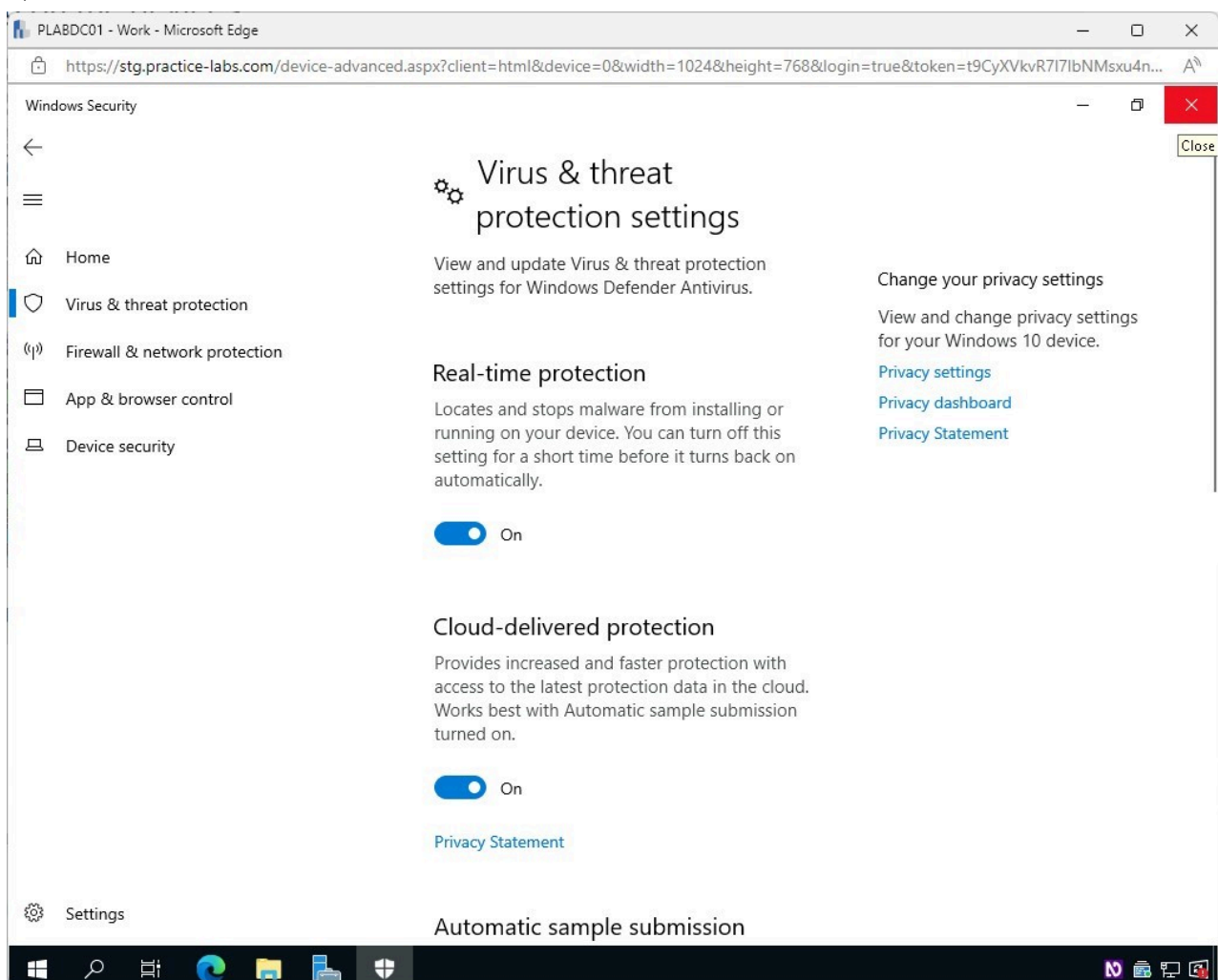


Figure 2.8 Screenshot of PLABDC01: Displaying closing the Windows Security - Virus & threat protection settings window.

Step 9

Close the **Windows Security** window.

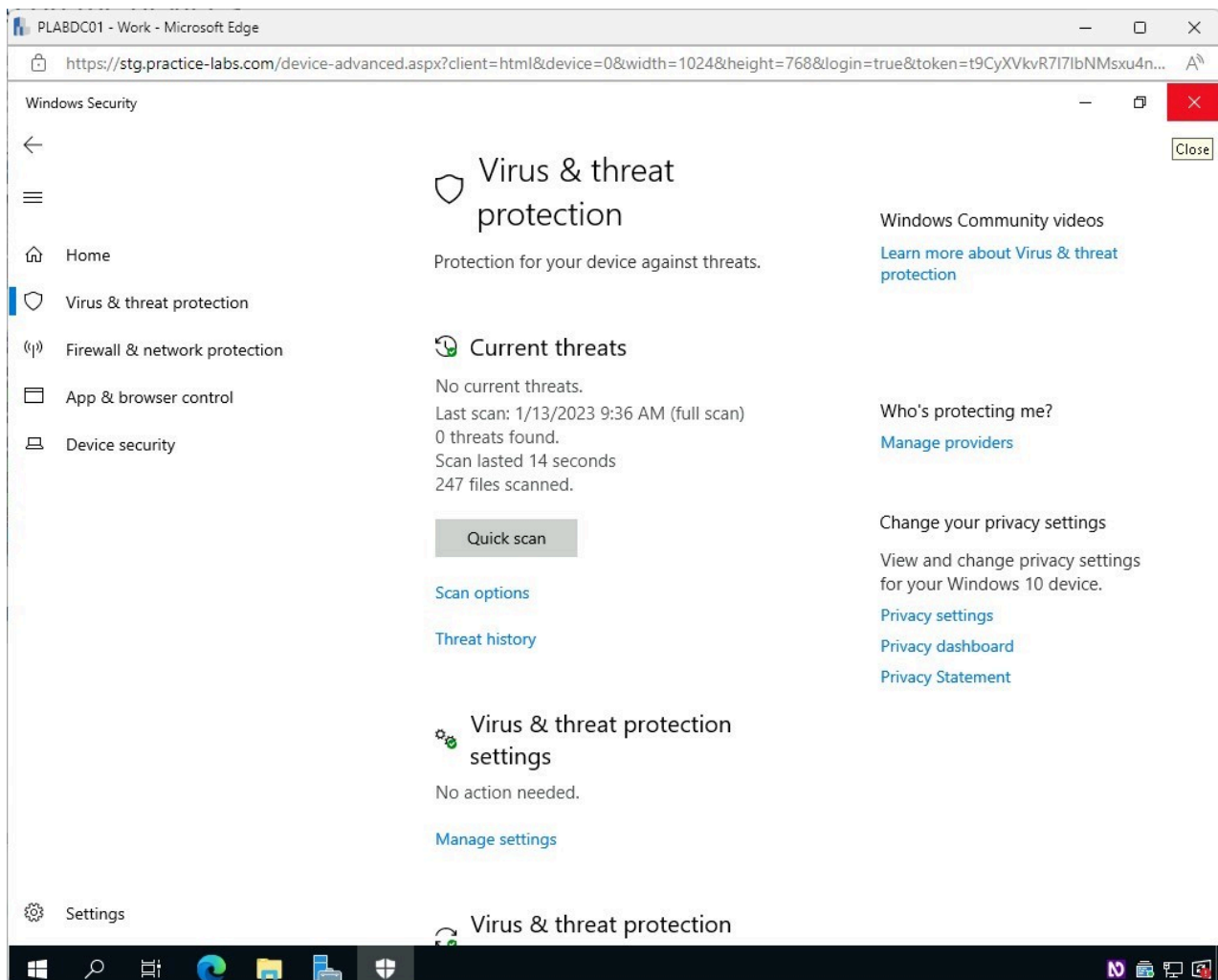


Figure 2.9 Screenshot of PLABDC01: Displaying closing the Windows Security window.

Third-Party Antivirus and Antimalware

Although Microsoft Defender Antivirus does a good job, it does have its shortcoming in lacking endpoint protection. Some users and businesses will go beyond the protections provided by Microsoft and utilize 3rd party Antivirus/Antimalware software. There are many providers of service who all have varying degrees of ability, customer satisfaction and pricing categories. There are free options that will often utilize ads and limited features to entice users to upgrade to paid versions.

Host Intrusion Detection System (HIDS) and Host Intrusion Prevention System (HIPS)

As the name implies, a host-based intrusion detection system (or host-based IDS) keeps tabs on the host computer. This is achieved by using specialized 3rd party software or hardware to monitor network activity for signs of abnormality and reporting them to the appropriate individuals. Your most crucial security systems' activity may be better understood with the help of a HIDS, but it will not provide you with a full picture of your security. With this resource,

you may monitor your network traffic to the host for any suspicious activity and take appropriate action to prevent damage. Data from your HIDS log should be easily cross-referenced with other critical security metrics and the most current threat intelligence.

Host Intrusion Prevention Systems (or host-based IPS) are similar to HIDS. They add an additional feature that can alert the user, log malicious behavior for later examination, reset the connection, drop malicious packets, and block further traffic from the suspect IP address, among other possible responses. Depending on the host intrusion prevention system, users may be able to submit logs of malicious activity and suspicious code fragments directly to the vendor for examination and probable identification.

There are two approaches that are typically used for HIDS/HIPS, and they are often used in tandem. Host Intrusion Detection/Prevention Systems that rely on Signatures look for something particular, whether it be a known pattern, an identifiable attacker, or a specific intrusion. As a result of their reliance on a definitions database, most IDSs require regular updates to stay up to date with the latest and most common cyber threats. Intruder detection systems of this sort work well if their databases are regularly updated. But attackers may tweak their attacks just enough to make databases fall behind in real time.

The second kind of system is an Anomaly-Based Host Intrusion Detection/Prevention System which relies more on studying trusted behavior and using machine learning techniques to highlight harmful activity. Consequently, there will be more false positives, as the algorithm will occasionally identify perfectly normal actions as suspicious. An anomaly-based IDS can help you spot network probing that occurs before a full-fledged assault. The effectiveness of an IDS relies not just on widespread deployment across the network but also on thorough IT admin training.

Exercise 3 - Hardware Hardening

Tools, strategies, and best practices in hardening can help an organization accomplish its goals of bolstering security, decreasing vulnerability, and shrinking the attack surface of its systems. These experts might fortify all types of technological components, including applications, systems, infrastructure, firmware, software, and hardware. For instance, they can close any unused or duplicate accounts, functions, apps, ports, rights, access, etc., that could be exploited by malicious actors.

In this exercise, you will learn to disable AutoRun and AutoPlay. Setting BIOS password and boot order will also be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Disable AutoRun and AutoPlay

After completing this exercise, you should have further knowledge of:

- Setting BIOS Password
- Setting Boot Order

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Disable AutoRun and AutoPlay

To assist in defending the network, you should disable unnecessary physical ports and prevent threat actors from plugging into a switch with port security. These kinds of equipment should not be freely available and should be kept behind closed doors, but what about our computers? They are likely to be visible and easily accessible. You can disable the Windows AutoRun feature that allows programs from external devices to be launched as soon as they are connected to the computer.

It is a broader setting that governs what happens when you insert a USB drive or a CD/DVD into a computer drive. AutoPlay is a Windows feature that works in conjunction with AutoRun. AutoPlay suggests that you listen to music, watch videos, or view images. These measures of convenience come at a cost to security. AutoRun can expose your computer to malware. Because malware can use the AutoRun option to spread a virus from an external device to your device, many users disable it.

In this task, you will learn how AutoRun and AutoPlay can be disabled.

Step 1

Ensure you are connected to **PLABDC01**.

Restore the **Server Manager** window from the **Taskbar**.

In the **Server Manager** window, click the **Tools** menu and select **Group Policy Management**.

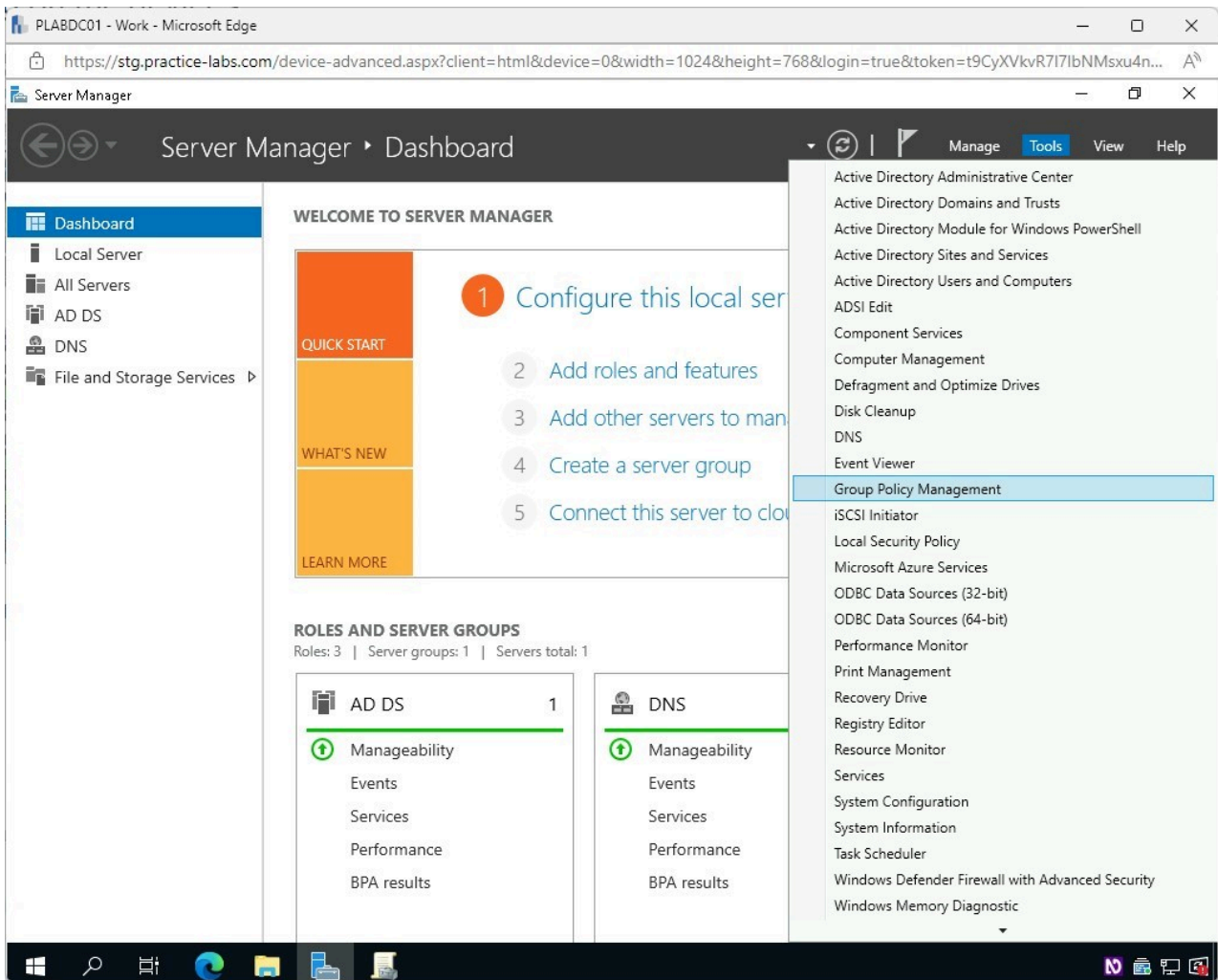


Figure 3.1 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Group Policy Management is selected.

Step 2

In the **Group Policy Management** window, you will create a new **Group Policy Object** to link the rule that will be created.

In the left pane, expand **Forest: practicelabs.com > Domains > practicelabs.com**.

Right-click on **Group Policy Objects** and select **New**.

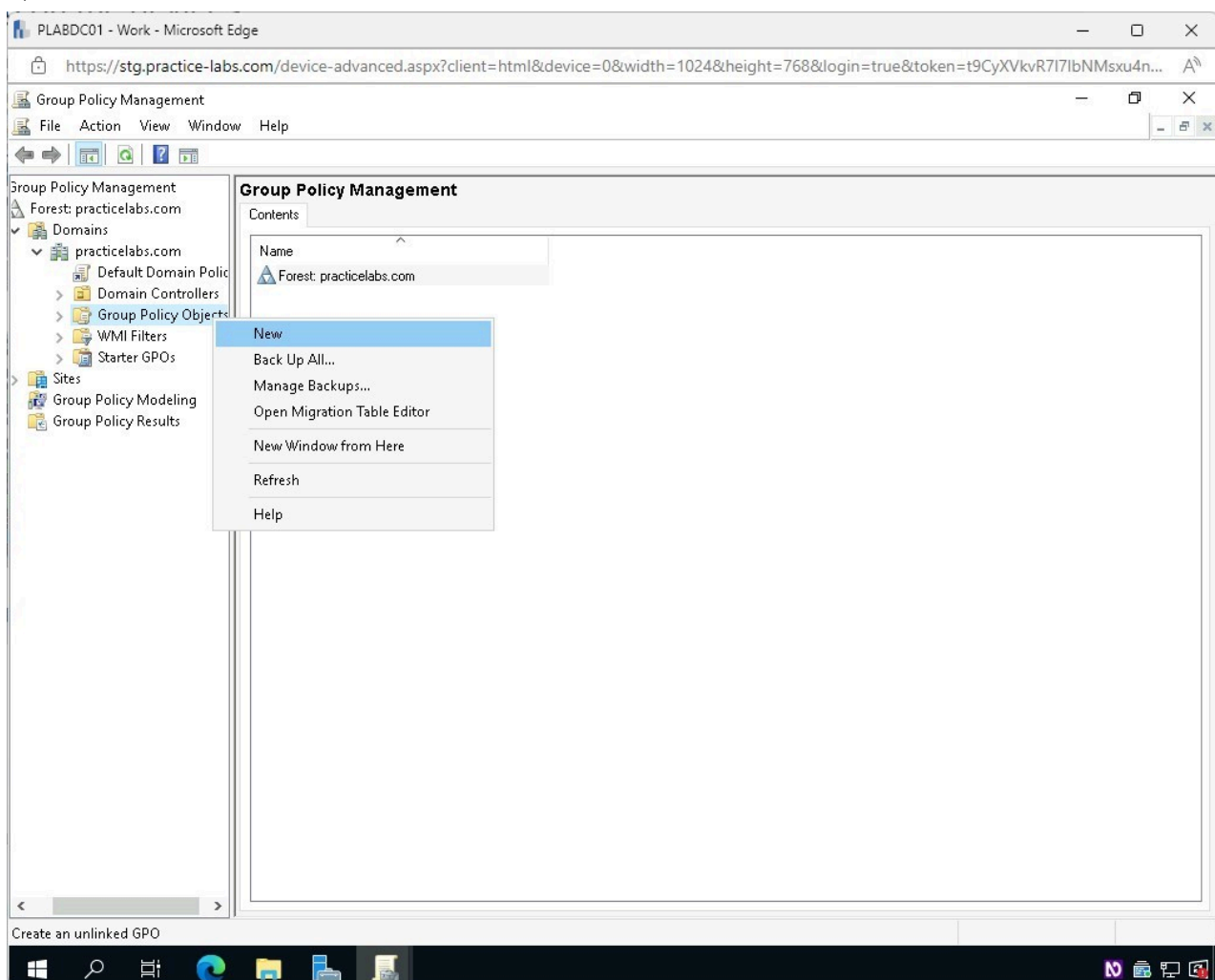


Figure 3.2 Screenshot of PLABDC01: Displaying the Group Policy Management window. Group Policy Objects is right-clicked, and New selected.

Step 3

In the **New GPO** window, type the following for the **Name** field and click **OK**:

Disable Autorun

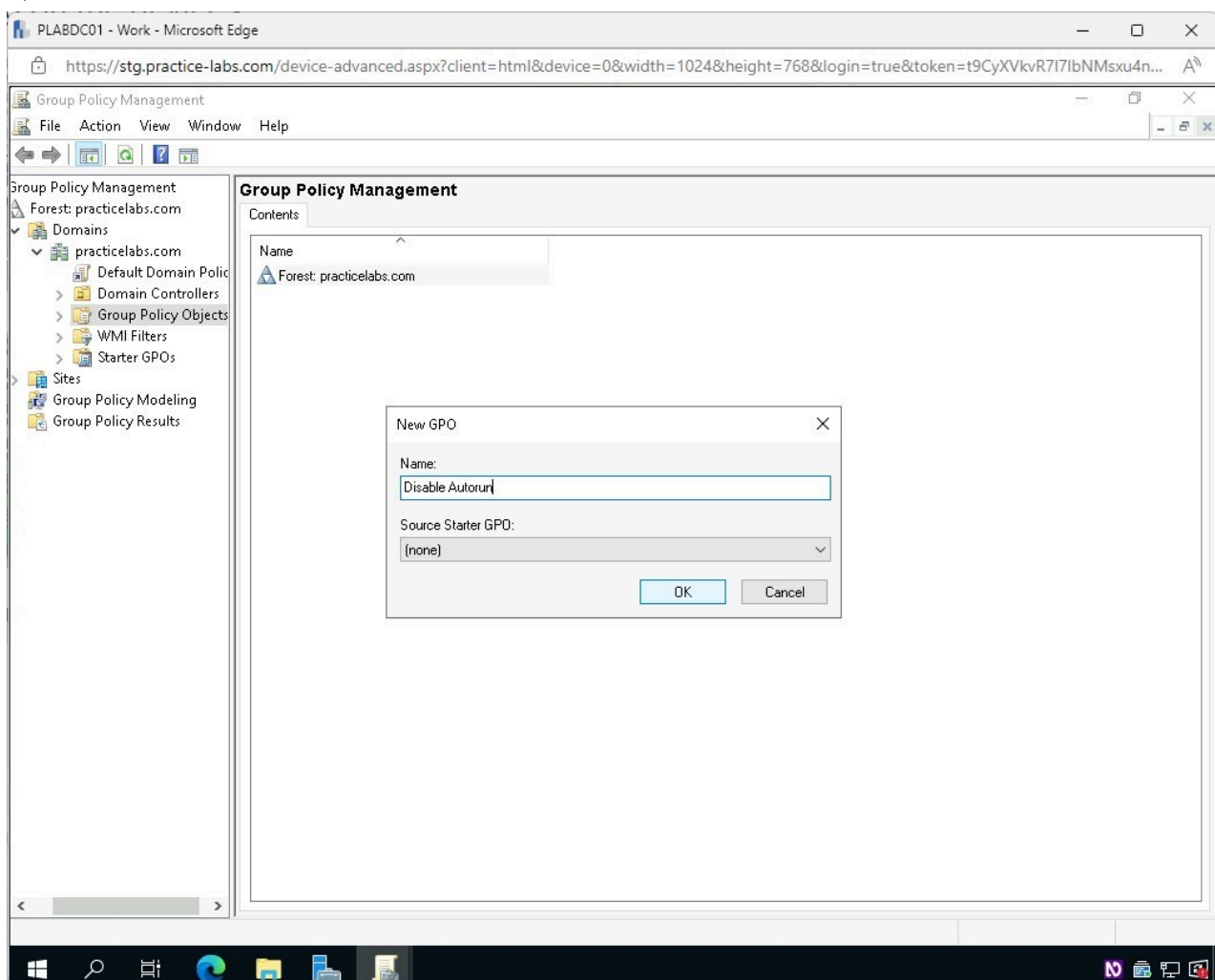


Figure 3.3 Screenshot of PLABDC01: Displaying the New GPO window, showing the required Name entered and the OK button selected.

Step 4

Back on the **Group Policy Management** window, expand the **Group Policy Objects** node.

Right-click on **Disable Autorun** and select **Edit**.

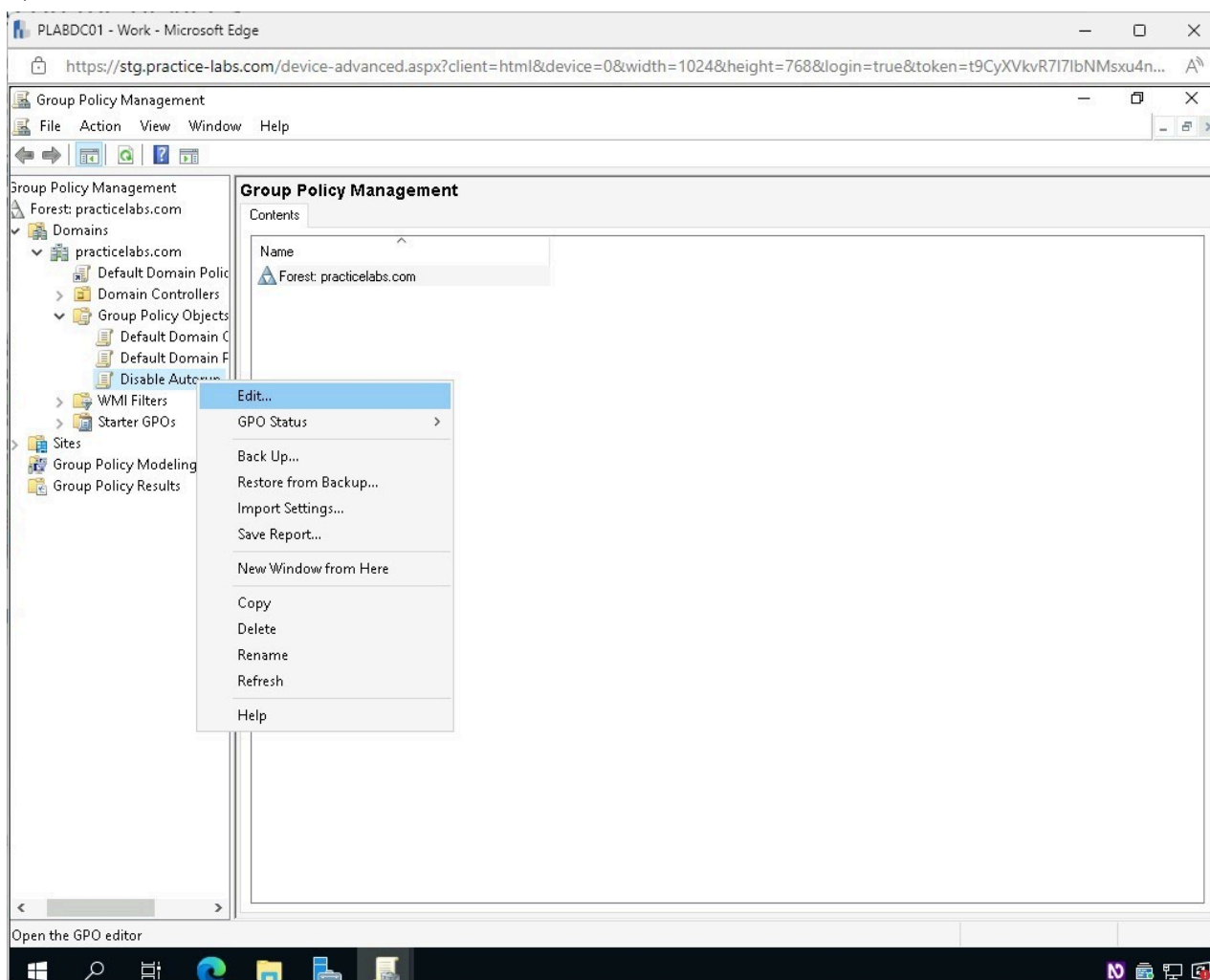


Figure 3.4 Screenshot of PLABDC01: Displaying the Group Policy Management window. Disable Autorun is right-clicked, and Edit selected.

Step 5

In the **Group Policy Management Editor** window, expand the following nodes in the left pane:

Computer Configuration > Policies > Administrative Template > Windows Component.

Select **Autoplay Policies**.

In the right-most pane, double-click on **Turn off Autoplay**.

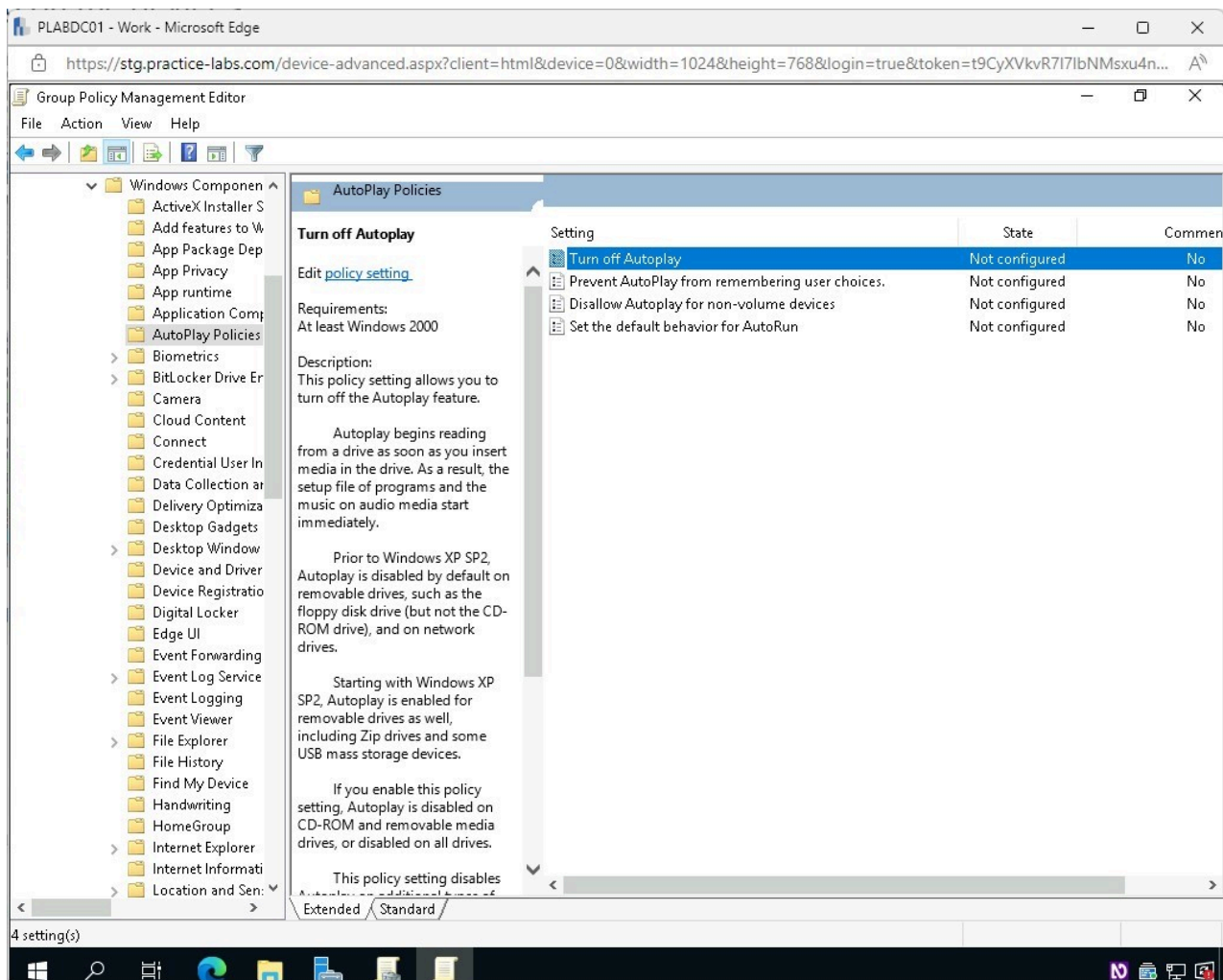


Figure 3.5 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Turn off Autoplay selected.

Step 6

In the **Turn off Autoplay** window, read the information available for this setting.

Click **Next Setting**.

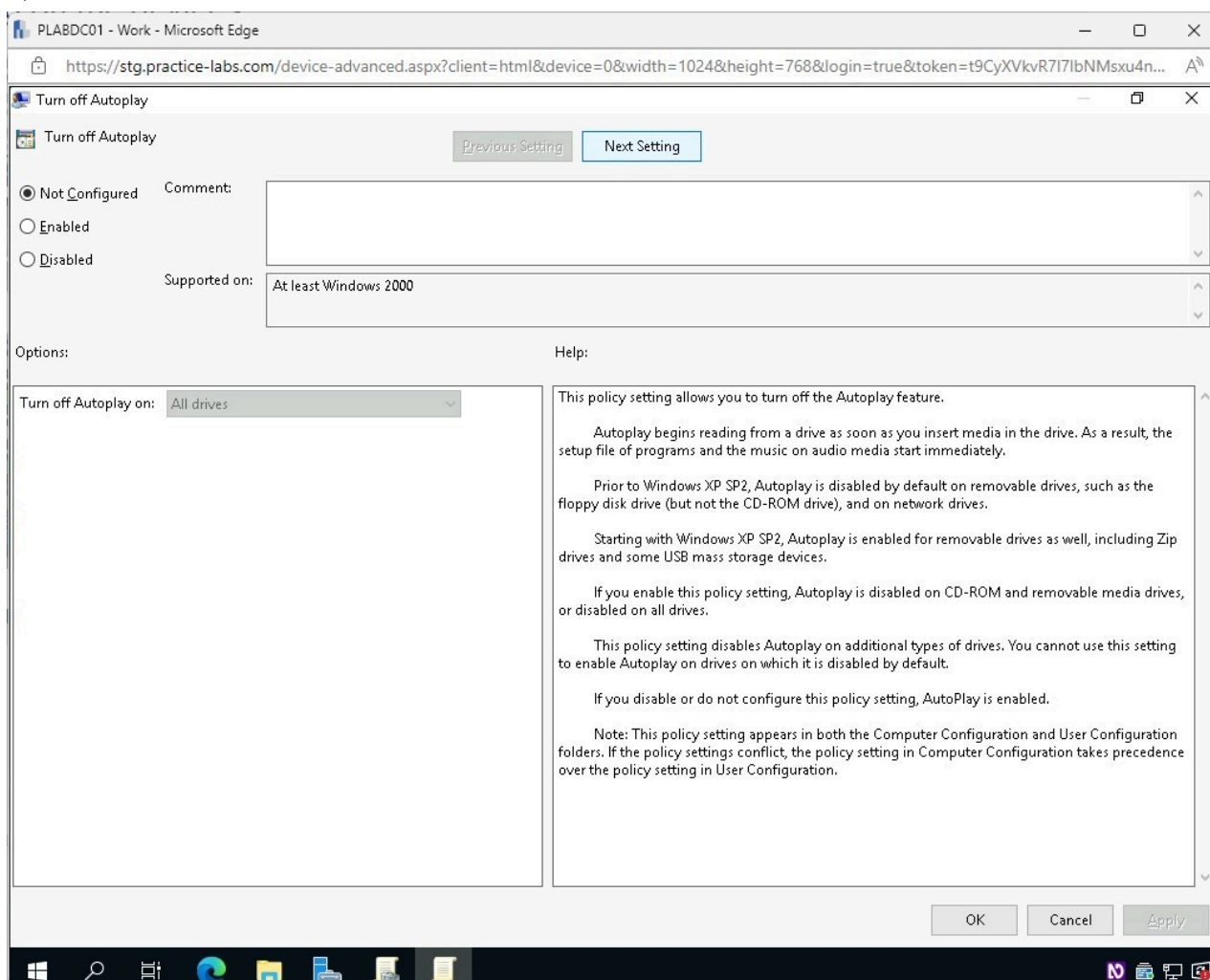


Figure 3.6 Screenshot of PLABDC01: Displaying the Turn off Autoplay window, showing Next Setting selected.

Step 7

In the **Prevent Autoplay from remembering user choices** window, read the information available on this setting.

Once done, click **Next Setting**.

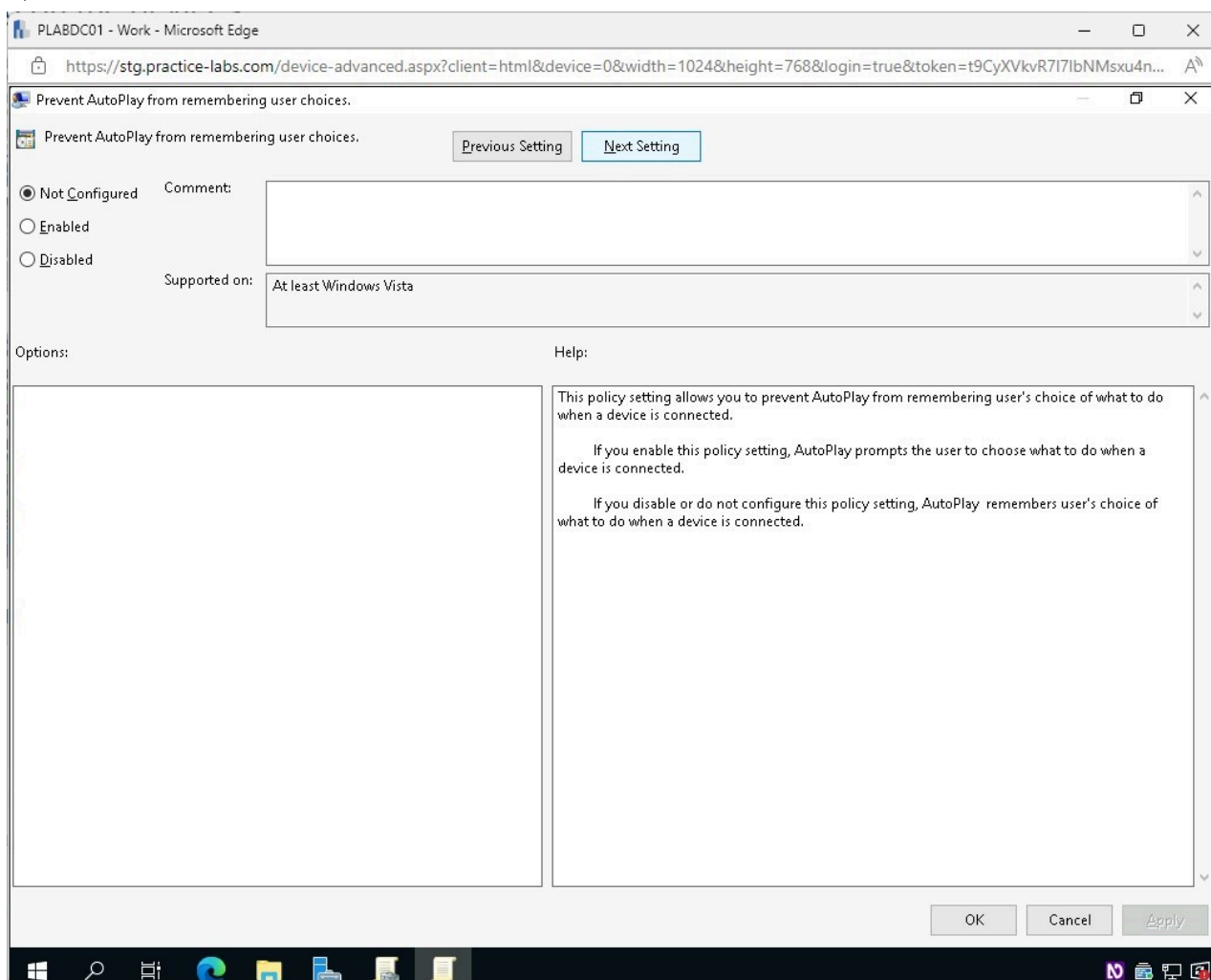


Figure 3.7 Screenshot of PLABDC01: Displaying the Prevent AutoPlay from remembering user choices window, showing Next Setting selected

Step 8

In the **Disallow Autoplay for non-volume devices** window, read the information on this setting and click **Next Setting**.

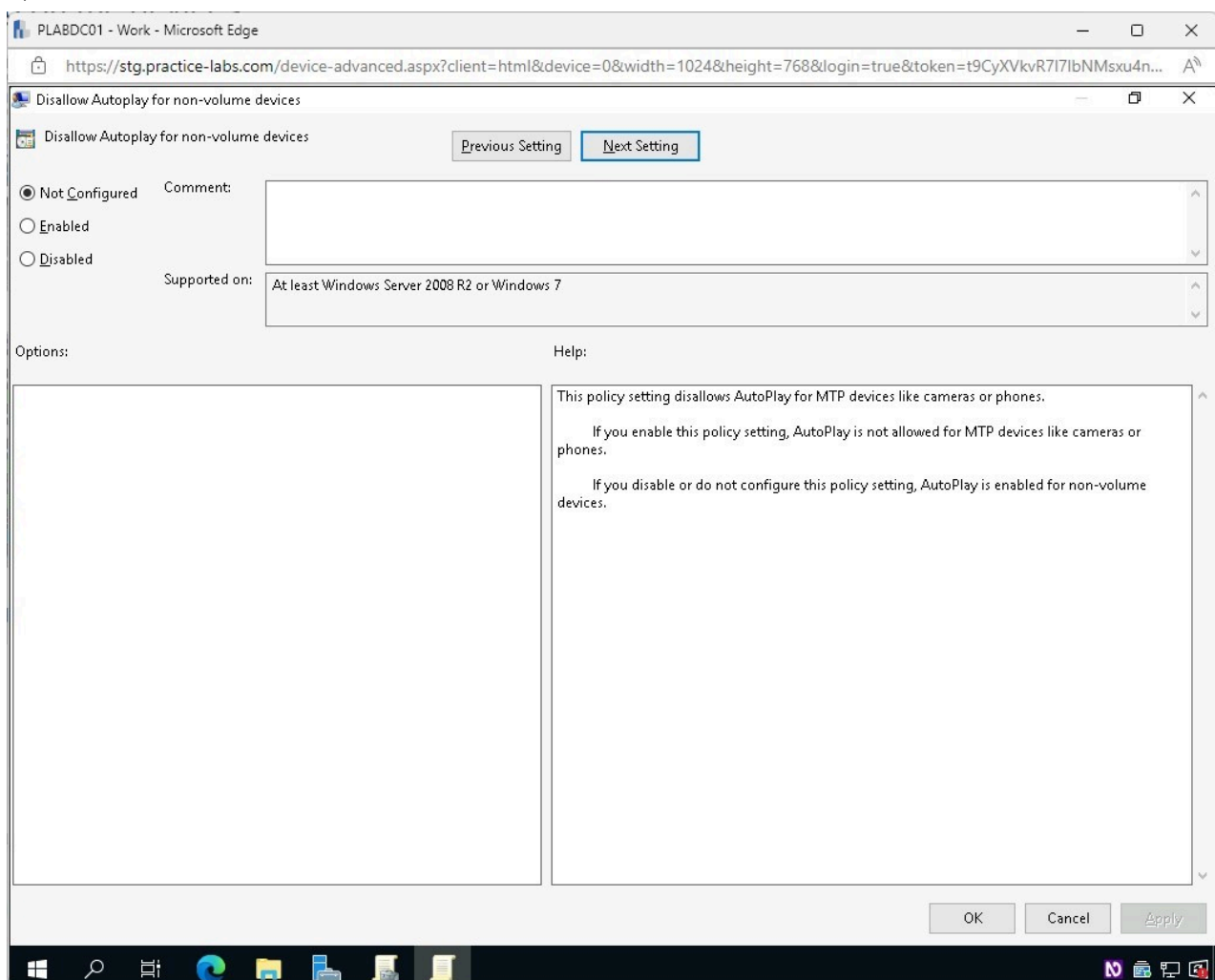


Figure 3.8 Screenshot of PLABDC01: Displaying the Disallow Autoplay for non-volume devices window, showing Next Setting selected.

Step 9

The **Set the default behavior for AutoRun** window is displayed.

Read the details on this setting, and click **OK**.

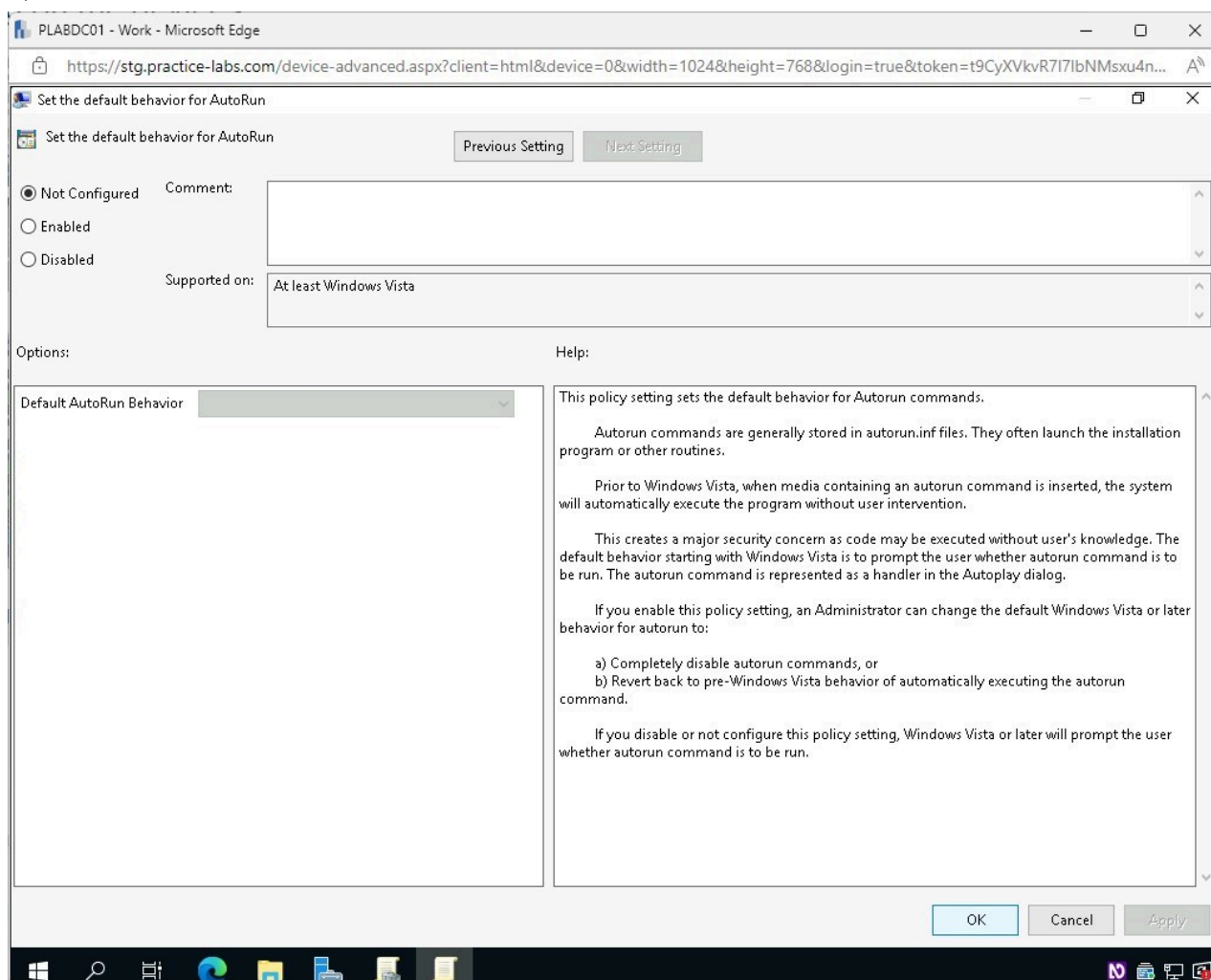


Figure 3.9 Screenshot of PLABDC01: Displaying the Set default behavior for AutoRun window, showing OK selected.

Step 10

Close the **Group Policy Management Editor** and **Group Policy Management** windows.

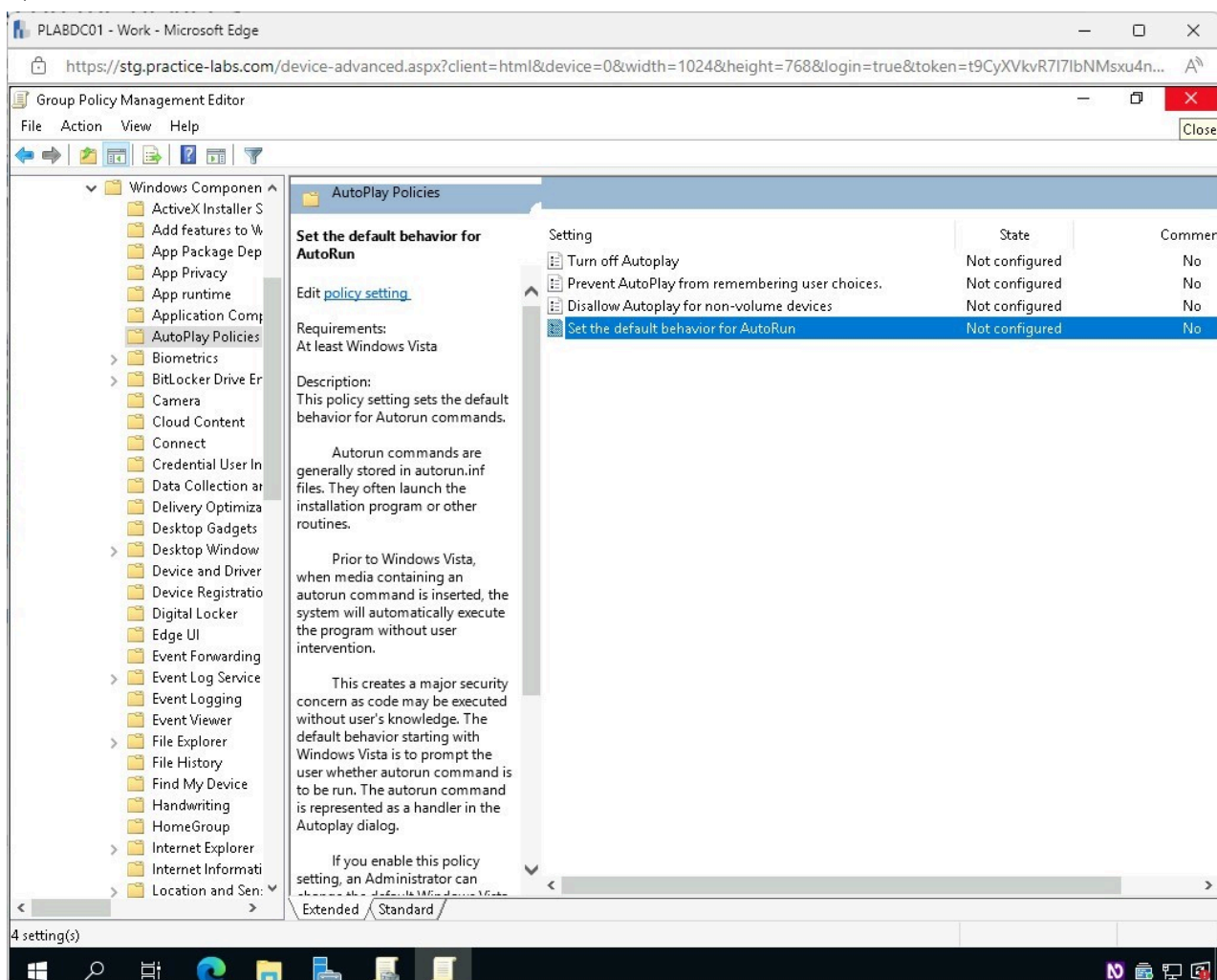


Figure 3.10 Screenshot of PLABDC01: Displaying closing the Group Policy Management Editor window.

Setting BIOS Password

BIOS is an abbreviation for Basic input/output system, and it is the firmware that is used to start the computer. UEFI, which stands for Unified Extensible Firmware, is a BIOS update. Essentially, one of these programs is in charge of the computer's boot up process. The menu is accessible when the computer boots up. Each manufacturer is different, but the start-up splash page will appear once the computer boots up, and the desired button to be pushed will be displayed on the screen. While this screen is displayed, typically the enter key, backspace, F2, or F10 can be pressed to access the BIOS/UEFI menu.

Password protection can be enabled to help protect the computer during the boot process. If this feature is not enabled, skilled threat actors can bypass the logon screen entirely by gaining access to sensitive data via the computer's BIOS or UEFI. To help prevent this, a password can be set in BIOS or UEFI to require users to authenticate for access.

In these menus, the many crucial settings to the computer's operation can be configured.

- ACPI can be enabled or disabled

- Activate or deactivate the computer logo
- Activate or deactivate the Quick Power On Self Test (POST)
- Alter the Boot Order
- Alter the Boot Up Status of NumLock
- Alter the Fan Speed Settings
- Alter the System Voltages
- BIOS Caching can be enabled or disabled
- BIOS Control of System Resources can be enabled or disabled
- Check the CPU and System Temperatures
- In a hurry (Update) BIOS
- Load BIOS Configuration Defaults
- Look at the Fan Speeds
- Look at the System Voltages
- Make a BIOS password
- Modify the ACPI Suspend
- Modify the CD/DVD/BD Drive Settings
- Modify the CPU Settings
- Modify the Date and Time
- Modify the Floppy Drive Settings
- Modify the Hard Drive Settings
- Modify the Memory Settings
- Modify the Power Button's Function
- Modify the Power-On Settings
- On multi-display setups, you can change which display is initialized first
- Onboard audio can be enabled or disabled
- Onboard Floppy Controller can be enabled or disabled
- Onboard IEEE1394 can be enabled or disabled
- Onboard Serial/Parallel Ports can be enabled or disabled
- Onboard USB can be enabled or disabled
- RAID can be enabled or disabled
- Reset the Extended System Configuration Data (ESCD) (ESCD)
- Toggle the CPU Internal Cache on or off
- Unlock a BIOS Password
- View the Memory Installed

Setting Boot Order

A boot sequence describes the initial set of operations carried out by a computer every time it is turned on. For a user to be able to interact with their computer, the system must first perform a

series of self-checks and diagnostics, during which the computer activates the required hardware components and loads the proper operational software. The boot sequence, often known as boot options or boot order, specifies the order in which various devices are probed to locate and load an operating system. It also details the sequence in which each device on the computer is verified. This process can be locked in the BIOS to ensure that a proper device is used to load the operating system. Common options for a computer to boot from are USB, Optical media, Network, Solid-state/flash drives, Internet-based, External/hot-swappable drives, and Internal hard drives (partition).

Exercise 4 - Application Hardening

Application hardening, like hardening systems, requires uninstalling unnecessary features and services, limiting user access, and ensuring that the software is always patched. Application security is crucial to keep programs secure and accessible to users. During development, code must be reviewed often. To maintain safety compliance, an ongoing review is required. Once end users install apps, they must maintain the most recent version. Application hardening and updating have benefits such as minimizing financial loss, protecting brand image, increasing software sales, closing security vulnerabilities and having a better application.

In this exercise, installing updates, patches and deployment will be discussed. You will also learn about change management and the importance of disabling unneeded services, roles, or features.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Installing Updates, Patches and Deployment
- Change Management
- Disabling Unneeded Services, Roles, or Features

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Install Updates, Patches and Deployment

When reviewing applications and discovering vulnerabilities in the code, developers will create patches to resolve the problem. Businesses should test these updates in a sandbox and deploy them to all systems necessary once satisfied. A single computer is relatively simple to manage, but managing all the application updates on a network with hundreds or thousands of users is a laborious ordeal. Fortunately, you may find third-party products to assist in lightening the load and expand the capabilities of Microsoft WSUS and SCCM to work with other software.

The administrators will be able to see a comprehensive update list for critical, security, definition, third-party, and service pack software with the use of these tools. Aside from the panel in the third-party program, you may get these update notifications by email. Several actions are available to administrators, including selecting the update, specifying the group of computers to update, accepting or rejecting the update, and scheduling the deployment.

Change Management

When making any changes to the networks and operating systems, the changes being conducted should be recorded. When applying updates and patches, systems will need to go offline to restart and initialize the changes. This will require downtime, so any updates and patches are subject to the change management process.

Change Management is a subset of IT Service Management that streamlines the implementation of requested modifications to your IT systems. Change management assists in the process of requesting, prioritizing, authorizing, approving, scheduling, and implementing alterations to the network. A well-executed IT Change Management approach aids in risk mitigation and service continuity regardless of the complexity of the change being implemented. If the update or patch fails, there are steps in place to ensure the continuity of service.

Disable Unneeded Services, Roles, or Features

Using weaknesses and bugs in an application's code, threat actors can get into a system without permission in many ways. Developers can make verification necessary by limiting access for roles like "guest accounts." By limiting access to these features, you can make it impossible for unauthorized services or ports to be exposed. Your system is more likely to be attacked if it has a lot of services, roles, and features. For input validation and authentication, application developers should always use standard procedures. So, hackers and threat actors can't take advantage of the software and use it for malicious purposes.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Server and Application Hardening Techniques** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Server and Host OS Hardening
- Exercise 2 - Host Security
- Exercise 3 - Hardware Hardening
- Exercise 4 - Application Hardening

You should now be able to:

- Configure Services
- Apply Driver Updates
- Apply Firewall Configuration
- Explore Windows Defender Antivirus
- Disable AutoRun and AutoPlay

You should now have further knowledge of:

- Installing only Required Software
- Third-Party Antivirus and Antimalware
- Host Intrusion Detection System (HIDS) and Host Intrusion Prevention System (HIPS)
- Setting BIOS Password
- Setting Boot Order
- Installing Updates, Patches and Deployment
- Change Management
- Disabling Unneeded Services, Roles, or Features

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.